A BACTERIAL GUANYLYL CYCLASE BRIDGES ANTIPHAGE RESPONSE WITH MULTIPLE CELLULAR PROCESSES FROM UROPATHOGENIC ESCHERICHIA COLI

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THESIS

Submitted in partial fulfillment of the requirements
for the degree of Master of Science in Microbiology
in the Graduate College of the University of Illinois
at Urbana-Champaign, 2026

Urbana, Illinois

Notice on Revisions

A Bacterial Guanylyl Cyclase Bridges Antiphage Response with Multiple Cellular Processes from Uropathogenic Escherichia coli

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Dedication

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Abstract

This thesis defines a unified antiphage defense in Uropathogenic *Escherichia coli* in which a guanylyl cyclase sensor, a ClyA-like pore-forming effector, and bacteriophage proteases are wired into a single regulatory axis. In Chapter 2, *in silico* analyses establish that the KTE100 cyclase is an unprecedented guanylyl cyclase–HEPN fusion whose active site simultaneously encodes nucleotide binding and HEPN-like nuclease potential, reinforced by flanking EXXX[RK] motifs that consolidate a bifunctional RNA-sensing platform. The cognate effector assembles as a three-transmembrane, hexameric ClyA-like pore-forming toxin with a conserved N-glycosylation motif and clustered post-translational modification candidates, defining a structurally novel effector class within bacterial innate immunity.

Chapter 3 shows that ectopic expression of antiphage guanylyl cyclases from *E. coli* and *Pseudomonas fluorescens* rewires cell architecture and lifestyle decisions even in the absence of phage. Cyclase expression elevates optical density without increasing viable cell counts and drives a gram-positive-like staining phenotype in *E. coli*, consistent with cGMP-dependent thickening of the cell wall or exopolysaccharide coat. By competing with diguanylyl cyclases for GTP, the *E. coli* cyclase enhances swimming motility and suppresses biofilm biomass in a manganese-dependent manner, indicating that basal cGMP production depresses c-di-GMP signaling and favors planktonic over sessile lifestyles while leaving the ClyA-like effector below its toxic activation threshold.

In Chapter 4, phage assays show that the cyclase–effector pair constitutes a *bona fide* abortive infection system against phages P1, lambda, and T6. Escaper phages reproducibly acquire mutations in head-processing proteases and capsid genes, and chimeric cyclases with swapped N-termini localize phage sensitivity to an intrinsically disordered N-terminal spine enriched for protease and phosphorylation motifs. These findings support a model in which phage-mediated remodeling of this spine relieves autoinhibition, drives cGMP above a critical threshold, and licenses ClyA-like pore formation to terminate the infected cell.

Table of Contents

Dedication....ii
Acknowledgements....iii
Abstract....iv
CHAPTER 1: INTRODUCTION....1
1.1 Historical Context: Bacteriophage Discovery and the Origins of Molecular Biology....1
1.2 The Red Queen Hypothesis and the Arsenal of Bacterial Antiphage Defenses....2
1.3 Cyclic Oligonucleotide-Based Antiphage Signaling Systems (CBASS): Architecture and Logic....3
1.3.1 The CD-NTase Family: Structural Conservation and Second-Messenger Diversity....3
1.3.2 Effector Diversity and Mechanisms of Abortive Infection....4
1.3.3 Cyclic Oligonucleotide Recognition and Linkage Specificity....5
1.3.4 Regulatory Mechanisms and Signal Integration in CBASS Immunity....5
1.4 A Novel CBASS-Like System in Uropathogenic *Escherichia coli*: Guanylyl Cyclase and ClyA Pore-Forming Toxin....8
1.4.1 Distinguishing Features: Conservation Versus Novelty....8
1.4.2 Criteria for Establishing Novelty....9
1.5 HEPN Domains in Cyclic Nucleotide Synthases: A Focused Literature Analysis 9
1.5.1 HEPN Domain Structure and Canonical Functions....10
1.5.2 HEPN Domain Architectures in Bacterial Defense Systems....10
1.5.3 Critical Finding: No Evidence for HEPN Domains in Cyclic Nucleotide Synthases....11
1.5.4 Interpreting the RDSTKH Motif: Plausible Explanations....12
1.6 Activation Mechanisms in Bacterial Antiphage Defense: Proteolytic Triggers and the SOS Response....12
1.6.1 Phage-Encoded Proteases as Determinants of Immune Activation....13
1.6.2 Candidate Proteolytic Triggers: Bridging Phage Morphogenesis and Host Defense....13
1.7 Thesis Gap and Experimental Objectives....14
Abbreviations....16
References....16
CHAPTER 2: *IN SILICO* ANALYSES OF AN ANTIPHAGE CYCLASE AND EFFECTOR OF UROPATHOGENIC *ESCHERICHIA COLI* STR. KTE100....22
2.1 Abstract....22
2.2 Introduction....23
2.3 Results....24
2.3.1 The cyclase and effector co-occur in Uropathogenic Enterobacteriaceae…24
2.3.2 The sensor harbors a bifunctional guanylyl cyclase–HEPN domain....25

2.3.3 Candidate post-translational modification residues localize to regions of sequence divergence....27
2.3.4 The cyclase dimer engages GTP at the dimerization interface and accommodates RNA in the active-site cleft....28
2.3.5 The effector is a ClyA-like α-pore-forming toxin with candidate PTM residues....29
2.4 Discussion....30
2.5 Materials and Methods....33
2.5.1 Database and sequence searches....33
2.5.2 Multiple sequence alignment and domain analysis....33
2.5.3 Post-translational modification site prediction....33
2.5.4 Structural prediction and visualization....33
2.6 Figures....35
Figure 2.1. The genetic neighborhood of the KTE100 cyclase and effector defense operon....35
Figure 2.2. Sequence-conservation architecture of the KTE100 cyclase....37
Figure 2.3. GTP-binding architecture and HEPN-domain placement of the KTE100 cyclase dimer....39
Figure 2.4. The KTE100 cyclase dimer accommodates a single-stranded RNA substrate at the dimerization interface....41
Figure 2.5. Sequence-conservation architecture of the KTE100 antiphage effector....42
Figure 2.6. Structural prediction of the KTE100 antiphage α-pore-forming effector....43
Abbreviations....48
References....48
CHAPTER 3: AN ANTIPHAGE GUANYLYL CYCLASE REGULATES CELL WALL SYNTHESIS AND MOTILITY IN *ESCHERICHIA COLI*....55
3.1 Abstract....55
3.2 Introduction....55
3.3 Results....57
3.3.1 Ectopic expression of antiphage guanylyl cyclases increases optical density without increasing cell number....57
3.3.2 Guanylyl cyclase expression induces a gram-positive-like cell wall phenotype in *E. coli*....58
3.3.3 Ectopic expression of the guanylyl cyclase increases the swimming distance of *E. coli*....59
3.3.4 *E. coli* guanylyl cyclase expression diminishes biofilm formation....60
3.4 Discussion....61
3.5 Materials and Methods....63
3.6 Figures....66

Figure 3.1. Antiphage guanylyl cyclases elevate the optical density of *Escherichia coli* without expanding viable cell populations....66
Figure 3.2. Ectopic guanylyl cyclase expression drives a gram-positive-like cell wall phenotype in *Escherichia coli*....67
Figure 3.3. The *Escherichia coli* antiphage guanylyl cyclase enhances swimming motility....68
Figure 3.4. The *Escherichia coli* guanylyl cyclase suppresses biofilm formation in a manganese-dependent manner....69
3.7 Abbreviations....70
3.8 References....70
CHAPTER 4: THE MOLECULAR TRIPWIRE — A PUTATIVE AUTOINHIBITORY DOMAIN OF A GUANYLYL CYCLASE IS LIKELY RELIEVED DURING BACTERIOPHAGE INFECTION TO INDUCE ANTIPHAGE ACTIVITY....76
4.1 Abstract....76
4.2 Introduction....77
4.3 Results....79
4.3.1 The cyclase and effector are jointly required to defend against phages P1, lambda, and T6....79
4.3.2 Complete defense ablates or attenuates P1, lambda, and T6 progeny release....79
4.3.3 KTE100 Type-I CBASS-like defense protects from P1, lambda, and T6 in liquid culture at low MOI....80
4.3.4 Bacteriophages mutate structural and protease genes to evade antiphage defense....81
4.3.5 Expression of full-length lambda GpC does not activate abortive infection....82
4.3.6 Mutated residues likely modulate protease scaffold autocleavage....82
4.3.7 The *E. coli* cyclase amino-terminus is a phage-sensing module (preliminary, n = 1)....83
4.3.8 The cyclase N-terminus harbors localized redundant cleavage sites and a putative TBK1-like phosphorylation loop....85
4.4 Discussion....87
4.5 Materials and Methods....90
4.6 Author's Note on Data Maturity and Independent Completion....94
4.7 Figures....96
Figure 4.1. The KTE100 guanylyl cyclase–effector operon defends from phages P1, lambda, and T6....96
Figure 4.2. Ectopic expression of phage parent genes does not recapitulate abortive infection....98
Figure 4.3. P1 Pro escaper mutations alter pentameric scaffold without disrupting the catalytic triad....100
Figure 4.4. The cyclase amino-terminus determines phage sensitivity (preliminary, n = 1)....102

Figure 4.5. The *E. coli* cyclase harbors a putative TBK1-like phosphorylation/cleavage loop on its N-terminus.....104
Figure 4.6. Two non-mutually-exclusive models for guanylyl cyclase activation by P1 phage infection.....105
4.8 Abbreviations.....112
References.....113
FUTURE DIRECTIONS.....122

CHAPTER 1

INTRODUCTION

By: Jeffrey Wyatt-Wolfgang Marciniak-Haynosch

1.1 Historical Context: Bacteriophage Discovery and the Origins of Molecular Biology

The independent discoveries of bacteriophages by Twort in 1915 and d'Hérelle in 1917 marked one of the most consequential inflection points in twentieth-century biology, reframing viruses of bacteria from anomalous filterable agents into a tractable system that would, within decades, lay the experimental foundation of molecular biology.[6,9,23,39] Through the work of Luria, Delbrück, Lederberg, and Ptashne, bacteriophages — most notably the temperate coliphage lambda — became the genetic substrate from which the principles of mutation, recombination, lysogeny, and gene regulation were forged.[9,23,31] The fluctuation analysis of Luria and Delbrück established the random nature of mutation,[23] while subsequent dissection of the lambda genetic switch by Ptashne defined the cis-trans logic that underlies all subsequent models of transcriptional control.[31] A century after their initial description, bacteriophages remain at the crucible of microbiological discovery, now driving fundamental advances in CRISPR-based genome editing, abortive infection biology, and — central to this thesis — the elucidation of bacterial innate immunity.[5,7,21,28]

The clinical resurgence of multidrug-resistant pathogens has renewed interest in bacteriophages as therapeutic agents, both as adjuncts to and alternatives for conventional antibiotics.[14,41] Yet the therapeutic potential of phage-mediated bacterial eradication is fundamentally constrained by the breadth and sophistication of bacterial antiphage defense systems, which have co-evolved with their viral predators across an evolutionary timescale measured in billions of years.[7,21,28] These defenses — from the classical restriction-modification systems and CRISPR-Cas adaptive immunity to the more recently characterized cyclic oligonucleotide-based antiphage signaling systems (CBASS) at the center of this thesis — collectively reflect the depth of prokaryotic innate immunity.[5,21,28] Understanding the molecular logic of bacterial antiphage defense is

therefore not merely a theoretical exercise; it is a prerequisite for the rational design of phage therapeutics and a window onto the deep evolutionary origins of innate immunity itself, the latter of which this thesis directly engages through the characterization of a previously uncharacterized Type I CBASS-like operon in Uropathogenic *Escherichia coli* strain KTE100.

1.2 The Red Queen Hypothesis and the Arsenal of Bacterial Antiphage Defenses

Bacteria and their bacteriophage predators are bound in perpetual evolutionary conflict, a dynamic articulated most cogently by the Red Queen Hypothesis: each combatant must continuously adapt merely to retain its relative fitness in the contest.[13,21] This deep antagonism has selected for a remarkably sophisticated arsenal of antiphage defense mechanisms, broadly partitioned into innate and adaptive immunity.[7,21,24,28] Innate defenses — encompassing restriction-modification (RM) systems, abortive infection (Abi) modules, retrons, Gabija, and CBASS — provide rapid, genetically encoded responses to phage invasion that act without the requirement for prior exposure.[7,21,28,41] Adaptive immunity, exemplified by the CRISPR-Cas systems, confers heritable, sequence-specific immunological memory through the integration of phage-derived spacer sequences into chromosomal CRISPR arrays.[22,24,26]

CRISPR-Cas systems are organized into two major classes: Class 1 systems (Types I, III, and IV) employ multi-subunit effector complexes, while Class 2 systems (Types II, V, and VI) deploy single-protein effectors such as Cas9, Cas12, and Cas13.[24] Despite their effectiveness, CRISPR-Cas systems impose nontrivial fitness costs, require prior infection for spacer acquisition, and are vulnerable to anti-CRISPR proteins encoded by co-evolving phages.[4,26,44] Critically, multidrug-resistant *Enterococcus* strains exhibit a striking inverse correlation between functional CRISPR-Cas loci and antibiotic resistance, suggesting that selective pressure from antimicrobials inadvertently favors strains with compromised adaptive immunity, thereby permitting promiscuous acquisition of mobile genetic elements that confer resistance.[30,44] This observation underscores the indispensability of complementary innate defense strategies that operate independent of spacer-based memory and provides part of the conceptual scaffold motivating the present investigation.

1.3 Cyclic Oligonucleotide-Based Antiphage Signaling Systems (CBASS): Architecture and Logic

CBASS comprises an expansive family of innate immune systems whose unifying biochemical signature is the use of cyclic oligonucleotide second messengers to couple phage detection to the execution of abortive infection.[5,8,11,15,28,40] The canonical CBASS architecture is a two-component sentinel: a cGAS/DncV-like nucleotidyltransferase (CD-NTase) that synthesizes a cyclic dinucleotide or trinucleotide signal upon detection of an infection-associated cue, paired with a cognate effector that is allosterically activated by the second messenger to execute cell death or dormancy — sacrificing the infected cell to deny progeny virion release and protect the surrounding clonal population.[5,8,21,28] This logic mirrors, with profound mechanistic conservation, the cGAS-STING pathway of metazoan innate immunity, supporting the now-widely accepted view that bacterial CBASS is the evolutionary progenitor of the eukaryotic cGAS-STING axis.[13,15,28]

1.3.1 The CD-NTase Family: Structural Conservation and Second-Messenger Diversity

CD-NTases adopt a structurally conserved Pol-β-like nucleotidyltransferase fold and constitute a distinct third family of cyclic dinucleotide synthases, evolutionarily separate from both the dimeric GGDEF c-di-GMP synthases and the DAC/DisA c-di-AMP synthases that govern bacterial cell-cycle and stress signaling.[15,40] Systematic biochemical interrogation of the CD-NTase family has revealed an astonishing diversity of signaling species that vastly exceeds the canonical 3′,3′-cyclic GMP-AMP (3′,3′-cGAMP) originally characterized in *Vibrio cholerae* DncV.[5,11,28,40] The current second-messenger repertoire spans cyclic dipurines (3′,3′-cGAMP, c-di-GMP, c-di-AMP); hybrid purine-pyrimidine cyclic dinucleotides such as cyclic UMP-AMP (cUMP-AMP) produced by *E. coli* CdnE;[40] cyclic dipyrimidines including cyclic di-UMP (c-di-UMP) from *Legionella pneumophila* CdnE02;[38,40] and cyclic trinucleotides such as cyclic AMP-AMP-GMP (cAAG) synthesized by *Enterobacter cloacae* Ec-CdnD02 and the cyclic triadenylate (cAAA) of HORMA-CD-NTase systems.[16,40,43] The recent definition of cyclic CMP and cyclic UMP as *bona fide* CBASS second messengers further expands a chemical lexicon

in which the prokaryotic immune system encodes an information channel of remarkable combinatorial complexity.[38]

The CD-NTase family clusters into approximately eight clades (A–H), unified by their location within conserved operons that lie predominantly on mobile genetic elements adjacent to genes encoding effector-like proteins.[28,40] This syntenic architecture facilitates the horizontal transfer of complete functional CBASS modules across phylogenetically distant bacterial lineages, accelerating the evolutionary reach of cyclic-nucleotide–based immunity throughout the prokaryotic biosphere.[21,28]

1.3.2 Effector Diversity and Mechanisms of Abortive Infection

CBASS effectors are the operational arm of the system, translating second-messenger occupancy into terminal cellular outcomes that prevent phage propagation. A genome-wide census across more than 38,000 bacterial and archaeal genomes identified four major CBASS types and at least six effector subtypes that drive abortive infection through membrane deterioration, indiscriminate nucleic acid degradation, or related catastrophic lesions.[7,8,28] Among the nuclease effectors, NucC is a homotrimeric endonuclease that, upon binding cyclic trinucleotide, assembles into a homohexamer competent for non-specific double-strand DNA cleavage and complete destruction of the bacterial chromosome during infection.[16] Cap4, the founding member of an expansive family of more than 2,000 SAVED domain–containing effectors, functions as a promiscuous DNA endonuclease that is activated through ligand-induced oligomerization,[22] while CARF-HEPN fusion proteins — paradigmatic of Type III CRISPR-Cas immunity — couple N-terminal CARF (CRISPR-associated Rossmann fold) sensing of cyclic oligoadenylate (cOA) ligands to allosteric activation of C-terminal HEPN (higher eukaryotes and prokaryotes nucleotide-binding) ribonucleases that degrade cellular RNA.[1,25,29,35]

Among phospholipase effectors, CapV from *V. cholerae* binds 3′,3′-cGAMP directly and degrades membrane phospholipids, releasing free fatty acids and compromising envelope integrity to lethal effect.[5,33] Co-expression of *dncV-capV* is acutely toxic to *Escherichia coli* and abolishes colony formation, demonstrating the strict functional coupling between CD-NTase product and downstream effector activation.[33] Although less

extensively characterized in the canonical CBASS literature, pore-forming effectors are recognized as a major and structurally diverse effector class,[8,28] raising the testable hypothesis that ClyA-like α-pore-forming toxins could function as *bona fide* CBASS effectors — a possibility directly examined in this thesis.

1.3.3 Cyclic Oligonucleotide Recognition and Linkage Specificity

Effector activation in CBASS is mediated by specialized cyclic oligonucleotide–binding domains that confer exquisite specificity for distinct second messengers. SAVED domains — striking evolutionary fusions of two CARF subunits co-opted from Type III CRISPR immunity — enable discrimination between 2′-5′ and 3′-5′ phosphodiester linkages and are predicted to recognize at least 180 distinct nucleotide species, defining a recognition lexicon of unprecedented combinatorial breadth in prokaryotic immunity.[22] This linkage-specific recognition, long appreciated in the mammalian cGAS-STING signaling axis,[13] has emerged as a fundamental organizing principle of bacterial CBASS immunity.[22,28] The CARF domain superfamily itself comprises ten major families that differ in architecture, in their associated effectors, and in the presence or absence of intrinsic ring-nuclease activity that cleaves cyclic oligoadenylates to attenuate or terminate dormancy and death signaling.[25]

1.3.4 Regulatory Mechanisms and Signal Integration in CBASS Immunity

CBASS systems integrate a remarkable diversity of regulatory inputs to prevent spurious activation, gate immune readiness to cellular context, and calibrate signal strength to the severity of viral threat. The foundational activation logic was established by Cohen and colleagues, who demonstrated that phage infection of *V. cholerae* triggers DncV-dependent cGAMP synthesis and the consequent activation of the CapV phospholipase effector — establishing the canonical sensor–messenger–effector paradigm of CBASS immunity.[5] Building on this framework, Millman and colleagues classified CBASS into four major types and identified WYL-domain transcription factors as accessory regulators that modulate CBASS operon expression in a signal-dependent manner, providing a transcriptional layer of immune control.[28]

Beyond transcriptional regulation, CBASS activation is governed at the level of CD-NTase post-translational modification: Ledvina and colleagues identified an E1–E2 ubiquitin-like fusion protein that covalently primes the CD-NTase for activation, the first documented post-translational priming mechanism in bacterial antiviral immunity and a striking functional parallel to ubiquitin-mediated signaling in eukaryotic innate immunity.[18] CBASS systems are further coupled to upstream physiological signals that report on cellular state. Lau and colleagues demonstrated that DNA damage activates CBASS immune signaling through a conserved pathway intertwined with the SOS response, revealing that phage-induced genomic stress is itself a sufficient trigger for cyclic nucleotide synthesis, independent of direct phage-protein detection.[17] In *V. cholerae*, Severin and colleagues established that the DncV-based CBASS system is gated by both quorum sensing inputs and folate depletion, coupling antiviral readiness to bacterial population density and metabolic status — ensuring that the steep fitness cost of abortive infection is incurred only under conditions in which communal benefit is maximized.[34]

At the level of the phage trigger itself, two complementary studies have defined the molecular features that activate CBASS sensors. Stokar-Avihail and colleagues conducted a large-scale genetic screen across diverse phage–host pairings and identified specific phage structural proteins — including major capsid and scaffolding proteins — as direct immune activators, establishing that CBASS systems are tuned to detect conserved features of virion assembly rather than the earliest events of infection.[37] In a striking departure from the protein-trigger paradigm, Banh and colleagues demonstrated that the *Staphylococcus schleiferi* 2142-05 Type I CBASS system is activated not by a protein but by a discrete ∼400-nucleotide non-coding RNA — termed the cGAS-activating bacteriophage RNA (cabRNA) — that is transcribed across the small (TerS) and large (TerL) terminase genes of the infecting phage.[2] The cabRNA is a phage-derived viral RNA bound directly by the bacterial CD-NTase (CdnE12) and is both necessary and sufficient to license cyclic nucleotide synthesis; this single observation expanded the repertoire of CBASS triggers to include nucleic-acid ligands and cemented RNA-based pattern recognition as a *bona fide* axis of bacterial antiviral immunity.[2]

Protein-level regulation is further extended by HORMA-domain proteins, which recognize infection-associated peptides, bind and activate CD-NTases to generate cyclic triadenylate (cAAA), and thereby drive activation of the NucC endonuclease effector; this signaling is temporally attenuated by a Pch2/TRIP13-like AAA+ ATPase that disassembles the active HORMA–CD-NTase complex, imposing temporal control on second-messenger synthesis.[16,43]

The mechanistic basis for CD-NTase activation at the protein-trigger level was most recently and definitively addressed by Hobbs and Kranzusch in a study published independently of, and immediately prior to, the work presented in this thesis.[12] Using biochemical reconstitution of phage T4 infection *in vitro*, they demonstrated that the T4 prohead protease gp21 is sufficient to activate the *E. coli* CD-NTase EcCdnG *in vivo*, and that this activation is strictly dependent on the gp21 catalytic active site — establishing that CD-NTase activation occurs in response to proteolytic activity itself, not simple protein–protein complex formation.[12] Mechanistically, activation co-occurs with cleavage of a ∼30–amino acid surface-exposed disordered loop centered between the two lobes of the EcCdnG bilobed architecture, a loop that functions as a molecular tripwire: substitution of trypsin cleavage sites within this loop abolished protease-dependent immune signaling, while alanine scanning of the loop's I/LxE motif attenuated defense against T4 by more than 100-fold without compromising immunity against other phages.[12] Structure-guided phylogenetic analysis revealed that analogous activation loops are present in approximately 24% of all CBASS operons across Clades A, D, and G, and that 7 of 9 tested representative CD-NTases — including the major model enzymes *V. cholerae* DncV and *E. cloacae* CdnD — responded to foreign protease activity, defining proteolytic cleavage of CD-NTase activation loops as a widespread mechanism of antiviral sensing.[12] Collectively, these findings — and their convergence with the genetic and structural data presented in this thesis, which were obtained prior to publication of the Hobbs and Kranzusch study — argue that protease-dependent activation-loop cleavage is a conserved organizational principle that operates across diverse CBASS architectures, and they raise the central mechanistic question of whether an analogous mechanism governs activation of the biochemically distinct guanylyl cyclase explored in the chapters that follow.

1.4 A Novel CBASS-Like System in Uropathogenic *Escherichia coli*: Guanylyl Cyclase and ClyA Pore-Forming Toxin

Uropathogenic *Escherichia coli* (UPEC) strains are the primary etiological agents of urinary tract infections (UTIs) and have evolved an unusually diverse repertoire of virulence and defense strategies tuned to the urinary tract niche. Genomic analysis of clinical UPEC isolates — including *E. coli* str. KTE100 (NCBI Accession: PRJNA193500), obtained from The Broad Institute (Boston, MA) as part of the UTI Defensins Study — has revealed a putative antiphage operon whose architecture is consistent with canonical CBASS logic but whose biochemical components are unmistakably distinct. The KTE100 operon encodes a putative cyclic nucleotide synthase (Locus tag: A1WK_02874), annotated as a hypothetical protein but predicted to function as a guanylyl cyclase that produces cyclic GMP (cGMP) as its second messenger, paired with a putative effector (Locus tag: A1WK_02875) predicted to encode a ClyA-like α-pore-forming toxin.

1.4.1 Distinguishing Features: Conservation Versus Novelty

This system displays the hallmarks of canonical CBASS logic — a two-component sensor–messenger–effector architecture in which a cyclic nucleotide synthase produces a diffusible signal that activates a lethal effector — yet diverges from previously characterized systems on three biochemically critical axes. First, with respect to second-messenger identity: if experimentally validated, monomeric cGMP would represent a striking departure from characterized CBASS second messengers, which are predominantly cyclic dinucleotides or trinucleotides spanning purine, pyrimidine, and mixed compositions.[5,16,18,28,32,33,34,38,40,43] The canonical role of cGMP in bacterial signaling is largely confined to the c-di-GMP networks that govern biofilm regulation and motility,[19] making its proposed deployment as an antiphage second messenger conceptually novel.

Second, with respect to synthase family: the putative synthase is classified as a guanylyl cyclase rather than a CD-NTase. CD-NTases are defined by a Pol-β-like nucleotidyltransferase fold and constitute a structurally distinct lineage from both the bacterial and eukaryotic adenylate and guanylyl cyclases.[15,40] Functional validation of the

predicted KTE100 guanylyl cyclase as a CBASS-compatible synthase would therefore expand the known enzymatic repertoire of bacterial cyclic nucleotide–based immunity beyond the CD-NTase family. Third, with respect to effector architecture: ClyA (cytolysin A; also designated HlyE) is an extensively characterized pore-forming toxin of *E. coli* that oligomerizes upon membrane engagement to generate transmembrane pores.[3] The predicted structure of the water-soluble monomeric KTE100 effector reveals the distinctive helical bundle and β-tongue topology characteristic of the ClyA architecture.[3] Although pore-forming proteins are recognized as a major CBASS effector class,[8,28] characterized examples derive from distinct protein families; deployment of a ClyA-like effector by UPEC clinical isolates would therefore represent a novel synthase–effector pairing. The identification of this defense system specifically within Uropathogenic clinical isolates rather than environmental or commensal strains may further reflect selective pressures unique to the urinary tract — a niche in which phage predation, biofilm formation, and mammalian innate immune evasion intersect.

1.4.2 Criteria for Establishing Novelty

Definitive characterization of the KTE100 system requires validation across three complementary levels. Bioinformatic validation entails domain-architecture analysis of locus A1WK_02874 to confirm guanylyl cyclase classification and to identify any unusual domain fusions or accessory modules, together with phylogenetic placement relative to the CD-NTase and canonical guanylyl cyclase families. Biochemical confirmation requires *in vitro* enzymatic assays demonstrating cGMP synthesis by purified A1WK_02874 and the direct binding of cGMP to the ClyA-like α-pore-forming toxin effector A1WK_02875. Functional validation demands genetic complementation and phage challenge assays demonstrating that the operon confers antiphage immunity through abortive infection, together with site-directed mutagenesis of predicted catalytic residues and cyclic-nucleotide binding motifs. This thesis addresses all three levels of validation through the experimental aims described in Section 1.7.

1.5 HEPN Domains in Cyclic Nucleotide Synthases: A Focused Literature Analysis

An unexpected and provocative feature of the predicted UPEC guanylyl cyclase is the presence of a putative HEPN-like motif (^{81}RDSTKH^{86}) within or immediately adjacent to the cyclase domain. Because such an architecture would be unprecedented in characterized cyclic nucleotide synthases, this observation prompted a comprehensive literature analysis to determine whether HEPN domains or HEPN-like motifs have ever been characterized in any class of cyclic nucleotide synthase, including guanylyl cyclases, adenylate cyclases, CD-NTases, or other CBASS-related enzymes.

1.5.1 HEPN Domain Structure and Canonical Functions

The HEPN (higher eukaryotes and prokaryotes nucleotide-binding) superfamily encompasses an all-α-helical fold defined by a four-helix up–down bundle in which two structurally similar α-hairpins assemble into a compact core.[1] The canonical HEPN motif — RγX$_{3-6}$H, where γ denotes a negatively charged or polar residue and X is any amino acid — is the most strongly conserved sequence feature of the domain and serves as the primary determinant of a metal-independent ribonuclease active site.[1,29,35,42] The conserved arginine is thought to bind substrate or stabilize reaction intermediates, while the histidine is essential for nucleolytic catalysis.[1,35] HEPN active sites generate 2′,3′-cyclic phosphate products through nucleophilic attack by the ribose 2′-OH of substrate RNA, operating via a metal-independent mechanism that distinguishes them from canonical metal-dependent ribonucleases.[1,29]

Mutagenesis studies have repeatedly affirmed the catalytic indispensability of these residues. In *Pyrococcus furiosus* Csx1 and *Thermus thermophilus* Csm6, substitution of the conserved HEPN motif residues (R431A and H436A in Csx1; R415A and H422A in Csm6) abolishes or severely attenuates ribonuclease activity.[29,35] In *Shewanella oneidensis* SO_3166, mutation of either Arg97 or His102 within the RγX$_{3-6}$H motif likewise eliminates toxicity, confirming that the HEPN signature is the structural and catalytic linchpin of HEPN function.[42]

1.5.2 HEPN Domain Architectures in Bacterial Defense Systems

HEPN domains exhibit remarkable architectural plasticity and recur in diverse fusion contexts within prokaryotic defense systems.[1] The most prevalent CBASS-relevant

architecture is an N-terminal CARF cyclic-oligonucleotide sensor fused to a C-terminal HEPN ribonuclease effector.[1,25,29,35] Four distinct HEPN families are associated with CRISPR-Cas loci, and all share this core CARF-HEPN organization.[1] In the Csm6 and Csx1 families, the CARF domain senses cyclic oligoadenylates (cOA) generated by the Cas10 polymerase-cyclase during Type III CRISPR-Cas defense, allosterically activating the HEPN ribonuclease to cleave cellular RNA and halt viral replication.[1,25,29,35] Structural studies of *T. thermophilus* Csm6 establish that dimerization of HEPN domains assembles a composite ribonuclease active site, with the CARF domain providing allosteric rather than catalytic function.[29]

Beyond CARF-HEPN fusions, HEPN domains pair with minimal nucleotidyltransferases (MNT) in two-component toxin–antitoxin systems, in which the HEPN domain functions as the toxin and MNT as the antitoxin.[1,42] Additional architectures include the Swt1 family, in which HEPN is fused to PIN endoribonuclease domains; the RNase LS family, in which HEPN fuses to RNase H-like modules; and a series of fusions with HNH/Endo VII DNases, helicases, and AAA+ ATPases.[1] In all such architectures, however, the HEPN domain functions exclusively as an RNase or toxin module — never as part of the cyclic nucleotide synthase itself.

1.5.3 Critical Finding: No Evidence for HEPN Domains in Cyclic Nucleotide Synthases

Comprehensive literature review (1990–2026) yielded no evidence for HEPN domains or HEPN-like motifs within any class of cyclic nucleotide synthase. Specifically, no HEPN domains have been reported in bacterial or eukaryotic guanylyl or adenylate cyclases, nor in CD-NTase family cyclic dinucleotide or trinucleotide synthases — including DncV, CdnE, CdnD, CdnG, and other CBASS cyclases[5,11,28,32,40] — nor in GGDEF family c-di-GMP synthases or DAC/DisA family c-di-AMP synthases. The HEPN domain appears exclusively in effector proteins — ribonucleases and toxins — that respond to cyclic nucleotide signals, never in the synthases that produce them.[1,25] Within CBASS and Type III CRISPR systems, HEPN domains are architecturally segregated into dedicated effector modules (most notably CARF-HEPN fusions) that are functionally and genetically distinct from the cyclic-oligonucleotide synthases (CD-NTases,

Cas10).[1,5,25,28,32] Identification of an authentic, catalytically active HEPN-like domain within a cyclic nucleotide synthase would therefore constitute a fundamental departure from the architectural logic established by more than two decades of HEPN biology.

1.5.4 Interpreting the RDSTKH Motif: Plausible Explanations

Given the absence of any precedent for HEPN domains in cyclic nucleotide synthases, four hypotheses warrant explicit consideration. First, the sequence ^{81}RDSTKH^{86} superficially conforms to the RγX$_{3\text{-}6}$H consensus but may represent a convergent motif that performs an unrelated biochemical function. True HEPN domains are defined not by primary sequence alone but by a stereotyped four-helix bundle, dimeric assembly, and experimentally validated RNase activity[1,29] — none of which has yet been demonstrated for the KTE100 cyclase. Second, the cyclase may harbor a fused HEPN-like module that functions as an autoregulatory domain or as a discrete effector domain within a bifunctional protein, although no such architecture has been documented in the CBASS literature.[2,18,28,32,40,43] Third, the motif may lie within a propeptide or N-terminal extension that is removed by proteolytic cleavage, in direct analogy to bacterial protease zymogens that require N-terminal propeptide excision for activation.[36] This interpretation is independently supported by the identification of a likely protease cleavage site at the amino terminus of the KTE100 cyclase, described in detail in Chapter 4. Fourth, the predicted guanylyl cyclase gene may represent a chimeric assembly artifact, the misannotation of a CARF-HEPN effector, or an unusual domain fusion that requires direct experimental validation. Discriminating among these hypotheses is one of the central aims of the bioinformatic and biochemical work presented in Chapters 2 and 4.

1.6 Activation Mechanisms in Bacterial Antiphage Defense: Proteolytic Triggers and the SOS Response

A central question in any CBASS or CBASS-like system is the molecular mechanism by which phage infection triggers cyclic nucleotide synthase activation. Characterized regulatory inputs include HORMA-domain–mediated protein–peptide sensing,[43] direct DNA or RNA binding by the synthase itself,[2,17] quorum sensing and metabolite depletion,[34] and large-scale phage genetic screens that have identified structural

determinants of immune activation.[32,37] Proteolytic activation, by contrast, has historically received comparatively limited experimental attention as a mechanism of bacterial antiviral immunity, despite its centrality in eukaryotic innate immunity (e.g., caspase activation in inflammasome assembly) and its now-established role in CBASS biology.[12,32]

1.6.1 Phage-Encoded Proteases as Determinants of Immune Activation

Bacteriophage-encoded proteases — including T4 gp21 (prohead protease), P1 Pro, and lambda C protease — play indispensable roles in phage morphogenesis, cleaving structural protein precursors during virion assembly to liberate the mature capsid.[20,27,36] In principle, the same proteolytic activities could serve as inadvertent activators of host defense by cleaving bacterial sensor proteins encoded within the host genome. Direct support for this possibility was provided by Richmond-Buccola and colleagues in a large-scale Type I CBASS screen, which established that phage prohead proteases are key determinants of immune activation and immune evasion across diverse phage–host pairings.[32] This insight, together with the *in vitro* biochemical demonstration by Hobbs and Kranzusch that the T4 prohead protease gp21 is necessary and sufficient to activate the *E. coli* CD-NTase EcCdnG via cleavage of a surface-exposed activation loop,[12] has elevated phage proteases from morphogenetic accessories to *bona fide* pattern-recognition triggers of bacterial innate immunity. The work presented here identifies a likely protease cleavage site at the amino terminus of the KTE100 cyclase, suggesting that a phage-encoded protease — or a host protease activated during infection — may cleave an N-terminal inhibitory domain to liberate active cyclase and thereby trigger cGMP synthesis. This would represent a hitherto-uncharacterized variant of proteolytic CBASS activation, integrating proteolytic sensing with cyclic nucleotide signaling in a guanylyl cyclase rather than a CD-NTase.

1.6.2 Candidate Proteolytic Triggers: Bridging Phage Morphogenesis and Host Defense

Although direct biochemical evidence for the KTE100 system remains the principal experimental objective of this thesis, several candidate scenarios merit explicit

consideration. In the first scenario, T4 gp21, P1 pro, or lambda C may possess sufficiently relaxed substrate specificity to inadvertently cleave the cyclase N-terminal propeptide during virion morphogenesis, releasing active cyclase and thereby triggering cGMP synthesis as a danger signal that activates the ClyA-like effector. In the second scenario, phage infection–induced DNA damage activates the SOS response, inducing expression of host proteases (including ClpXP and other regulated proteases) that process the cyclase propeptide, integrating phage-induced genomic stress with cyclic nucleotide–based immunity in a manner conceptually parallel to the SOS-coupled CBASS activation mechanism described by Lau and colleagues.[17] In the third scenario, phage infection products — replication intermediates, scaffolding proteins, or mature capsid components — may directly engage the cyclase, inducing conformational changes that expose the protease cleavage site to endogenous or phage-encoded proteases. These scenarios are not mutually exclusive and may operate in parallel or in sequence during the phage lytic cycle. Experimental validation through site-directed mutagenesis of the predicted cleavage site, *in vitro* protease cleavage assays with purified cyclase and candidate proteases, and quantification of cGMP levels and ClyA activation in phage-infected versus uninfected cells is addressed in Chapters 2 and 4.

1.7 Thesis Gap and Experimental Objectives

The discovery of a putative cGMP–ClyA defense operon in Uropathogenic *E. coli* clinical isolates represents a potentially novel branch of cyclic nucleotide–based antiphage immunity. While this system adheres to the canonical CBASS sensor–messenger–effector logic, it deploys biochemically distinct components: a guanylyl cyclase rather than a CD-NTase, monomeric cGMP rather than cyclic dinucleotides or trinucleotides, and a ClyA-like α-pore-forming toxin in lieu of the structurally diverse but characterized CBASS effector classes. The presence of a putative HEPN-like motif (^{81}RDSTKH^{86}) within the cyclase is unprecedented; comprehensive literature analysis returned no evidence for HEPN domains in any class of cyclic nucleotide synthase, as HEPN domains function exclusively as ribonuclease effectors in CBASS and CRISPR-Cas systems and never as components of the signaling enzymes themselves.[1,25]

The central unanswered question is therefore the following: what phage-derived or infection-associated signal activates the KTE100 cyclase to initiate cGMP synthesis and ClyA-mediated abortive infection? Identification of a putative protease cleavage site at the amino terminus of the cyclase suggests a proteolytic activation mechanism, in which phage-encoded proteases (T4 gp21, P1 pro, lambda C) or SOS-induced host proteases cleave an N-terminal inhibitory domain to liberate the active cyclase. This hypothesis is supported by precedent in bacterial protease zymogen activation,[36] by the demonstration that prohead proteases are primary determinants of Type I CBASS activation and evasion,[32] and by the recent biochemical reconstitution of T4 gp21–dependent activation of EcCdnG.[12]

This thesis addresses this gap through three experimental aims. First, bioinformatic analysis of the cyclase and effector identifies sites of post-translational modification and domain architecture that inform regulatory function (Chapter 2). Second, phenotypic characterization of ectopically expressed cyclase identifies cellular processes — including motility, cell wall remodeling, and biofilm formation — that are modulated by cyclase expression or cGMP production (Chapter 3). Third, phage challenge assays, escaper phage sequencing, and chimeric cyclase experiments identify the proteolytic trigger that activates the cyclase and establish the structural basis for phage-sensing specificity (Chapter 4). Together, these aims construct a mechanistic model for how phage infection coordinates crosstalk between a guanylyl cyclase and a pore-forming effector to abort viral replication — and, in doing so, place the KTE100 system within the broader mechanistic landscape of cyclic nucleotide–based innate immunity.

Abbreviations

Abi	Abortive Infection
CBASS	Cyclic Oligonucleotide-Based Antiphage Signaling System
CD-NTase	Cyclic-oligonucleotide Dinucleotide-Nucleotidyltransferase
cGAMP	cyclic Guanosine Monophosphate–Adenosine Monophosphate
cGAS	cyclic GMP-AMP Synthase
cGMP	cyclic Guanosine Monophosphate
ClyA	Cytolysin A pore-forming toxin
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
HEPN	Higher Eukaryotes and Prokaryotes Nucleotide-binding domain
KTE100	*Escherichia coli* strain KTE100
MNT	Minimal Nucleotidyltransferase
NCBI	National Center for Biotechnology Information
RM	Restriction-Modification
SAVED	SMODS-Associated and Various Effector Domains
SOS	SOS DNA-damage response
STING	Stimulator of Interferon Genes
UPEC	Uropathogenic *Escherichia coli*

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CHAPTER 2

IN SILICO ANALYSES OF AN ANTIPHAGE CYCLASE AND EFFECTOR OF UROPATHOGENIC *ESCHERICHIA COLI* STR. KTE100

By: Jeffrey Wyatt-Wolfgang Marciniak-Haynosch

2.1 Abstract

I performed a comprehensive *in silico* characterization of a previously uncharacterized Type-I CBASS antiphage system encoded by uropathogenic *Escherichia coli* str. KTE100, with the explicit goal of resolving the functional and regulatory architecture of its sensor and effector proteins. The 264-residue sensor encodes a bifunctional guanylyl cyclase–HEPN domain fusion in which the primary nucleotide-binding signature (^{81}RDSTKH^{86}) simultaneously satisfies the canonical RγX$_{3–6}$H consensus of a metal-independent ribonuclease active site, an architectural pairing not previously reported for any cyclic nucleotide synthase.[2,21,27] The HEPN assignment is reinforced by four flanking secondary EXXX[R/K] motifs and by strict conservation of the bifunctional fold across nine independent Uropathogenic genomes, ruling out a sequence-level coincidence. The 183-residue effector, by contrast, is predicted to traverse the inner membrane via three transmembrane domains, to bear a single conserved N-linked glycosylation motif (^{120}NYT^{122}), and to oligomerize as a hexameric α-pore-forming toxin (α-PFT) — the first α-PFT, to my knowledge, implicated as a CBASS effector.[20,30,43] Candidate phosphorylation and acetylation residues are selectively partitioned into regions of inter-strain sequence divergence in both proteins, supporting a model in which post-translational modification fine-tunes activity at structurally peripheral sites while the conserved catalytic core remains invariant.

2.2 Introduction

Bacterial antiphage defense systems must resolve a stringent cost–benefit calculus: constitutive deployment of cytotoxic effectors imposes a substantial fitness penalty under non-infected conditions, yet failure to mobilize rapidly upon viral challenge is fatal to the clonal population.[8,18,23] Abortive infection systems — of which the cyclic oligonucleotide-based antiphage signaling systems (CBASS) are the paradigmatic example — resolve this tension through exquisitely gated activation, layered onto a sensor–messenger–effector architecture in which spurious second-messenger synthesis is precluded under homeostatic conditions and unleashed only in response to *bona fide* infection.[7,8,23] Post-translational modification (PTM) provides a uniquely well-suited regulatory medium for this task: it operates on second timescales, requires no transcriptional reprogramming, and is intrinsically reversible — properties that match the temporal demands of innate immunity.

The intracellular environment during phage infection is fundamentally non-equilibrium. Phage replication hijacks host transcription, translation, and nucleotide biosynthesis, and the resulting collision of host and viral macromolecular machinery places severe selective pressure on antiphage systems to integrate multiple, partially redundant infection cues.[14,18,28] The molecular logic by which a host cyclase distinguishes a productive viral signal from background noise — and how that signal is transduced into committed effector activation — remains an open mechanistic frontier.[3,29,39]

I hypothesized that the regulatory architecture of the KTE100 system would be encoded in the spatial partitioning of conserved and divergent residues: catalytic and structural elements would be invariant across orthologs, while regulatory residues subject to PTM would localize to inter-strain variable regions. This hypothesis predicts that conservation analysis, when integrated with PTM site prediction, should resolve the regulatory architecture of both the sensor and effector at single-residue resolution.

Here I apply a sequential pipeline of phylogenetic, conservation, domain-architecture, PTM, and AlphaFold3-based structural analyses to the KTE100 cyclase–effector pair, with two principal objectives. First, I delineate the phylogenetic distribution of the cyclase–effector module across Uropathogenic *Enterobacteriaceae* and characterize the conserved gene neighborhood that flanks the operon. Second, I define the

catalytic, regulatory, and structural architecture of each protein in atomic detail.[7,23,29] The analyses converge on two conceptually unprecedented features: a bifunctional guanylyl cyclase–HEPN fusion in the sensor, and a hexameric α-PFT in the effector — each bearing candidate PTM residues concentrated in regions of sequence divergence, consistent with my central hypothesis.

2.3 Results

2.3.1 The cyclase and effector co-occur in Uropathogenic Enterobacteriaceae

BlastP interrogation of the NCBI non-redundant database with the cyclase and effector sequences (1,000 hits each; E-value ≤ 0.05) returned 648 and 329 homologs, respectively. Cross-indexing the two result sets by accession identified nine genomes in which both genes co-occur, consistent with the pair constituting a genetically linked, co-inherited defense module rather than two functionally unrelated genes that happen to share a chromosomal locus.[7,23] Of the 648 cyclase homologs, 16 shared 95–100% identity with the KTE100 sequence; the predominant annotation was guanylyl/adenylyl cyclase. Of the 329 effector homologs, 19 shared 98–100% identity, but the overwhelming majority were annotated as hypothetical proteins — a signature consistent with an under-characterized defense module.

Parallel queries against UniProtKB (1,000 hits; E-value ≤ 0.01) returned 64 cyclase and 11 effector sequences. Among the 64 cyclases, approximately 15 shared ∼40% identity with KTE100 and all but two were annotated as adenylyl/guanylyl cyclases. Among the 11 effectors, two shared >40% identity; the remainder were annotated as DUF3307-, SLATT-domain-containing, or uncharacterized proteins. The asymmetric annotation status of the sensor (broad guanylyl/adenylyl cyclase homology) versus the effector (predominantly uncharacterized) is itself informative: it is consistent with the sensor being subject to long-range purifying selection on its enzymatic core, while the effector evolves under faster, niche-specific selective pressure — a pattern observed in canonical CBASS effectors.[8,23]

Enzyme Function Initiative gene neighborhood analysis (EFI-GNT) confirmed that the effector is encoded immediately upstream of the cyclase across all nine co-occurring

Uropathogenic genomes, with overall sequence identity to the KTE100 reference decreasing progressively down the organism list (Figure 2.1). Three discrete clusters of flanking genes — represented by orange (Cluster 1), green (Cluster 2), and gold (Cluster 3) arrows in Figure 2.1 — are consistently distributed across phylogenetically distinct Uropathogens. A string of seven small genes (≤150 residues each) is shared among KTE100, *Enterobacter asburiae*, and *Citrobacter portucalensis*; a three-gene antisense cluster is conserved in *Klebsiella pneumoniae*, *Enterobacter cloacae* complex, and *Salmonella enterica* subsp. *houtenae*; and the *Vibrio* species retain the largest complement of flanking genes (Figure 2.1). Whether these "traveler" genes function as accessory regulators — in the manner of WYL-domain transcription factors or HORMA-domain proteins documented in other CBASS systems[3,48] — or as genetically unlinked passengers riding the same mobile element remains an open question that warrants direct experimental dissection.

2.3.2 The sensor harbors a bifunctional guanylyl cyclase–HEPN domain

Submission of the 264-residue cyclase to the NCBI Conserved Domain (CD) database[19,22,44] classified residues 77–228 as a cyclase homology domain (cd07302; Nucleotidyl Cyclase III superfamily, AcyC COG2114; $E = 6.79 \times 10^{-10}$; bitscore: 58.66), with metal-coordinating residues at positions 79, 132, and 215 and a nucleotidyl-binding site spanning ^{79}DMRDST^{84} (Figure 2.2). No conserved domains were detected in the 183-residue effector, consistent with its uniform annotation as a hypothetical protein and motivating the structural reconstructions described in Section 2.3.5.

Sequence-conservation architecture was resolved by MAFFT alignment of ten phylogenetically representative cyclase orthologs (Figure 2.2).[39] Cyclases sharing ≥99% identity with KTE100 are denoted by a single asterisk; the most divergent ortholog (*P. fluorescens*) is denoted by double asterisks (Figure 2.2A). The consensus identity track and per-residue similarity coloring (Figure 2.2A–C) resolve three discrete medial blocks of high conservation: the nucleotidyl-binding signature ^{79}DMRDST^{84} (80–100% conservation; Figure 2.2B1), the downstream binding element ^{127}TEYLGDGILAFF^{138} (Figure 2.2B2), and a C-terminal block spanning residues 189–222 (Figure 2.2B3). In sharp contrast, residues 3–70 constitute the most divergent N-terminal region, with a

consensus gap at positions 29–48 relative to the KTE100 sequence (Figure 2.2C1). A second variable block at residues 158–169 diverges progressively as phylogenetic distance from KTE100 increases, with charged-to-neutral substitutions at positions 159, 161, 169, and 172 (Figure 2.2C2). The discrete partitioning of the cyclase into invariant catalytic and variable peripheral regions is the structural substrate on which my regulatory hypothesis rests.

Embedded within the conserved nucleotidyl-binding region, residues ^{81}RDSTKH^{86} satisfy the canonical HEPN consensus (RγX$_{3–6}$H), assigning the same eight-residue stretch the dual function of guanylyl cyclase nucleotide binding and HEPN-like RNA cleavage — an architectural pairing without precedent in the cyclic-nucleotide synthase literature.[2,11,16,27] Crucially, the assignment is corroborated by four flanking secondary HEPN motifs of the form EXXX[R/K] — ^{25}ENGGK^{29}, ^{35}ESLAK^{39}, ^{209}ECVFR^{213}, and ^{228}ERMQK^{232} (Figures 2.2 and 2.3) — each conserved across the nine co-occurring Uropathogenic genomes and architecturally consistent with the secondary EXXX[R/K] motifs documented in characterized HEPN ribonucleases.[2,47] AlphaFold3 modeling[1] places the primary HEPN motif on the interior of the dimerization interface, optimally positioned to act on substrates delivered into the active-site cleft (Figure 2.3). Notably, RNA-based pattern recognition by CBASS sensors has direct mechanistic precedent: Banh and colleagues established that the *Staphylococcus schleiferi* 2142-05 Type-I CBASS sensor (CdnE12) is licensed by direct binding of a discrete ∼400-nucleotide non-coding viral RNA — termed cGAS-activating bacteriophage RNA (cabRNA) — transcribed across the small (TerS) and large (TerL) terminase genes of the infecting phage; the cabRNA is both necessary and sufficient for cyclase activation, defining nucleic-acid sensing as a *bona fide* CBASS triggering modality.[3] The bifunctional cyclase–HEPN architecture identified here is fully consistent with this paradigm and is testable through direct RNA-binding assays.

SMART database analysis identified additional putative domains overlapping the guanylyl cyclase region: an exonuclease III domain, a DUF862 (permuted papain-fold of dsRNA viruses and eukaryotes), an eIF5C domain, and an aldehyde oxidase/xanthine dehydrogenase domain, with E-values spanning 3.7×10^{-4} to 1.6×10^{5} (Figure 2.2D). Given that the HEPN domain is an established RNA-cleavage activity,[2,27] the exonuclease

III and DUF862 annotations are most parsimoniously interpreted as the SMART database independently detecting facets of the same underlying nucleic-acid-processing function, rather than evidence of additional, biochemically distinct activities. The HEPN fusion — supported by both primary and four secondary motifs conserved across nine genomes — is the more parsimonious and better-substantiated interpretation.

2.3.3 Candidate post-translational modification residues localize to regions of sequence divergence

To resolve the regulatory architecture of the cyclase, the sequence was submitted to GPS-pPLM[49] (prokaryotic phosphorylation site prediction; comprehensive parameters, medium threshold, personal cutoff: 0.55) and STALLION[5] (lysine acetylation prediction; default parameters, cutoff: 0.55), and the predicted sites were cross-referenced with the MAFFT alignment to assess their positional context. GPS-pPLM identified two phosphorylation candidates: Ser99 (score: 0.65) and Ser257 (0.56), denoted by orange horizontal lines in Figure 2.3A. STALLION identified four acetylation candidates: Lys10 (0.71), Lys85 (0.58), Lys232 (0.70), and Lys254 (0.82), denoted by blue lines (Figure 2.3A). Consistent with my central hypothesis, all six predicted PTM sites localize to regions of inter-strain sequence divergence rather than to the conserved catalytic core (Figures 2.2C and 2.3A) — a partitioning that would be statistically unlikely under random sampling and that supports a model in which PTM regulates peripheral architecture without compromising catalytic invariance.

Two predicted PTM sites are of particular mechanistic interest. Lys85, a candidate for acetylation (0.58), is positioned ventral and offset to the dimerization interface and is immediately adjacent to the γ-phosphate of bound GTP (Figure 2.3D), implicating acetylation at this site as a candidate switch governing nucleotide engagement. Lys254 (0.82) and Ser257 (0.56) cluster proximal to the C-terminal cap/hinge defined by Glu251–Glu253–Lys254 — a fork-like third prong relative to the active site (Figure 2.3B). Crucially, the secondary HEPN motif ^{228}ERMQK^{232} harbors the Lys232 acetylation site (0.70), raising the testable hypothesis that PTM may simultaneously regulate the secondary HEPN RNA-processing activity and the primary cyclase function from spatially distinct but mechanistically coupled positions.

2.3.4 The cyclase dimer engages GTP at the dimerization interface and accommodates RNA in the active-site cleft

AlphaFold3[1] was used to predict the structure of the cyclase homodimer in complex with two GTP molecules and a physiologically relevant complement of divalent cations (six Mn^{2+}, four Fe^{3+}, four Mg^{2+}, two Zn^{2+}; Figure 2.3D). The complex prediction was high-confidence: ipTM = 0.87, pTM = 0.88, overall rank score = 0.93 (values >0.8 indicate a reliable protein–protein complex model)[1,42]. As with all *in silico* predictions, the configuration captured here represents a low-energy state under server conditions and may differ from the conformations adopted during active phage infection or homeostasis; experimental validation is required.

GTP-binding residues were mapped onto the primary sequence and tertiary structure (Figure 2.3A, D). The guanine ring is coordinated by Glu128, with Leu130, Leu135, and Val211 providing hydrophobic stabilization; γPO_4-binding residues (Met80–Thr84) overlap directly with the primary HEPN domain (sky blue; Figure 2.63A–B). Lys216 of monomer A coordinates the α-PO_4, while Asp132 of monomer B contributes stabilizing contacts across the interface (Figure 2.3D), establishing GTP engagement as a *bona fide* inter-monomer event that requires the dimer to be assembled. The C-terminal cap/hinge is defined by the negative charges of Glu251 and Glu253, with Lys254 forming a fork-like third contact (Figure 2.3C). The surface electrostatic map of the dimer reveals a predominantly negatively charged dorsal face and a positively charged ventral face (Figure 2.3C), consistent with the ventral surface attracting negatively charged substrates such as nucleoside triphosphates and nucleic acids. A bird's-eye view of the dimer reveals a yin-yang-like clasped architecture, while the side profile shows a proportional 50:50 split at the dimerization interface (Figure 2.3B). Both the primary HEPN domain (^{81}RDSTKH^{86}; sky blue) and the secondary HEPN motif ^{209}ECVFR^{213} (gold) are positioned ventral to and within the dimerization interface (Figure 2.3F), supporting a model in which GTP processing and nucleic-acid engagement converge on a shared structural platform.

To directly test whether the dimerization interface accommodates nucleic-acid substrates, AlphaFold3 was employed to predict the dimer in complex with a 12-uracil RNA strand (Figure 2.4; ipTM = 0.86, pTM = 0.88, overall rank = 0.91). The RNA

inserts into the active-site cleft at the same ventral interface that engages GTP, with the spines (black arrows, Figure 2.4B1) flanking the substrate channel — a topology consistent with RNA being delivered into the HEPN active site from the same surface that recruits the nucleotide substrate. Comparative AlphaFold3 modeling with single-base homopolymeric RNA and DNA strands (12-Cs, 12-Gs, 12-As, 12-U/Ts; Figure S2.1) and with mixed-composition strands (alternating G/Cs, alternating A/[U/T]s, arbitrary nucleotides; Figure S2.2) identified a strong preference for RNA over DNA across nucleotide compositions, with the highest-confidence engagement observed for pyrimidine-rich RNA substrates (Figures S2.1, S2.2). These models nominate the cyclase as a candidate sensor of viral RNA — a hypothesis that aligns directly with the cabRNA paradigm established by Banh and colleagues for *Staphylococcus schleiferi* 2142-05 Type-I CBASS[3] — and motivate direct biochemical RNA-binding and ribonuclease assays in the chapters that follow.

2.3.5 The effector is a ClyA-like α-pore-forming toxin with candidate PTM residues

The 183-residue effector was subjected to the same analytical pipeline as the cyclase. MAFFT alignment of nine orthologs resolved the conservation architecture of the effector across the co-occurring Uropathogens (Figure 2.5A); asparagine-linked glycosylation motifs (NX[ST], where X is any residue except proline) are denoted by green arrows (Figure 2.5A).[6,33] Protter topology prediction[25] identified three transmembrane domains (Ala21–Leu39, Leu59–Leu79, Phe155–Tyr178) and a single conserved N-linked glycosylation motif (^{120}NYT^{122}) at the extracellular face, conserved in seven of nine effector orthologs examined (Figures 2.65B2 and 2.65A1, B).

GPS-pPLM[49] predicted six phosphorylation candidates at the 0.55 cutoff: Lys3, Lys4, Tyr9, Lys95, Thr148, and Tyr149 (Figure 2.6A2). Thr148 carried the highest probability (0.86) and was mapped as an orange residue onto the monomer structure (Figure 2.66B). STALLION[5] identified Lys144 as the highest-confidence acetylation candidate (0.84; Figure 2.6A2). Thr148 and Lys144 are in close spatial proximity near the effector's C-terminus (Figure 2.6B), while the N-linked glycosylation site (Asn120; green) sits beneath the β-tongue, distal from both PTM sites (Figure 2.6B). Although Tyr121 is theoretically phosphorylatable, GPS-pPLM did not predict it as a site —

consistent with N-linked glycosylation at the flanking Asn120 being the dominant modification at this motif.[6,33]

AlphaFold3[1] reconstruction of the effector monomer (Figure 2.6B) reveals a predominantly α-helical architecture punctuated by a β-tongue (winter green), three transmembrane helices (salmon), and the mapped PTM sites. The highest-confidence oligomeric prediction is a hexameric pore (Figure 2.6C, D; ipTM = 0.74, pTM = 0.75, overall rank = 0.78), defined by a central α-helical channel surrounded by a β-sheet ring (Figure 2.66D1). Dimensions measured in ChimeraX: complex length ~70 Å (Figure 2.6C1), complex diameter ~60 Å (Figure 2.66D1), pore lumen ~20 Å (Figure 2.6D2). Surface electrostatic maps from both lateral and apical viewpoints define candidate regions of charge-mediated regulation and substrate gating (Figures 2.6C3, D3).

An α-helix-dominated monomer bearing a β-tongue, assembling into a hexameric ring pore, is the diagnostic architecture of a ClyA-like α-pore-forming toxin (α-PFT).[20,30,43] While β-PFTs have been documented as effectors in bacterial antiphage defense systems,[10] this analysis represents, to my knowledge, the first identification of an α-PFT as a candidate antiphage effector. Equally without precedent in the antiphage literature is N-linked glycosylation of an antiphage effector. Together, these features establish the KTE100 effector as a structurally and biochemically distinct class of antiphage effector, whose candidate post-translational modifications represent a novel axis of regulation that warrants direct biochemical interrogation.

2.4 Discussion

The most consequential finding of this study is the identification of a bifunctional guanylyl cyclase–HEPN domain fusion (^{81}RDSTKH^{86}) at the catalytic core of the KTE100 sensor, in which a single eight-residue stretch satisfies both the canonical primary nucleotide-binding signature of the guanylyl cyclase and the RγX$_{3–6}$H consensus of a metal-independent ribonuclease.[2,11,27] This interpretation does not rest on any single line of evidence: the primary motif assignment is corroborated by four flanking secondary EXXX[R/K] motifs, all of which are conserved across the nine independently sampled Uropathogenic genomes and which mirror the secondary HEPN architecture documented in characterized RNA-processing prokaryotic enzymes.[2,16,47] Co-localization

of a HEPN motif with a guanylyl cyclase active site is unprecedented in the cyclic-nucleotide synthase literature, but it is not without functional analogy: HEPN domains are increasingly recognized as architecturally promiscuous nucleotide-sensing and nuclease modules that integrate into structurally diverse host proteins.[21,27]

My hypothesis that regulatory residues preferentially occupy regions of sequence divergence is supported by the PTM predictions for both proteins. For the cyclase, all six candidate PTM sites (Ser99 and Ser257 for phosphorylation; Lys10, Lys85, Lys232, and Lys254 for acetylation) localize outside the conserved catalytic core. Three of these sites are mechanistically suggestive: the clustering of Lys254 and Ser257 near the C-terminal fork (Glu251–Glu253–Lys254), the placement of Lys232 within the secondary HEPN motif ^{228}ERMQK^{232}, and the positioning of Lys85 immediately adjacent to the γ-phosphate of GTP — collectively suggest a coordinated PTM network that could simultaneously modulate cyclase catalysis and HEPN ribonuclease activity from spatially distinct but functionally coupled positions. For the effector, the spatial proximity of Thr148 and Lys144 near the C-terminus (and their distance from the Asn120 N-glycosylation site) implies that phosphorylation and acetylation may regulate effector function through a mechanism distinct from glycosylation, with the latter serving as a constitutive structural marker.

The effector's predicted assembly as a ClyA-like α-PFT is mechanistically significant.[20,30,43] ClyA-family toxins do not require a cell-surface receptor to localize to the inner membrane and oligomerize through membrane contact alone.[30] This permits a parsimonious activation model: PTM of soluble effector monomers under homeostatic conditions could function as a brake against inadvertent oligomerization, while demodification or proteolytic processing upon phage infection could release this brake and license pore assembly. Such a model is concordant with the activation logic of Type-I CBASS systems generally, in which sensor proteolysis is the upstream trigger.[29] Empirical validation through site-directed mutagenesis at the predicted modification sites and liposome disruption assays will be required to test this model directly — work that is undertaken in subsequent chapters.

Both the cyclase and effector are phylogenetically restricted to Uropathogens within *Enterobacteriaceae — Salmonella*, *Klebsiella*, *Citrobacter*, *Vibrio*, and *Enterobacter* —

a distribution that suggests this defense module confers a fitness advantage specifically within the urinary tract niche. The three conserved flanking gene clusters identified by EFI-GNT represent an underexplored axis of this system; whether they encode co-regulated accessory functions analogous to the WYL-domain transcription factors or HORMA-domain regulators documented in other CBASS systems[3,48] — or, alternatively, whether they constitute an independent operon riding the same mobile element — is a question that warrants direct genetic dissection. The integration of a guanylyl cyclase–HEPN sensor with a ClyA-like α-PFT effector, taken together with the cabRNA precedent established by Banh and colleagues,[3] positions the KTE100 system as a candidate exemplar of a new branch of cyclic-nucleotide-based bacterial innate immunity, in which RNA sensing and α-pore-forming effector deployment converge in a single regulatory architecture.

The post-translational regulatory potential of the KTE100 system is further supported by the PTM repertoire of *E. coli* itself. *E. coli* encodes at minimum two tyrosine kinases (Etk and Wzc), several serine/threonine kinases (AceK, HipA, SrkA, YegL), and acyltransferases (PlsB, LpxA, Lnt) that could, in principle, target the predicted modification sites of either protein. How the cellular PTM landscape shifts during phage infection — and whether phage-encoded effectors deliberately hijack or disrupt this signaling network — represents a productive direction for future investigation.

The central limitation of this study is the absence of empirical validation. Computational predictions, while indispensable for hypothesis generation, cannot substitute for biochemical and genetic experiments.[1,42] Four future priorities follow directly from the analyses presented here: (i) confirming guanylyl cyclase catalytic activity and cGMP synthesis *in vitro*; (ii) testing HEPN-domain RNA-cleavage activity and its modulation by the identified PTM sites; (iii) characterizing effector oligomerization in the presence and absence of the candidate PTM modifications; and (iv) establishing the proteomic PTM landscape of both proteins under homeostatic and phage-infection conditions. The present chapter establishes the hypotheses; the subsequent chapters of this thesis address them experimentally.

2.5 Materials and Methods

2.5.1 Database and sequence searches

Protein sequences were submitted to NCBI BlastP under default parameters with the number of hits set to 1,000 and the E-value cutoff set to 0.05. Parallel searches were performed against UniProtKB BLAST (align.uniprot.org) under default parameters with the number of hits set to 1,000 and the E-value cutoff set to 0.01. Conserved domains were identified by submitting protein sequences to the NCBI Conserved Domain database under default parameters.[19,22,44] The cyclase and effector sequences were submitted to the Enzyme Function Initiative Gene Neighborhood Tool (EFI-GNT) to visualize the genomic context of the defense operon across related organisms.

2.5.2 Multiple sequence alignment and domain analysis

Ten phylogenetically representative cyclase orthologs and nine effector orthologs were aligned using the MAFFT online server under default parameters. Alignment visualizations were generated in Geneious Prime. Protein sequences were submitted to the SMART (Simple Modular Architecture Research Tool) database under default parameters to identify additional domain features.

2.5.3 Post-translational modification site prediction

Phosphorylation site prediction was performed using GPS-pPLM (https://pplm.biocuckoo.cn/)[49] under comprehensive parameters with a medium threshold and a personal cutoff of 0.55. Lysine acetylation site prediction was performed using STALLION[5] under default parameters with a personal cutoff of 0.55. Transmembrane topology and extracellular glycosylation motif prediction was performed using Protter.[25]

2.5.4 Structural prediction and visualization

Protein and protein–ligand complex structures were predicted using the AlphaFold3 server[1] under default parameters. Divalent cation and ligand inputs are specified for each

prediction in the Results. Confidence thresholds: ipTM and pTM > 0.8 indicate a high-confidence complex; the overall rank score is computed as a weighted sum of ipTM, pTM, and additional structural metrics.[1,42] CIF files produced by the AlphaFold3 server were visualized and quantitatively measured in UCSF ChimeraX.

2.6 Figures

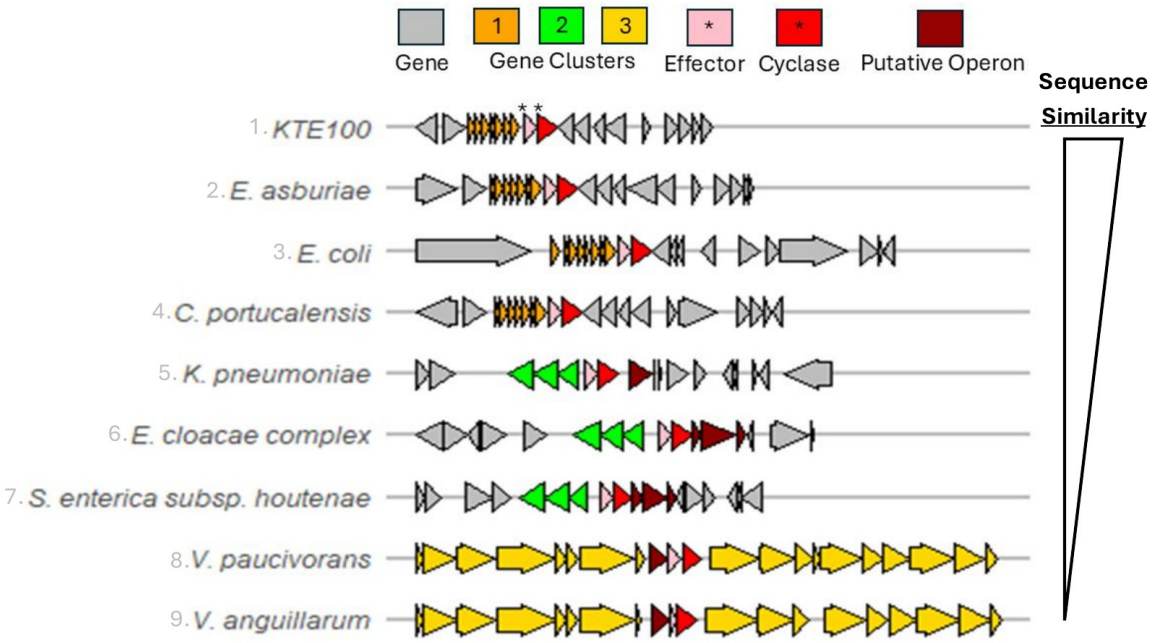


Figure 2.1. The genetic neighborhood of the KTE100 cyclase and effector defense operon. The cyclase–effector pair from *E. coli* str. KTE100 (row 1) is shown adjacent to eight homologous genetic neighborhoods identified across Uropathogenic species, ranked by overall sequence similarity to KTE100 (descending). Color key: gray, non-clustered flanking genes; orange (Cluster 1), green (Cluster 2), and gold (Cluster 3), three discrete clusters of conserved flanking genes; pink, KTE100 effector; red, KTE100 cyclase; dark red, additional putative operon-associated genes flagged with asterisks in row 1. Genetic neighborhoods were defined using the Enzyme Function Initiative Gene Neighborhood Tool (EFI-GNT). Sequence similarity to the KTE100 reference decreases progressively down the rows, and the conservation of three discrete flanking gene clusters across phylogenetically distinct Uropathogens defines a co-inherited genetic context for the defense module.

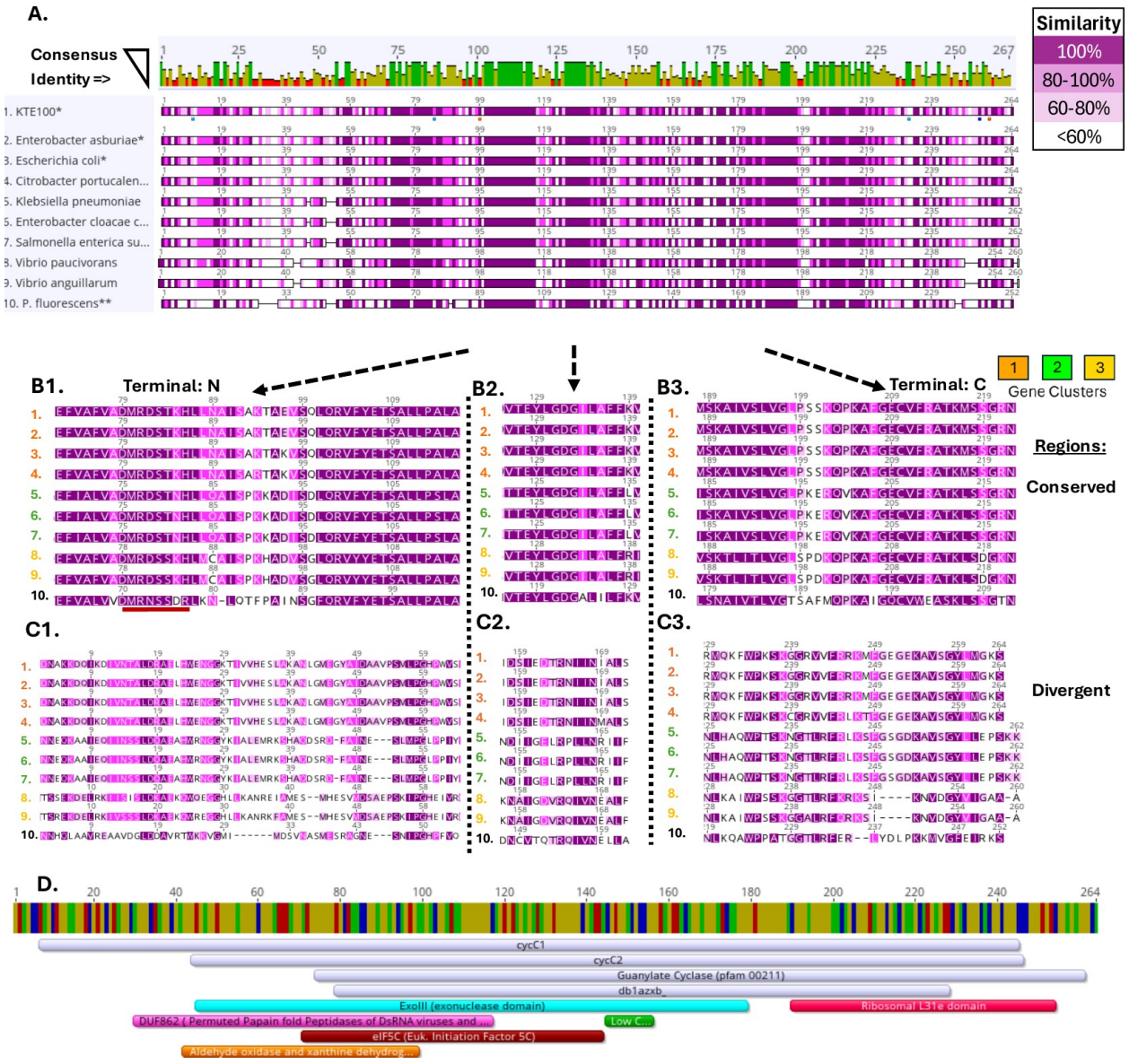

A.
Consensus Identity =>
Similarity
100%
80-100%
60-80%
<60%
1. KTE100*
2. Enterobacter asburiae*
3. Escherichia coli*
4. Citrobacter portucalen...
5. Klebsiella pneumoniae
6. Enterobacter cloacae c...
7. Salmonella enterica su...
8. Vibrio paucivorans
9. Vibrio anguillarum
10. P. fluorescens**
B1.
Terminal: N
B2.
B3.
Terminal: C
1
2
3
Gene Clusters
Regions:
Conserved
Divergent
C1.
C2.
C3.
D.
cycC1
cycC2
Guanylate Cyclase (pfam 00211)
db1azxb_
ExoIII (exonuclease domain)
Ribosomal L31e domain
DUF862 (Permuted Papain fold Peptidases of DsRNA viruses and ....
Low C...
eIF5C (Euk. Initiation Factor 5C)
Aldehyde oxidase and xanthine dehydrog...

Figure 2.2. Sequence-conservation architecture of the KTE100 cyclase. (A) MAFFT alignment of cyclase orthologs from ten representative strains. Single asterisk: orthologs with >99% identity to the KTE100 cyclase; double asterisk: the most divergent ortholog (*P. fluorescens*). Consensus identity track: green, 100% identical; gold, 30–99% identity; red, <30% identity. Per-residue similarity coloring: dark purple, 100%; medium purple, 80–100%; light purple, 60–80%; unshaded, <60%. Predicted lysine-acetylation residues are denoted by blue marks; predicted phosphorylation residues are denoted by orange marks. **(B)** Conserved blocks of the cyclase. **(B1)** Residue-level resolution of the conserved N-terminal block, including the ^{79}DMRDST^{84} nucleotidyl-binding signature. **(B2)** Conserved central block (^{127}TEYLGDGILAFF^{138}). **(B3)** Conserved C-terminal block (residues 189–222). **(C)** Divergent regions of the cyclase. **(C1)** Divergent N-terminal region (residues 3–70). **(C2)** Divergent central block (residues 158–169). **(C3)** Divergent C-terminal region. **(D)** SMART-database domain architecture. Vertical color bars at top denote per-residue physicochemical properties (red/blue, charged residues; green, polar uncharged; gold, hydrophobic); horizontal bars beneath identify candidate domain assignments. Numbered cluster colors correspond to the gene-cluster assignments in Figure 2.1. Phosphorylation sites were predicted with GPS-pPLM; lysine-acetylation sites were predicted with STALLION.

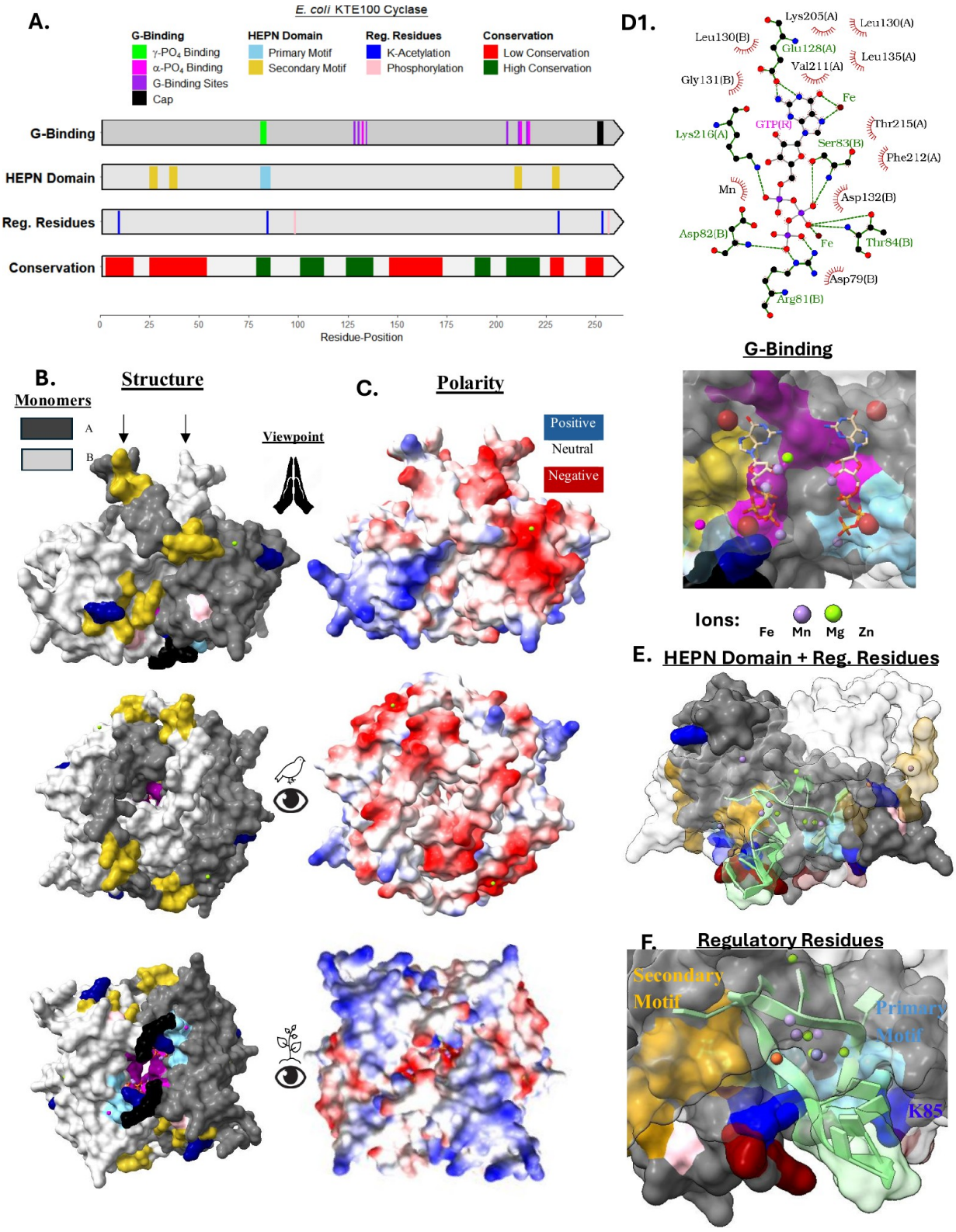

A.
E. coli KTE100 Cyclase
G-Binding
γ-PO4 Binding
α-PO4 Binding
G-Binding Sites
Cap
HEPN Domain
Primary Motif
Secondary Motif
Reg. Residues
K-Acetylation
Phosphorylation
Conservation
Low Conservation
High Conservation
G-Binding
HEPN Domain
Reg. Residues
Conservation
0 25 50 75 100 125 150 175 200 225 250
Residue-Position
D1.
Lys205(A)
Leu130(A)
Leu130(B)
Glu128(A)
Leu135(A)
Val211(A)
Gly131(B)
Fe
GTP(R)
Thr215(A)
Lys216(A)
Ser83(B)
Phe212(A)
Mn
Asp132(B)
Asp82(B)
Fe
Thr84(B)
Asp79(B)
Arg81(B)
G-Binding
B.
Monomers
A
B
Structure
Viewpoint
C.
Polarity
Positive
Neutral
Negative
Ions:
Fe
Mn
Mg
Zn
E.
HEPN Domain + Reg. Residues
F.
Regulatory Residues
Secondary Motif
Primary Motif
K85

Figure 2.3. GTP-binding architecture and HEPN-domain placement of the KTE100 cyclase dimer. (A) Linear-gene maps of the cyclase, color-coded by functional class: top, GTP-binding residues; second, primary (sky blue) and secondary (gold) HEPN motifs; third, regulatory residues (lysine-acetylation sites in blue, phosphorylation sites in pink); bottom, residue conservation (red, low; green, high). **(B)** Surface representation of the predicted cyclase homodimer (monomer A in dark gray, monomer B in light gray) viewed from the side (top), bird's-eye (middle), and ground-level (bottom) perspectives. The dimerization interface is indicated by the inset glyphs. **(C)** Surface electrostatic maps of the dimer at the same three viewpoints (red, negatively charged; blue, positively charged; white, neutral). **(D)** GTP-coordination network. **(D1)** Two-dimensional topology map of GTP localization at the dimerization interface, with monomer-A and monomer-B residues annotated. **(D2)** Surface representation of the dimer with two GTP molecules bound deep in the dimerization cleft. **(E)** Lateral view of the HEPN/RNA-binding residues, with a 12-cytosine RNA strand modeled within the dimerization interface. **(F)** Ground-level view of the HEPN domain showing the regulatory residues Lys85, Lys254, Ser256, the primary HEPN motif (^{81}RDSTKH^{86}), and the secondary HEPN motif (^{209}ECVFR^{213}). Color and ion legends as in panel A. Spheres: Fe (gray), Mn (purple), Mg (green), Zn (magenta). The GTP–cyclase complex was predicted with AlphaFold3 (ipTM = 0.87, pTM = 0.88, overall rank = 0.93); the cytosine–cyclase complex was predicted with AlphaFold3 (ipTM = 0.86, pTM = 0.87, overall rank = 0.91). The complex was modeled and rendered in UCSF ChimeraX. Acetylation sites were predicted with STALLION; phosphorylation sites were predicted with GPS-pPLM.

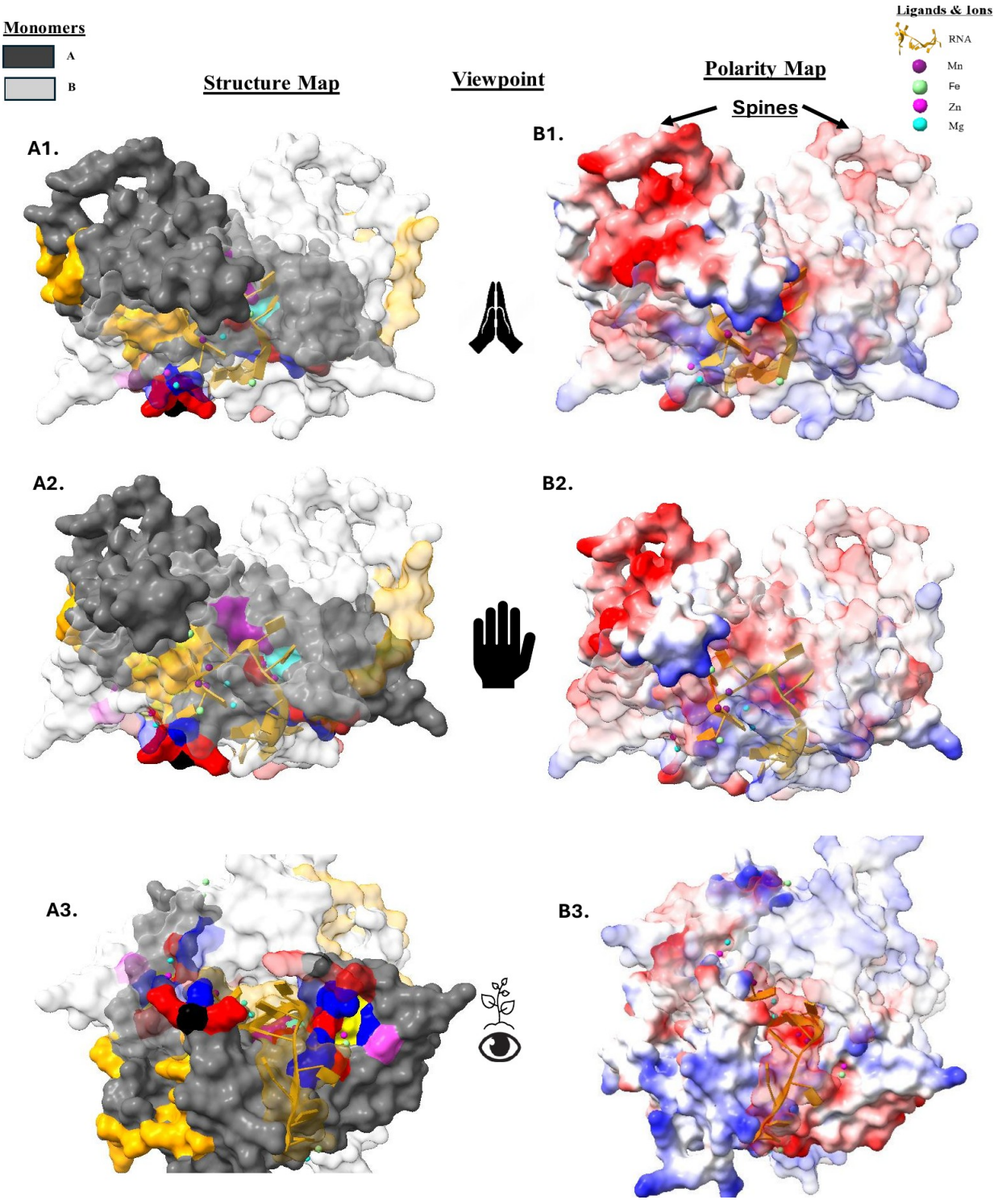

Monomers
A
B
Structure Map
Viewpoint
Polarity Map
Spines
Ligands & Ions
RNA
Mn
Fe
Zn
Mg
A1.
B1.
A2.
B2.
A3.
B3.

Figure 2.4. The KTE100 cyclase dimer accommodates a single-stranded RNA substrate at the dimerization interface. (A1–A3) Structural representation of the cyclase dimerization interface in complex with a 12-uracil RNA strand, viewed from the side (**A1**, top), lateral profile (**A2**, middle), and ground-level (**A3**, bottom) perspectives. Monomer A, dark gray; monomer B, light gray. **(B1–B3)** Surface electrostatic maps of the same complex at corresponding viewpoints (red, negatively charged; blue, positively charged; white, neutral). The "spines" of the dimerization interface (black arrows in B1) flank the active-site cleft into which the RNA strand inserts. The complex was predicted with AlphaFold3 (ipTM = 0.86, pTM = 0.88, overall rank = 0.91) and rendered in UCSF ChimeraX. Spheres: Mn (purple), Fe (light green), Zn (magenta), Mg (cyan).

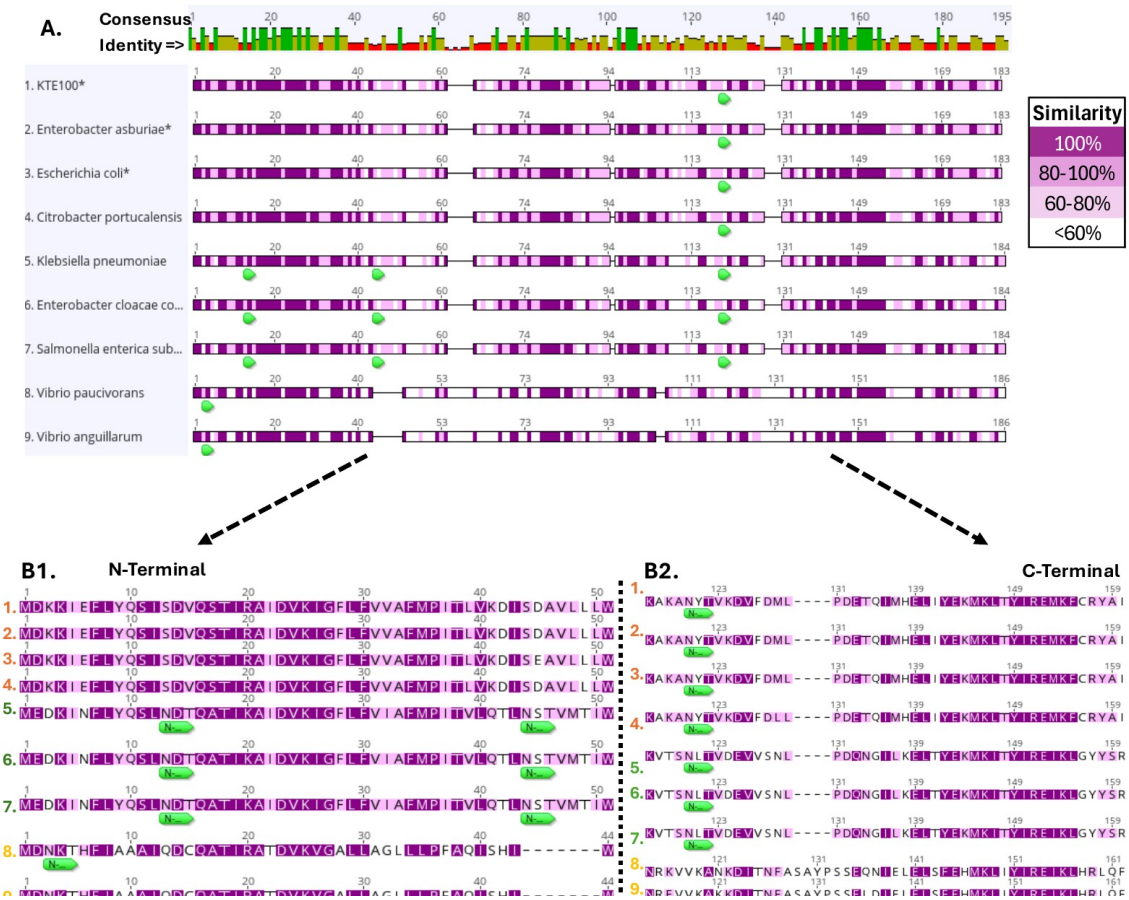


Figure 2.5. Sequence-conservation architecture of the KTE100 antiphage effector. (A) MAFFT alignment of effector orthologs from nine representative strains. Single asterisk: orthologs with >99% identity to the KTE100 effector; double asterisk: the most divergent ortholog. The *P. fluorescens* effector was excluded from the alignment due to its incompatible length. Consensus identity track: green, 100% identical; gold, 30–99% identity; red, <30% identity. Per-residue similarity coloring: dark purple, 100%; medium purple, 80–100%; light purple, 60–80%; unshaded, <60%. Green arrows denote asparagine-linked glycosylation motifs (NX[ST], where X is any residue except proline). **(B1)** Residue-level resolution of the divergent N-terminal block. **(B2)** Residue-level resolution of the conserved C-terminal block, including the ^{120}NYT^{122} N-glycosylation motif present in seven of nine orthologs. The discrete partitioning of the effector into a conserved transmembrane core and divergent termini parallels the conservation architecture of the cyclase (Figure 2.2).

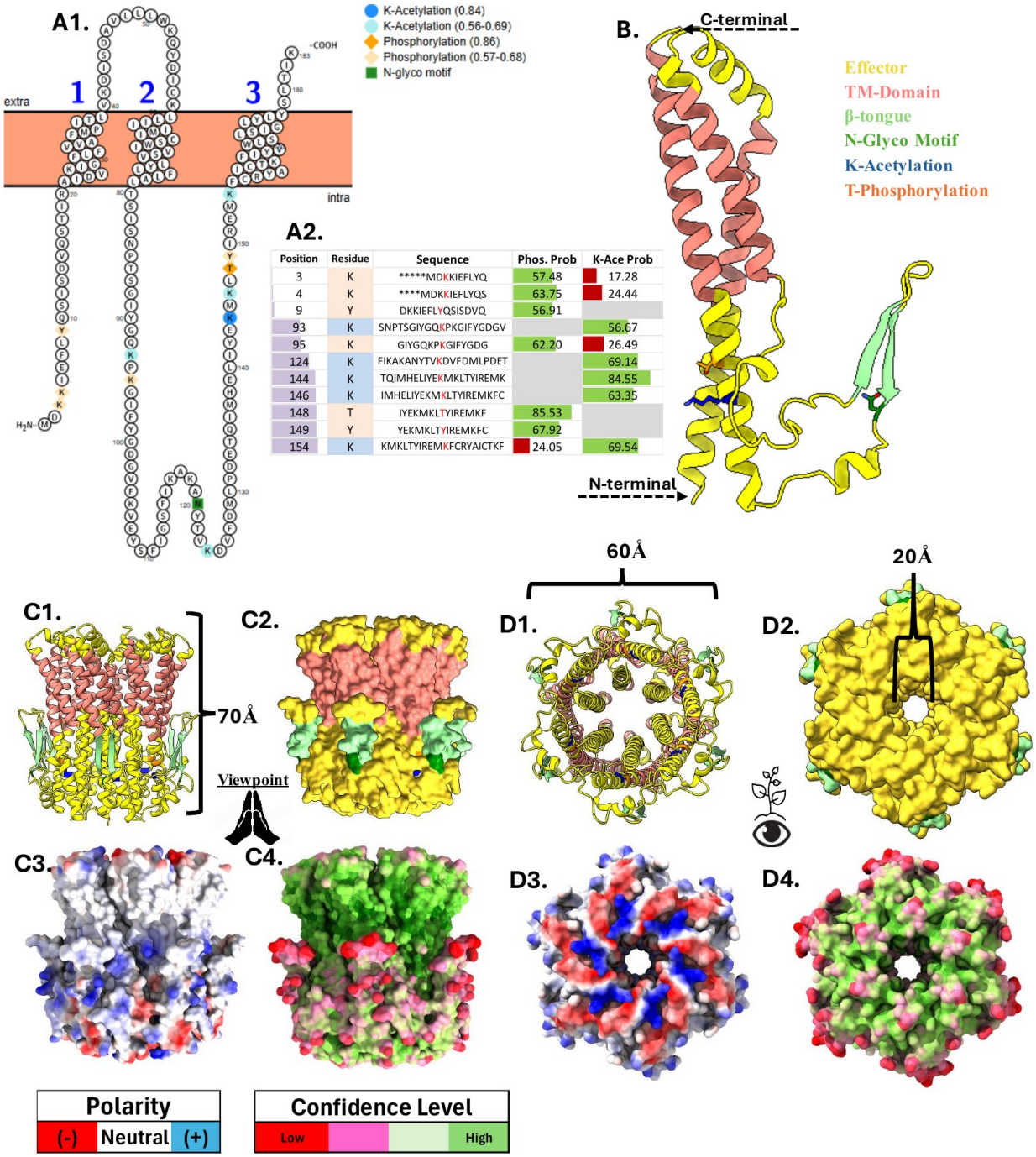


Figure 2.6. Structural prediction of the KTE100 antiphage α-pore-forming effector. (A1) Predicted topological localization of the effector monomer in the inner membrane, with three transmembrane domains. **(A2)** Per-residue probabilities of phosphorylation and lysine acetylation; asterisks denote terminal residues, red coloring denotes residues of mechanistic interest. **(B)** Effector monomer ribbon prediction (yellow), with the β-tongue (pale green), three transmembrane helices (salmon), and the highest-probability PTM residues (Asn120, Lys144, Thr148) shown as atom displays. **(C)** Hexameric pore (lateral view): **(C1)** ribbon, **(C2)** surface, **(C3)** electrostatic, and **(C4)** per-residue confidence representations. **(D)** Hexameric pore (apical view): **(D1)** ribbon, **(D2)** surface, **(D3)** electrostatic, and **(D4)** per-residue confidence

representations. Polarity legend: red, negative; white, neutral; blue, positive. Confidence legend: red, <50%; pink, 50–70%; light green, 70–90%; dark green, >90%. Symbol legend: filled blue circle, lysine-acetylation site (probability 56–69%); large filled blue circle, lysine-acetylation site (probability 84%); orange diamond, phosphorylation site (57–68%); filled orange diamond, phosphorylation site (84%); filled green box, N-linked glycosylation motif. Structures were predicted with AlphaFold3 (in the absence of metal ions) and rendered in UCSF ChimeraX (hexamer ipTM = 0.74, pTM = 0.75, overall rank = 0.78). Acetylation sites were predicted with STALLION; phosphorylation sites were predicted with GPS-pPLM. Complex dimensions measured in ChimeraX: lateral length ∼70 Å; apical diameter ∼60 Å; pore lumen ∼20 Å.

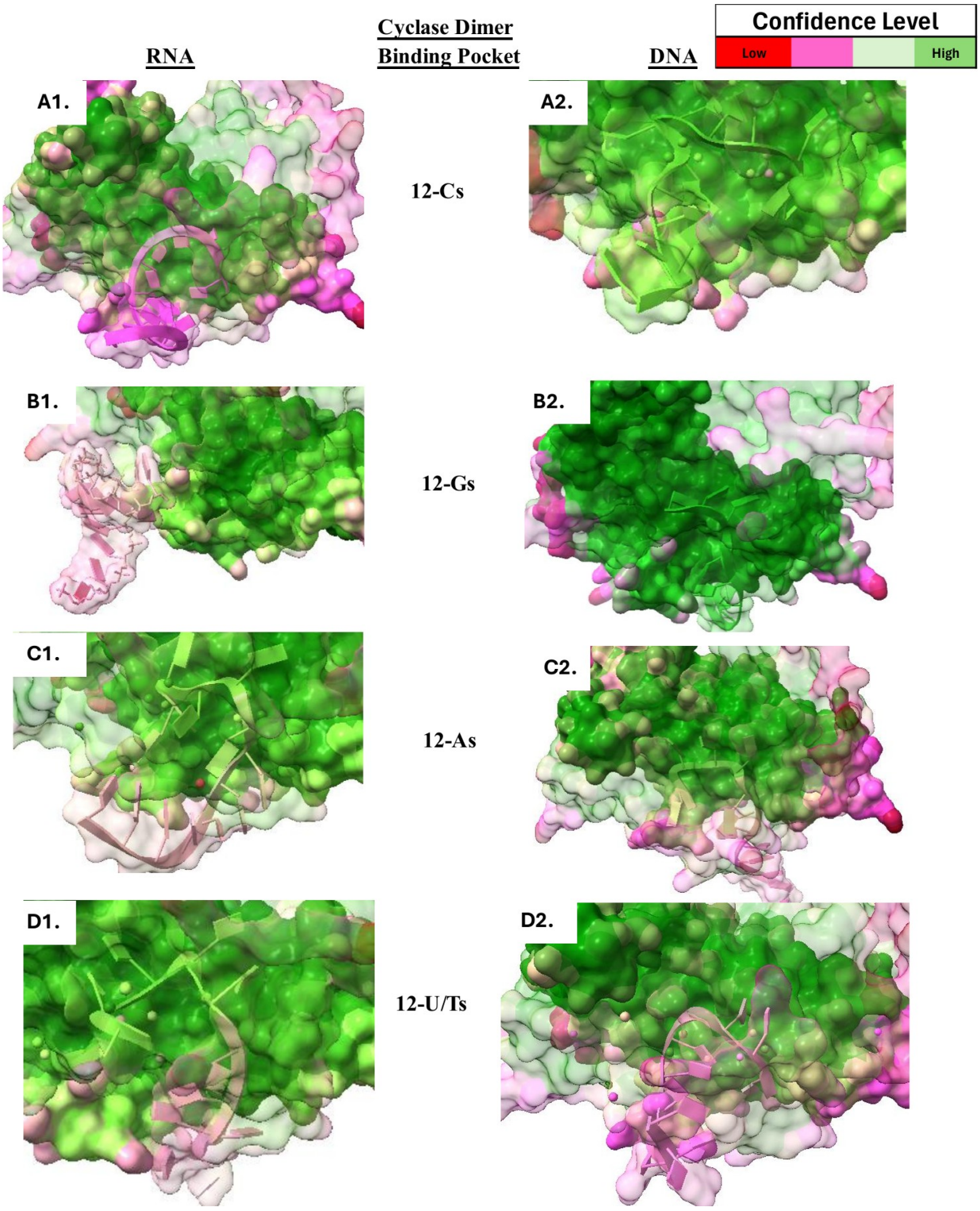


Figure S2.1. The KTE100 cyclase dimer engages homopolymeric nucleic-acid substrates with RNA preference. Predicted complexes of the cyclase dimer with single-stranded homopolymeric nucleic-acid substrates, with RNA in the left column (**A1–D1**) and DNA in the right column (**A2–D2**). Substrates: **(A)** 12-cytosines; **(B)** 12-guanosines; **(C)** 12-adenines; **(D)** 12-uracils (**D1**) or 12-thymines (**D2**). Per-residue confidence is rendered as a heat map: red, <50%;

pink, 50–70%; light green, 70–90%; dark green, >90%. Across all four homopolymeric compositions, RNA substrates engage the dimerization interface with higher confidence than the corresponding DNA substrates, identifying the cyclase as a candidate viral-RNA sensor. Complexes were predicted with AlphaFold3 and rendered in UCSF ChimeraX.

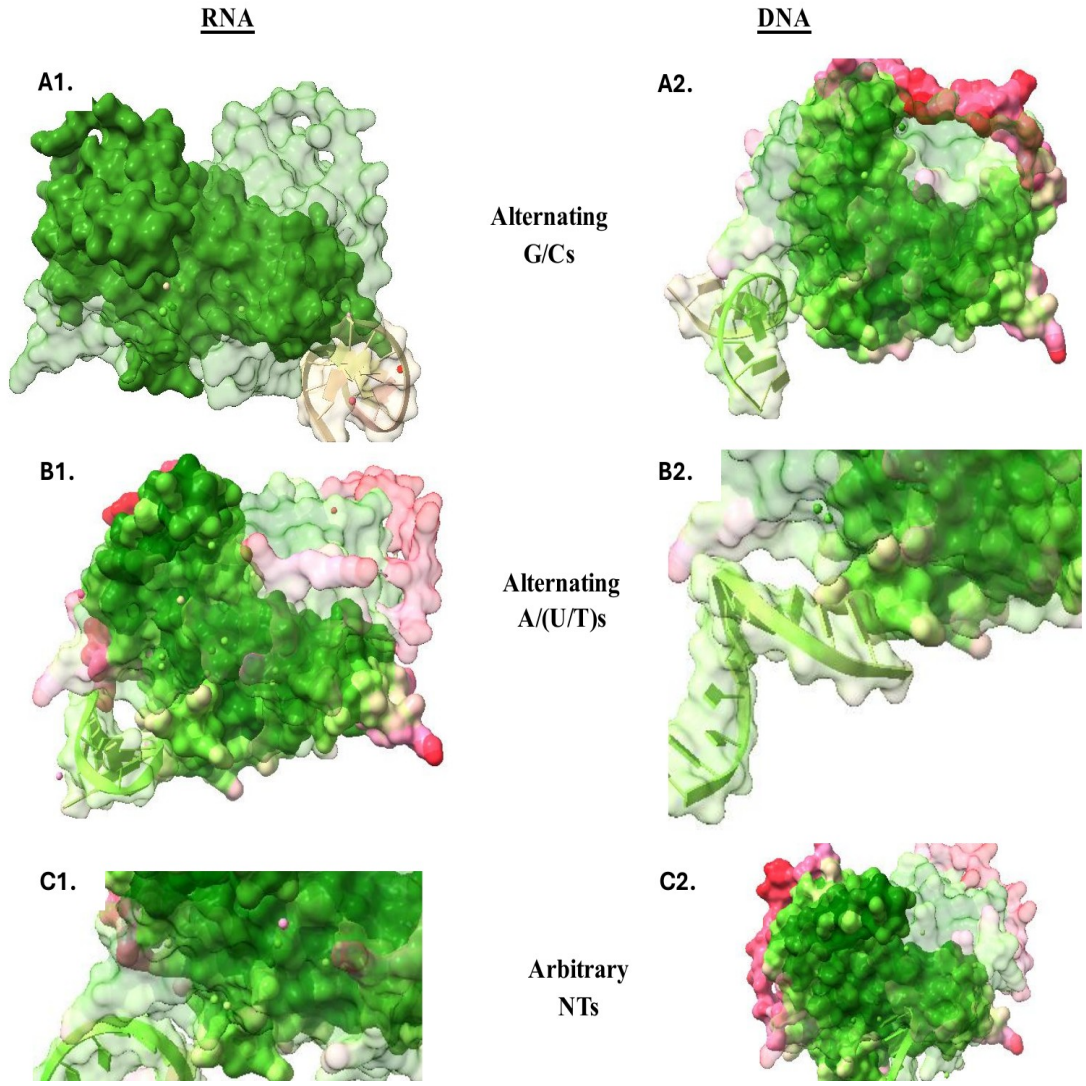


Figure S2.2. The KTE100 cyclase dimer engages mixed-composition nucleic-acid substrates with RNA preference. Predicted complexes of the cyclase dimer with single-stranded mixed-composition nucleic-acid substrates, with RNA in the left column (**A1–C1**) and DNA in the right column (**A2–C2**). Substrates: **(A)** alternating G/Cs; **(B)** alternating A/(U or T)s; **(C)** arbitrary nucleotide compositions. Per-residue confidence is rendered as a heat map: red, <50%; pink, 50–70%; light green, 70–90%; dark green, >90%. The complementary modeling of homopolymeric (Figure S2.1) and mixed-composition (this figure) substrates collectively supports preferential RNA engagement at the cyclase dimerization interface. Complexes were predicted with AlphaFold3 and rendered in UCSF ChimeraX.

Abbreviations

CBASS	Cyclic Oligonucleotide-Based Antiphage Signaling System
cGAS	cyclic GMP-AMP Synthase
cGMP	cyclic Guanosine Monophosphate
ClyA	Cytolysin A pore-forming toxin
EFI-GNT	Enzyme Function Initiative — Gene Neighborhood Tool
GTP	Guanosine Triphosphate
HEPN	Higher Eukaryotes and Prokaryotes Nucleotide-binding domain
KTE100	*Escherichia coli* strain KTE100
NCBI	National Center for Biotechnology Information
PFT	Pore-Forming Toxin
PTM	Post-Translational Modification
α-PFT	α-helical Pore-Forming Toxin

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CHAPTER 3

AN ANTIPHAGE GUANYLYL CYCLASE REGULATES CELL WALL SYNTHESIS AND MOTILITY IN *ESCHERICHIA COLI*

By: Jeffrey Wyatt-Wolfgang Marciniak-Haynosch

3.1 Abstract

Cyclases function as molecular alarm systems, empowering bacteria to interpret and respond to environmental insults including toxins, temperature flux, nutrient scarcity, and viral invasion.[11,17,24,32,16] While the physiological roles of adenylyl and diguanylyl cyclases in bacteria are firmly established,[15,17,30,45] the cellular functions of bacterial guanylyl cyclases remain incompletely understood.[16,24,39] Here I demonstrate that CBASS-like antiphage guanylyl cyclases from *Escherichia coli* and *Pseudomonas fluorescens* orchestrate pleiotropic effects on cellular architecture and behavior. Compound light microscopy revealed that ectopic guanylyl cyclase expression induces a gram-positive-like morphological transition in *E. coli*, in stark contrast to strains expressing the α-pore-forming toxin (PFT) effector, which retained canonical gram-negative features. Concomitantly, cyclase overexpression enhanced motility by 24% and reduced biofilm biomass by 68%, indicating cyclase-mediated prioritization of planktonic over sessile lifestyles. Critically, these phenotypic transitions occurred in the absence of PFT effector oligomerization, a process strictly confined to phage-induced conditions. These findings establish that a bacterial antiphage guanylyl cyclase rewires cell wall architecture and lifestyle commitment, bridging phage resistance to the most fundamental decisions in bacterial growth strategy.

3.2 Introduction

Bacteria deploy cyclases to translate environmental cues into pleiotropic cellular responses through the synthesis of small-molecule second messengers.[15,16,23,30] A specialized subset of these enzymes is integral to defense against bacteriophages.[12,25,27] Foremost

among these systems are the Cyclic oligonucleotide-Based Antiphage Signaling System (CBASS) and the Pyrimidine Cyclase System for Antiphage Resistance (PYCSAR), which operate as coordinated two-protein partnerships in which a sentinel cyclase couples to a cognate effector.[11,27,47] The cyclase senses infection and synthesizes a small-molecule second messenger that activates the downstream effector, committing the host to self-termination before the phage completes its replication cycle.[11,13,25,27,47]

The molecular trigger that licenses CBASS activation has long remained elusive,[37] but recent work by Banh and colleagues defined a discrete RNA ligand. Within the Type I CBASS system of *Staphylococcus schleiferi* strain 2142-05, a 400-nucleotide CBASS-activating bacteriophage RNA (cabRNA) — derived from the phage terminase small subunit (*terS*) and terminase large subunit (*terL*) genes — directly engages the host CdnE03 cyclase and promotes cGAMP synthesis to initiate the antiphage cascade.[4] This finding establishes a viral RNA as the proximal trigger of CBASS-encoded cGAS-like activity and reframes how prokaryotic immunity discriminates self from non-self at the molecular level.

The best-studied CBASS cyclase, DncV, principally synthesizes cGAMP but is biochemically promiscuous and capable of producing multiple second messengers,[21] whereas adenylyl and guanylyl cyclases generally exhibit far stricter specificity for their respective nucleoside triphosphates.[46] PYCSAR cyclases synthesize cCMP or cUMP from CTP or UTP, respectively.[47] Dinucleotide cGAMP,[11,47] trinucleotide cAAA,[50] and pyrimidine-derived cCMP and cUMP signals[47] together populate a structurally diverse niche of antiphage second messengers.

The role of guanylyl cyclases in bacterial physiology has been the subject of decades of debate.[3,5,14,18,24,29] Cyclic GMP was first detected in *E. coli* and *Bacillus licheniformis* by Bernlohr and colleagues,[5] and NO-sensitive guanylyl cyclases — the best-characterized class — were subsequently shown to govern cGMP-dependent signaling across both eukaryotes and prokaryotes.[3,14,29,31] Structural studies of Cya2 from *Synechocystis* confirmed *bona fide* guanylyl cyclase activity in a prokaryotic enzyme,[33,36] and bioinformatic analyses have substantially expanded the inventory of putative bacterial guanylyl cyclases.[39] Functionally, cGMP has been implicated in cyst

development in *Rhodospirillum centenum*,[26] phytopathogenesis in *Xanthomonas campestris*,[2] virulence-associated physiology in *Ralstonia solanacearum*,[22] and broader nucleotide second-messenger cross-talk.[16]

Here I delineate functional roles for antiphage guanylyl cyclases in *E. coli*. I employed three structurally homologous cyclases: two guanylyl cyclases — one from Uropathogenic *E. coli* str. KTE100 (ecC) and one from *P. fluorescens* (pfC) — and a uridylate cyclase from *Pseudomonas koreensis* (pkC; PYCSAR). I first identified a robust optical density phenotype upon ectopic cyclase expression, then narrowed my focus to ecC and demonstrated that its expression remodels cell wall architecture and biases the planktonic-to-sessile transition.

3.3 Results

3.3.1 Ectopic expression of antiphage guanylyl cyclases increases optical density without increasing cell number

Ectopic expression of the KTE100 antiphage defense operon in *E. coli* produced a striking elevation in optical density at 600 nm (OD_{600}) (Figure 3.1A). To determine whether this elevation reflected an increase in cell number, colony-forming units per milliliter (CFU/mL) were measured in parallel. Two additional antiphage cyclases — pfC from *P. fluorescens* and pkC from *P. koreensis* — sharing overall structural homology with ecC (Figure 3.1B) but distinct primary sequences and cognate effectors, were included for comparison. PfC has been demonstrated to synthesize cGMP,[36,39] while pkC operates as a PYCSAR uridylate cyclase that synthesizes cUMP.[47]

Within the first three hours of subculture, *E. coli* ectopically expressing ecC alone or the complete antiphage defense operon (ecC+ecE) exhibited a ~160% increase in OD_{600} relative to the empty vector (EV) control (Figure 3.1Ac). Cells expressing pfC showed a ~120% elevation in OD_{600} relative to EV, while pkC- and ecE-expressing cells showed no significant OD_{600} increase over the same interval (Figure 3.1Ac). Across 24 hours, pfC-expressing cells maintained an elevated OD_{600}, whereas ecC-expressing cells returned to EV-comparable levels (Figure 3.1Ab). Cells overexpressing ecE alone displayed a marked overall decrease in OD_{600} at 24 hours (Figure 3.1Ab, 3.1Ac).

CFU/mL was quantified by spot assay at 0, 3, and 24 hours post-subculture (Figure 3.1Ad). At three hours, CFU/mL for all conditions was equivalent to the EV control (Figure 3.1Ad). At 24 hours, CFU/mL remained unchanged for all conditions except ecE-expressing cells, which exhibited a ~10-fold decrease relative to EV — consistent with unregulated effector overexpression imposing a fitness cost on the host (Figure 3.1Ad). Taken together, the OD_{600} elevation observed at three hours for ecC- and pfC-expressing cultures cannot be ascribed to enhanced proliferation, as CFU/mL remained constant throughout (Figure 3.1Ac, 3.1Ad). Downstream multiplicity-of-infection (MOI)-based assays therefore require no normalization, since viable cell counts are equivalent to the negative control.

3.3.2 Guanylyl cyclase expression induces a gram-positive-like cell wall phenotype in E. coli

Because elevated OD_{600} without a corresponding increase in CFU/mL cannot be explained by additional cells, I hypothesized that the cell walls of ecC- and pfC-expressing cultures had thickened. The gram stain differentiates gram-positive from gram-negative bacteria on the basis of cell wall architecture: gram-positive organisms retain the crystal violet–iodine (CVI) complex within their thick peptidoglycan layer during decolorization, whereas gram-negative organisms shed the complex because their thin peptidoglycan — sandwiched between two lipid bilayers — cannot retain it.[6,43,48] Pathological thickening of the gram-negative cell wall would therefore manifest as a darker, gram-positive-like phenotype in *E. coli*, which ordinarily stains pink. Cultures were grown to late-log phase, fixed to glass slides, gram-stained, and observed by compound light microscopy at 100x oil immersion. Empty-vector controls were co-fixed to each slide to provide an internal reference, accounting for the three EV replicates depicted (Figure 3.2).

As predicted, EV control, ecE-, and pkC-expressing cells displayed the canonical pink phenotype of gram-negative bacteria (Figure 3.2). In stark contrast, cells ectopically expressing ecC or pfC stained dark purple, indicative of a gram-positive-like cell wall (Figure 3.2). To control for decolorization artifacts, ecC+ecE-expressing and EV-expressing cells were mixed and co-stained on a single slide; the resulting field contained

both pink (EV) and dark purple (ecC+ecE) cells, confirming that the phenotype is specific to cyclase-expressing cells and is not a staining artifact (Figure 3.2). Notably, purple aggregates in the EV control were observed only at dense cell clusters and not among free-floating individual cells; by contrast, free-floating ecC- and pfC-expressing cells remained uniformly dark purple (Figure 3.2). This pattern indicates that the CVI complex is stably retained within the cell walls of cyclase-expressing cells rather than passively trapped by physical crowding.

A parsimonious explanation for this phenotype is that ectopic cGMP production by ecC drives a cyst-like cell wall remodeling program. In *R. centenum*, cGMP is the master regulator of cyst development,[26] a differentiation program that produces cells with substantially thickened, polysaccharide-rich outer coats that retain the CVI complex more efficiently than vegetative gram-negative cells. Analogous cGMP-dependent wall remodeling has been documented in *X. campestris*, where a cGMP-binding GGDEF-domain protein drives exopolysaccharide production and phytopathogenesis,[2] and in *R. solanacearum*, where 2′,3′-cGMP regulates virulence-associated physiology including biofilm formation and cell surface properties.[22] Under a cyst-like differentiation model, thickened cell walls would (i) scatter light more efficiently per cell, accounting for the elevated OD_{600} without increased CFU/mL (Figure 3.1Ac, 3.1Ad); (ii) retain the CVI complex during decolorization, producing the gram-positive-like phenotype (Figure 3.2); and (iii) reflect a cGMP-specific developmental program rather than a generic cyclic nucleotide response, since pkC-expressing cells — which produce cUMP — did not exhibit the phenotype. Direct confirmation through intracellular cGMP quantification, peptidoglycan mass spectrometry, and transmission electron microscopy of cell wall ultrastructure will be required to establish the underlying mechanism definitively.

3.3.3 Ectopic expression of the guanylyl cyclase increases the swimming distance of E. coli

Cyclic di-GMP is a critical second messenger that drives cells from a motile to a sessile state,[19,35,44,45] and is synthesized from two GTP molecules by diguanylyl cyclases bearing GGDEF domains.[1,16] Because cGMP is synthesized from a single GTP molecule by guanylyl cyclases, the two enzyme classes compete for the same intracellular GTP

pool. I therefore reasoned that if ecC produces cGMP at a basal level under ectopic expression, the resulting depletion of the GTP pool would suppress c-di-GMP synthesis and hold cells in a motile state.[1,45]

As predicted, ectopic expression of ecC — either alone or as the complete ecC+ecE module — increased the swimming distance of *E. coli* by 0.6 cm (~24%) relative to the EV control (Figure 3.3). Cells expressing ecE alone swam an equivalent distance to the EV control, confirming that the motility phenotype is specific to cyclase activity and is not an effector artifact (Figure 3.3). The assay was repeated with pfC and pkC. Cells expressing pfC swam an equivalent distance to the EV control, while pkC-expressing cells exhibited a ~12% increase in swimming distance relative to EV (Figure 3.3). These data indicate that ecC produces sufficient cGMP at baseline to detectably deplete the GTP pool and reduce c-di-GMP synthesis,[7,21] while pfC produces enough cGMP to drive cell wall remodeling (Section 3.3.2) but not enough to measurably alter motility under these conditions.

3.3.4 E. coli guanylyl cyclase expression diminishes biofilm formation

Elevated c-di-GMP promotes biofilm activity by stimulating exopolysaccharide synthesis and surface adhesion.[28,30,45] Because ecC and diguanylyl cyclases compete for the same GTP pool, I reasoned that ecC overexpression would diminish the GTP available for c-di-GMP synthesis and thereby suppress biofilm formation. To modulate cyclase activity directly, I exploited the well-documented dependence of guanylyl cyclases on divalent cations,[27,36] particularly manganese,[36] by growing ectopically expressing strains in media supplemented with 0.1 μM $MnCl_2$ (LBMMB) or 0.1 mM $MnCl_2$ (LBMMB+).

OD_{600} was monitored over six hours (Figure 3.4A–C). Growth was equivalent among cyclase-expressing strains and the EV control under both manganese conditions (Figure 3.4A, 3.4B). However, 0.1 μM $MnCl_2$ rescued growth in strains expressing only ecE relative to the EV control, suggesting that low-concentration manganese suppresses ecE toxicity, possibly by interfering with effector oligomerization (Figure 3.4A, 3.4B). This growth rescue was abolished at 0.1 mM $MnCl_2$, consistent with higher manganese concentrations being toxic to *E. coli* (Figure 3.4C).

Biofilm was quantified by crystal violet staining of glass tube walls following overnight ambient incubation, solubilization in 30% methanol/10% glacial acetic acid, and measurement at OD_{550} (Figure 3.4D). Without supplemental manganese, ecC-expressing cells produced minimal biofilm relative to EV controls, consistent with the GTP depletion model (Figure 3.4D). At 0.1 μM $MnCl_2$, biofilm production was further diminished in cells expressing ecC alone, confirming that manganese-dependent cyclase activity diverts GTP toward cGMP synthesis at the expense of c-di-GMP biosynthesis (Figure 3.4D). Notably, biofilm production was rescued upon co-expression of ecC and ecE, suggesting that the effector functionally antagonizes the biofilm-suppressive effect of cyclase activity alone (Figure 3.4D). At 0.1 mM $MnCl_2$, overall OD_{600} and biofilm production were diminished across all conditions, consistent with manganese toxicity at this concentration (Figure 3.4C, 3.4D).

Together, these data demonstrate that manganese supports ecC-dependent cGMP synthesis at low concentrations but becomes broadly toxic at higher concentrations. The biofilm suppression observed exclusively in ecC-expressing cells affirms that ecC produces cGMP at a basal level under these conditions, and the effector-dependent rescue suggests that co-expression of ecC and ecE establishes a regulatory equilibrium that restores near-normal biofilm capacity.

3.4 Discussion

Ectopic expression of the KTE100 antiphage defense operon (ecC+ecE) in *E. coli* elevated OD_{600} without a corresponding increase in CFU/mL across 24 hours (Figure 3.1Ad), excluding enhanced proliferation as an explanation. I therefore concluded that the OD_{600} elevation reflects a per-cell increase in light scattering, most plausibly attributable to peptidoglycan accumulation or impaired peptidoglycan recycling.

Gram staining substantiated this interpretation: ecC- and pfC-expressing cells adopted a gram-positive-like phenotype within the gram-negative *E. coli* host (Figure 3.2). Motility and biofilm assays together provided biochemical corroboration for cGMP production, since both phenotypes are mechanistically tethered to GTP pool dynamics: elevated cGMP synthesis depletes the substrate available for diguanylyl cyclases, tipping

the c-di-GMP balance downward and favoring motility over sessile growth (Figures 3.3, 3.4D).[7,28,45] Together, these results indicate that ecC produces more cGMP at baseline than pfC, since pfC is sufficient to drive cell wall remodeling but does not measurably affect motility or biofilm formation under these conditions.

Critically, co-expression of ecC and ecE did not cause colony loss (Figure 3.1Ad). I therefore propose that a threshold cGMP concentration must be reached to license ecE oligomerization into a functional pore, and that the basal cGMP levels achieved under ectopic expression in the absence of phage infection fall below this threshold. Overexpression of ecE alone was toxic, as evidenced by a ~10-fold drop in CFU/mL at 24 hours (Figure 3.1Ad), consistent with unregulated α-PFT effector activity in the absence of a cyclase-dependent brake. The partial rescue of ecE toxicity by 0.1 μM $MnCl_2$, but not 0.1 mM $MnCl_2$, is consistent with low-concentration manganese interfering with effector oligomerization, although biochemical confirmation will be required (Figure 3.4A, 3.4B).

The gram-positive-like staining phenotype is the most mechanistically significant and least understood result of this study. For ecC to drive CVI complex retention in an *E. coli* host, the cell wall must be substantially and abnormally thickened,[6,43,48] since gram-negative cells ordinarily lack the dense peptidoglycan layer that traps CVI in gram-positive organisms. cGMP-mediated regulation of peptidoglycan biosynthesis is not yet established in *E. coli* and represents a novel finding. However, precedent for cGMP driving cyst-like cell wall differentiation in bacteria is unambiguous: in *R. centenum*, cGMP is the master regulator of cyst development, a program that yields cells with thickened polysaccharide-rich outer coats.[26] In *Synechocystis* strain PCC 6803, the guanylyl cyclase Cya2 mediates light-dependent cGMP production that controls cellular remodeling.[33] More recently, a 2′,3′-cGMP signaling pathway was shown to regulate cell surface properties and virulence-associated phenotypes in *R. solanacearum*,[22] and elevated cGMP levels were demonstrated to increase c-di-GMP synthesis through a cGMP-responsive GGDEF-domain protein in *X. campestris*.[2] Together, these precedents support the hypothesis that elevated intracellular cGMP produced by ecC may trigger a cyst-like differentiation program in *E. coli*, driving cell wall thickening through peptidoglycan accumulation or exopolysaccharide deposition analogous to the cyst coat

of *R. centenum*.[26] Under this model, thickened cell walls would scatter more light per cell, accounting for the elevated OD_{600} without increased CFU/mL (Figure 3.1Ac, 3.1Ad); trap the CVI complex during decolorization, yielding the gram-positive-like phenotype (Figure 3.2); and arise from a developmental program driven specifically by cGMP rather than by cyclic nucleotides broadly, consistent with the absence of the phenotype in pkC-expressing cells. Direct biochemical evidence — including intracellular cGMP quantification, peptidoglycan mass spectrometry, and transmission electron microscopy of cell wall ultrastructure — will be required to confirm this hypothesis.

3.5 Materials and Methods

Bacterial overnight culture preparation

Strains were streaked from −80 °C glycerol stocks onto LB plates supplemented with 100 µg/mL ampicillin and 25 µg/mL chloramphenicol (LB_{ac}) and incubated at 37 °C overnight. One isolated colony was inoculated into 2 mL of LB media supplemented with 100 µg/mL ampicillin, 25 µg/mL chloramphenicol, 0.1 mM $MnCl_2$, and 0.5 mM $MgCl_2$ in a 10 mL glass culture tube and grown overnight at 37 °C on a pinwheel rotator.

OD600 measurements and CFU/mL

Overnight cultures were back-diluted to an OD_{600} of 0.02 in 2 mL LB_{ac}MMB in a 10 mL glass culture tube and incubated at 37 °C on a pinwheel for three hours. OD_{600} was measured by 1:2 dilution in a 1 mL cuvette. Twenty microliters of culture at 0, 3, and 24 hours were serially diluted 10-fold (10^{-1}–10^{-8}) in a 96-well plate and spotted on room-temperature LB_{ac} plates. Spots were allowed to dry and plates were incubated at 37 °C overnight.

Microscopy slide preparation

Methods were adapted from UIUC MCB-101 Laboratory Exercises. Overnight cultures were back-diluted 1:100 in 2 mL LB_{ac}MMB in a 10 mL glass culture tube and incubated at 37 °C on a pinwheel for approximately four hours. OD_{600} was measured by 1:2 dilution. Strains were induced with 0.2% arabinose and 10 µM IPTG for at least 20 minutes at 37 °C. One microliter of each strain was transferred to a microscope slide and

spread with a 200 µL pipette tip. Spots were allowed to evaporate completely at room temperature, then heat-fixed by passage through a flame several times. Slides were sequentially stained with crystal violet (2 min), iodine (1 min), 98% ethyl alcohol (≤3 sec), and safranin (30 sec), with gentle water washes between steps. Slides were blot-dried with bibulous paper. Cells were visualized at 10× for focus, then at 100× oil immersion.

Motility assay

0.3% LB_{ac}MMB swimming agar was prepared in a prewarmed 125 mL glass bottle by combining: 22.5 mL of 0.8% LB top agar, 37.5 mL LB_{ac} media, 75 µL ampicillin (100 mg/mL), 1.9 µL chloramphenicol (100 mg/mL), 600 µL IPTG (100 mM), 600 µL arabinose (20%), 300 µL $MgCl_2$ (1 M), and 60 µL $MnCl_2$ (100 mM). Thirty milliliters of swimming agar were aliquoted into square petri dishes approximately one hour before the assay. Liquid cultures were back-diluted 1:100 in 2 mL LB_{ac}MMB, incubated at 37 °C for 2.5 hours, and induced with 20 µL each of 20% arabinose and 100 mM IPTG. Two microliters of each induced culture were inoculated into the center of each quadrant of the swimming agar plate. Plates were incubated at 37 °C for three hours, after which colony-forming spot assays were performed on LB_{ac}MMB agar containing inducers at the above concentrations and incubated overnight at 37 °C.

Biofilm assay

MG1655 strains expressing empty vectors, ecC+ecE, ecC alone, or ecE alone were subcultured in 300 µL of LB_{ac}MMB with appropriate inducers in 96-well plates supplemented with 0.1 µM or 0.1 mM $MnCl_2$. Plates were incubated ambiently at 37 °C in a plate reader for six hours with OD_{600} measurements every 10 minutes. Following growth, media and planktonic cells were removed by pipette and replaced with 300 µL crystal violet for 5 minutes at room temperature. Crystal violet was discarded and replaced with 300 µL of precipitating solution (30% methanol, 10% glacial acetic acid) and gently swirled. Endpoint OD_{550} was quantified in the plate reader. Data were normalized by subtracting the mean OD_{550} of cell-free media controls from each experimental condition. Glass culture tubes were also assessed for biofilm formation under static and shaking conditions at 37 °C across both manganese concentrations.

Three technical replicates of three independent biological replicates were performed for each condition.

3.6 Figures

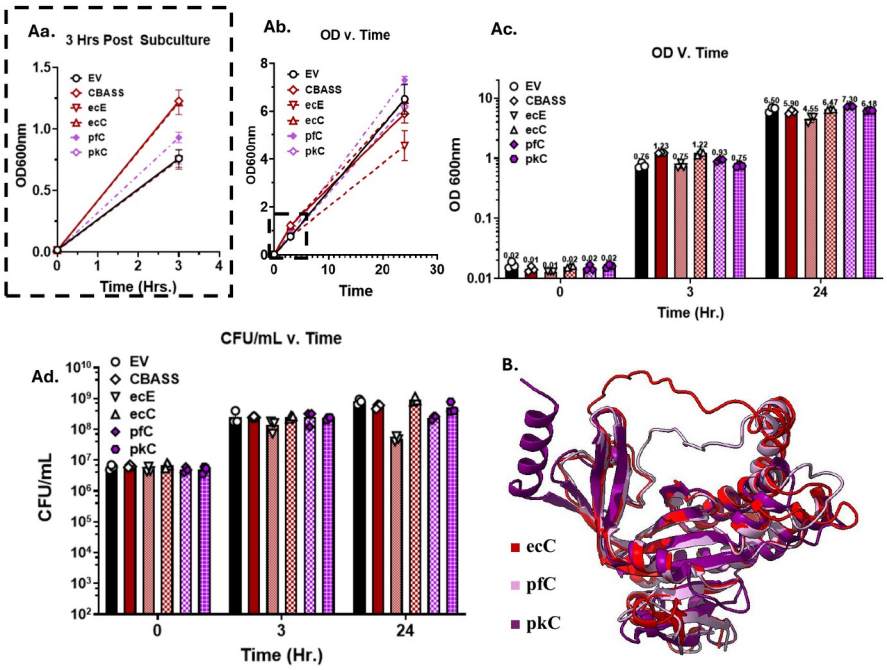


Figure 3.1. Antiphage guanylyl cyclases elevate the optical density of *Escherichia coli* without expanding viable cell populations. (**Aa**) Optical density at 600 nm (OD_{600}) of *E. coli* MG1655 ectopically expressing the complete antiphage operon (ecC+ecE), the pore-forming effector alone (ecE), the *E. coli* cyclase alone (ecC), the *Pseudomonas fluorescens* cyclase (pfC), or the *Pseudomonas koreensis* cyclase (pkC) over the first 3 h post-subculture. (**Ab**) Extended growth profile of the same cultures across 24 h. (**Ac**) Mean OD_{600} values at 0, 3, and 24 h. (**Ad**) Colony-forming units per milliliter (CFU/mL) for the same cultures at 0, 3, and 24 h, demonstrating that the OD_{600} elevation does not arise from increased cell division. (**B**) Superimposed AlphaFold 3-predicted monomer structures of ecC (red), pfC (lavender), and pkC (purple), illustrating overall structural homology among all three cyclases despite distinct primary sequences and cognate effectors. Data are representative of three independent biological replicates.

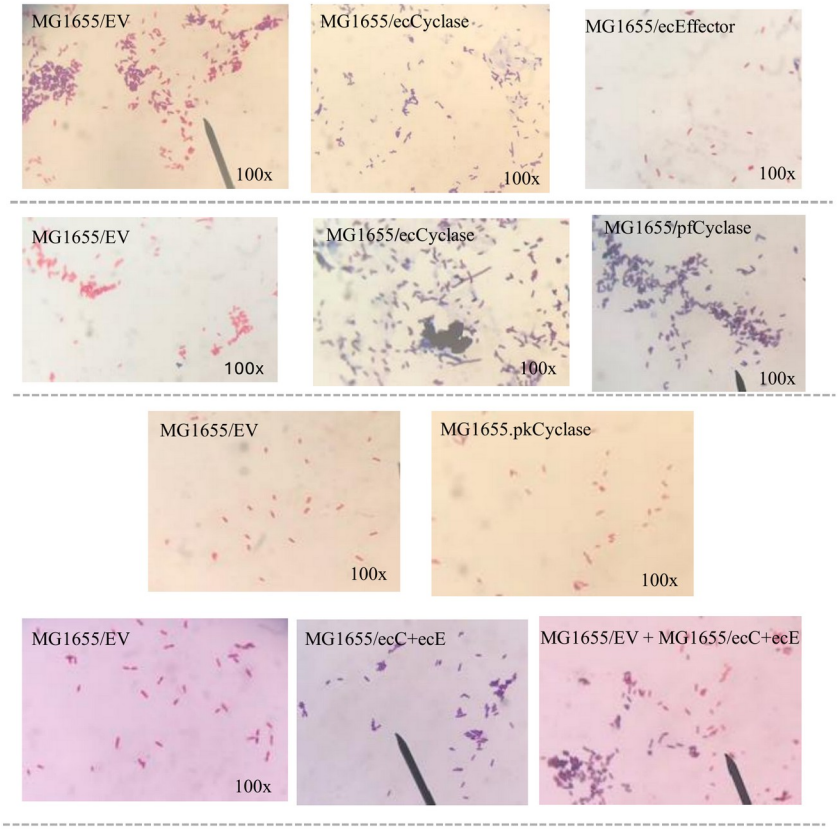


Figure 3.2. Ectopic guanylyl cyclase expression drives a gram-positive-like cell wall phenotype in *Escherichia coli*. Strains were revived from −80 °C glycerol stocks, grown for 6 h at 37 °C, induced for at least 20 min with 0.2% arabinose and 10 µM IPTG, and gram-stained on heat-fixed slides. Cells were imaged by compound light microscopy at 100× oil immersion. EV: empty vectors; ec: *E. coli*; pf: *Pseudomonas fluorescens*; pk: *Pseudomonas koreensis*. The bottom row depicts a co-stained slide containing both EV and ecC+ecE-expressing cells, demonstrating that the gram-positive-like staining is specific to cyclase-expressing cells and is not a decolorization artifact. Data are representative of three independent biological replicates.

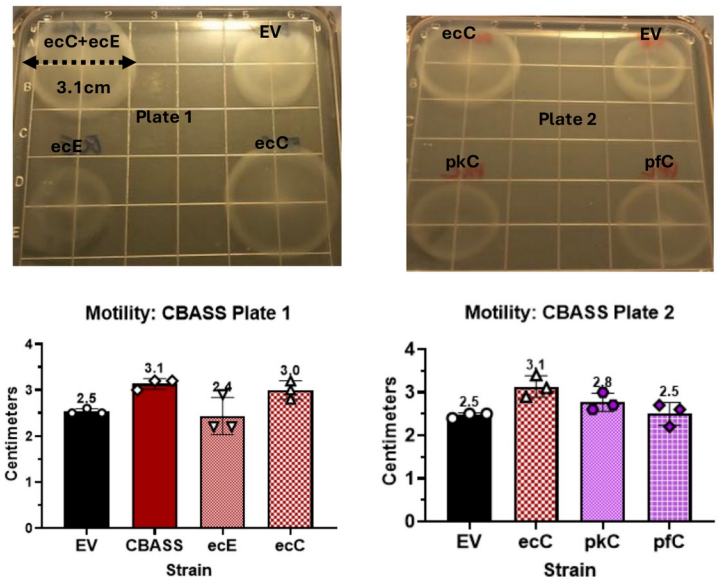


Figure 3.3. The *Escherichia coli* antiphage guanylyl cyclase enhances swimming motility. Strains were grown for 6 h at 37 °C from −80 °C glycerol stocks and induced for at least 20 min before inoculation onto 0.3% LB_{ac}MMB swimming agar containing inducers. Plates were incubated at 37 °C for 3 h, after which the diameter of swimming halos was quantified. EV: empty vectors; CBASS: complete ecC+ecE operon; ecE: *E. coli* effector alone; ecC: *E. coli* cyclase alone; pfC: *Pseudomonas fluorescens* cyclase; pkC: *Pseudomonas koreensis* cyclase. Data are representative of three independent biological replicates.

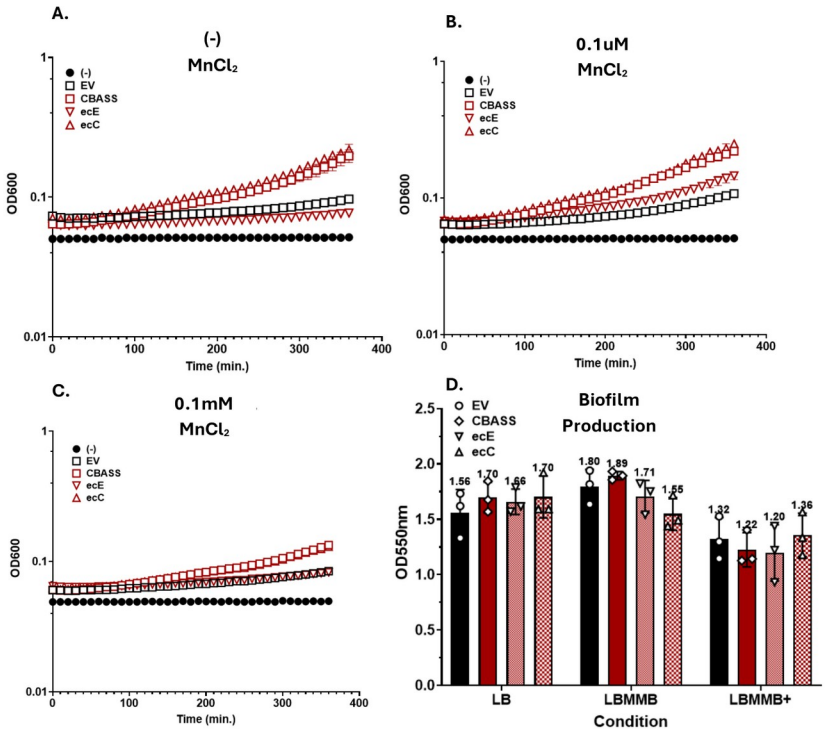


Figure 3.4. The *Escherichia coli* guanylyl cyclase suppresses biofilm formation in a manganese-dependent manner. Optical density of *E. coli* ectopically expressing empty vectors (□), the complete ecC+ecE operon (□), the effector alone (∇), the cyclase alone (Δ), or media-only controls (●), measured every 10 min over 6 h: (**A**) without supplemental $MnCl_2$, (**B**) with 0.1 µM $MnCl_2$, and (**C**) with 0.1 mM $MnCl_2$. (**D**) Endpoint biofilm biomass quantified at OD_{550} after crystal violet staining and solubilization in 30% methanol/10% glacial acetic acid. Data are representative of three independent biological replicates.

3.7 Abbreviations

CBASS	Cyclic Oligonucleotide-Based Antiphage Signaling System
CFU	Colony-Forming Units
cGAMP	cyclic Guanosine Monophosphate–Adenosine Monophosphate
cGAS	cyclic GMP-AMP Synthase
cGMP	cyclic Guanosine Monophosphate
EV	Empty Vector
GTP	Guanosine Triphosphate
IPTG	Isopropyl β-D-1-thiogalactopyranoside
KTE100	*Escherichia coli* strain KTE100
LB	Lysogeny Broth
MOI	Multiplicity Of Infection
OD600	Optical Density at 600 nm
PFT	Pore-Forming Toxin
PYCSAR	Pyrimidine Cyclase System for Antiphage Resistance
α-PFT	α-helical Pore-Forming Toxin

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CHAPTER 4

THE MOLECULAR TRIPWIRE — A PUTATIVE AUTOINHIBITORY DOMAIN OF A GUANYLYL CYCLASE IS LIKELY RELIEVED DURING BACTERIOPHAGE INFECTION TO INDUCE ANTIPHAGE ACTIVITY

By: Jeffrey Wyatt-Wolfgang Marciniak-Haynosch

4.1 Abstract

Antiphage defenses must reconcile two contradictory demands: instantaneous activation upon viral invasion and absolute quiescence under homeostatic conditions. Many such systems impose host cell death prior to phage-mediated lysis, rendering inadvertent activation an existential liability. CBASS systems address this constraint by transducing detection of phage components into the production of cyclic nucleotide second messengers that trigger downstream effectors only above a defined activation threshold. Phage proteases have been implicated in Type-I CBASS regulation, but how these proteases physically interact with the CBASS apparatus to drive activation remains incompletely resolved. Here I leveraged a combination of bioinformatics, classical phage characterization assays, escaper-phage selection, and chimeric cyclase fusion proteins to dissect how an antiphage defense composed of a guanylyl cyclase and a ClyA-like α-pore-forming toxin coordinates protection against phages P1, lambda, and T6. Through structural modeling, I identify a putative regulatory domain located within a region of low conservation on the cyclase amino terminus that sensitizes bacteria to phage infection. I propose two non-mutually-exclusive models for cyclase activation by phage proteases: (i) capsid-localized cleavage of an autoinhibitory N-terminal spine, and (ii) phosphorylation-mediated recruitment of phage proteases to the same spine. The work presented here was conducted independently and prior to the publication of Hobbs and Kranzusch (2026); the convergence of independent findings strengthens the central conclusion that proteolytic activation of CBASS cyclases is a conserved organizational principle of bacterial antiviral immunity.

4.2 Introduction

The capacity of bacteria to perceive and respond to viral threat is a non-negotiable prerequisite for survival in the phage-saturated environment of the prokaryotic biosphere. As a strategy to forestall population-level annihilation, bacteria self-destruct upon bacteriophage (phage) infection prior to the release of viral progeny — an altruistic logic that sacrifices the infected cell to protect the surrounding clonal population.[7,13,29,58] This response is engaged only at the terminal stages of the phage lifecycle, after all upstream defenses have been bypassed; it is therefore rapid but never indiscriminate. Multiple molecular safeguards prevent inadvertent self-destruction in the absence of authentic viral infection.

These safeguards include the bifurcation of sensor and effector modules into structurally and biochemically distinct proteins,[10,39,57] the use of unique transmission signals encoded by the cyclic nucleotide second messengers themselves,[15,30,57,66] regulation of sensor activity by host metabolite status,[53] and direct sensor recognition of phage-derived structural or proteolytic components.[4,27,49,68] Two intracellular cyclic nucleotide signaling systems exemplify this architecture: the cyclic-oligonucleotide-based antiphage signaling systems (CBASS)[10,39] and the pyrimidine cyclase system for antiphage resistance (PYCSAR)[21,57], identified across approximately 5,675 and 937 bacterial genomes, respectively.[39,57] Both systems comprise a minimal sensor cyclase that synthesizes a unique second messenger to activate a downstream self-destructive effector, with accessory genes within the operon further fine-tuning the infection response.[24,67]

The molecular triggers of CBASS activation are diverse and continue to be defined. Banh and colleagues established that the *Staphylococcus schleiferi* 2142-05 Type-I CBASS sensor (CdnE12) is licensed by direct binding of a discrete ∼400-nucleotide non-coding viral RNA — termed cGAS-activating bacteriophage RNA (cabRNA) — transcribed across the small (TerS) and large (TerL) terminase genes of the infecting phage; the cabRNA is necessary and sufficient for cyclase activation, expanding the repertoire of CBASS triggers to include nucleic-acid ligands.[4] Stokar-Avihail and colleagues identified phage structural components, including major capsid and

scaffolding proteins, as direct CBASS triggers in a large-scale genetic screen,[56] while Richmond-Buccola and colleagues demonstrated that phage prohead proteases are the primary determinants of Type-I CBASS activation and evasion across diverse phage–host pairings.[49] Lau and colleagues independently established that DNA damage activates CBASS signaling through a conserved pathway coupled to the SOS response.[27]

Type-I CBASS systems constitute approximately 42% of all identified CBASS systems.[39] Phage proteases and structural genes have been implicated as instigators of Type-I CBASS stimulation,[49,56] yet the mechanism by which the head-processing protease physically engages the cyclase to license antiphage activity remains incomplete. Here I identify a previously uncharacterized Type-I CBASS-like antiphage operon encoded by Uropathogenic *E. coli* str. KTE100 (ecC+ecE), composed of a guanylyl cyclase paired with a ClyA-like, three-transmembrane α-pore-forming toxin (α-PFT) as its destruction module.[6,50]

In this chapter, I identify two distinct defensive phenotypes — phage progeny abolition and progeny attenuation — conferred by ecC+ecE against phages P1, lambda, and T6. I demonstrate that escape from defense is achieved by phages through point mutations in protease and structural genes. Through chimeric cyclases that exchange the amino terminus between *E. coli* ecC and *P. fluorescens* pfC, I localize a putative regulatory module to a region of low conservation on the ecC N-terminus. Finally, I propose two non-mutually-exclusive models for cyclase activation by phage infection. The work in this chapter was conducted independently and completed prior to the publication of Hobbs and Kranzusch (2026), whose biochemical reconstitution of phage T4 prohead protease–mediated activation of the *E. coli* CD-NTase EcCdnG arrives at a mechanistically convergent conclusion through complementary methods. The independent convergence of these findings strengthens, rather than undermines, the central thesis that protease-dependent cleavage of a cyclase activation loop is a conserved organizing principle of cyclic nucleotide–based bacterial innate immunity.

4.3 Results

4.3.1 The cyclase and effector are jointly required to defend against phages P1, lambda, and T6

To determine whether the *E. coli* KTE100 guanylyl cyclase and ClyA-like effector (ecC+ecE) confer antiphage activity, the cyclase and effector genes were cloned onto two separate plasmids (pQE-60 and pBAD33, respectively) under independently inducible promoters (IPTG and arabinose). Two-plasmid combinations carrying either both antiphage genes or one gene paired with the matched empty vector were transformed into *E. coli* MG1655. Strains were grown to late-log phase, induced for at least 20 minutes, and embedded as lawns (final OD ∼0.2) in 0.8% LB top agar supplemented with 0.2% arabinose and 100 µM IPTG. High-titer phage lysates (HTL) were diluted 1:10 in LB and spotted onto each lawn (Figure 4.1A).

Phages were classified as ecC+ecE-sensitive when ≤20% of the phage population survived on ecC+ecE-expressing strains relative to empty-vector titers (Figure 4.1B). Phages P1, lambda, and T6 each exhibited reduced titers on ecC+ecE lawns (Figure 4.1A). Phage P1 produced a slight burn-through at the 10^{-1} dilution; lambda yielded turbid, difficult-to-count plaques on ecC+ecE lawns relative to defenseless strains; and T6 displayed an approximately 10-fold titer reduction with visible but smaller-diameter plaques (Figure 4.1A). While P1, lambda, and T6 infections were each successfully aborted, the magnitude of defense varied substantially between phages — a heterogeneity consistent with prior observations of CBASS systems against diverse phage challenges.[49,56]

To probe the breadth of ecC+ecE activity, a collection of ∼60 phages from the BASEL collection[32] was screened against putative guanylyl cyclase–like systems from *Pseudomonas fluorescens* (UniProt: A0A0W0H6C2) and a putative PYCSAR defense from *Pseudomonas koreensis* (UniProt: A0A341F9Z4). None of the ∼40 classic or BASEL phages tested[32] were sensitive to either system under agar-plating conditions (Figure S4.1).

4.3.2 Complete defense ablates or attenuates P1, lambda, and T6 progeny release

To determine whether ecC+ecE functions as a *bona fide* abortive infection system,[29] classical one-step growth burst-size assays were performed to quantify phage output.[12,14,22] If ecC+ecE acts as an abortive system, phage progeny release should be abolished. Phages P1 and lambda failed to release more than five mature viral progeny in *E. coli* ectopically expressing ecC+ecE (Figure 4.1C). The average P1 burst size was ∼326 PFU without and ∼2 PFU with ecC+ecE expression, a 163-fold reduction (Figure 4.1C). For lambda, the average burst size was ∼110 PFU without and ∼2 PFU with ecC+ecE expression, a ∼55-fold reduction (Figure 4.1C). For T6, burst size was ∼480 PFU without and ∼257 PFU with ecC+ecE expression, a ∼1.88-fold change (Figure 4.1C).

Burst size was calculated by averaging PFU across three time points after the first burst (∼45 minutes for P1 and lambda; ∼20 minutes for T6) divided by the average PFU at three time points before the first burst. Percent phage survival was determined by dividing the titer on the ecC+ecE-expressing strain by that on the empty-vector control and multiplying by 100. The data argues that phages with longer latent periods, such as P1 and lambda, are substantially more susceptible to ecC+ecE-mediated defense than T6, whose shorter latent period likely permits progeny release before abortive infection is fully engaged. Importantly, expression of ecC+ecE in the absence of phage infection did not stimulate inadvertent antiphage activity, indicating that the translated cyclase and effector are post-translationally restrained under homeostatic conditions — a phenotype consistent with the regulatory architecture proposed in Chapter 2.

4.3.3 KTE100 Type-I CBASS-like defense protects from P1, lambda, and T6 in liquid culture at low MOI

To corroborate the abortive infection phenotype in a more sensitive assay,[43] cultures were challenged with phages P1, lambda, or T6 at low or high multiplicities of infection (MOI) for four to twelve hours, depending on the phage latent period (Figure 4.1D). At low MOI, OD of cultures expressing a *bona fide* CBASS defense should recover from phage challenge.[10,40,43,57] Phage clearance was confirmed by quantification of phage titer at the conclusion of each time course (Figure 4.1E). In the absence of phage, *E. coli* grew normally under ectopic ecC+ecE expression, confirming that the defense module is not lethal to the host (Figure 4.1D). At low MOI (0.02), ecC+ecE-expressing cultures

eliminated >99% of P1 and lambda from the system (Figure 4.1D–E). Against T6, ecC+ecE extended culture survival at low MOI (Figure 4.1D) and removed >10% of the phage population (Figure 4.1E), consistent with the attenuated burst-size reduction observed in Section 4.3.2.

4.3.4 Bacteriophages mutate structural and protease genes to evade antiphage defense

Because ectopic expression of ecC+ecE does not stimulate antiphage activity in the absence of phage, I hypothesized that specific phage components physically engage ecC+ecE during infection. To identify these phage genes, P1 and lambda escaper phages were generated by overnight incubation of sensitive parent phages with *E. coli* MG1655 expressing ecC+ecE at high MOI in top agar enriched with divalent cations (5 mM MgCl + 0.1 mM MnCl; Figure 4.2).[49,56] Plaques were isolated, purified through three rounds of plaque-to-plaque passaging, confirmed for resistance, and submitted for sequencing.

P1 escapers from three independent biological replicates each carried single-nucleotide substitutions in the head-processing gene Pro: two replicates (P1E1A, P1E2) acquired N24S, and one replicate (P1E3) acquired R135C (Figure 4.2A; Figure S4.2a, S4.2b). The convergent targeting of Pro across all three replicates — through two chemically distinct substitutions — identifies this gene as the principal determinant of ecC+ecE-mediated immunity against P1.

Lambda escapers carried mutations distributed across multiple structural and regulatory genes: gpC/gpNu3 ($\Delta^{337/98}$GENASAAQPD$^{346/107}$), cII (N80K), gpE (P180S or V273G), gpV (D201E), gpZ (V164G), and the promoter region of gpE (E; Figure 4.2A; Figure S4.2a). The gpC/gpNu3 deletion was observed in only one technical replicate of one biological replicate; full-length gpNu3 was retained in the other two biological replicates (Figure S4.2a). Across all three biological replicates, however, mutations in cII (N80K) and gpV (D201E) were consistently identified (Figure S4.2a). Notably, the gpV tail-tube protein shares an evolutionary ancestor with phage protease genes,[19] suggesting that a single protease mutation — or multiple mutations distributed across protease-fold genes — can each confer resistance to ecC+ecE-mediated defense.

To phenotypically characterize escaper phages, infection dynamics were measured at low MOI. Canonical abortive infection systems protect cultures at low MOI, and parent

P1 and lambda confirmed this expectation (Figure 4.2B–C). Escaper phages collapsed cultures at low MOI, as expected for defense-resistant variants. The lambdaE4B escaper carried an adenine-to-thymine mutation in the promoter region of gpE — the essential major capsid protein.[19,20] Reduced gpE expression — dampened but not ablated — in conjunction with additional mutations in cII or gpV, likely accounts for residual ecC+ecE activation and the partial defense observed against this variant.[4,27]

4.3.5 Expression of full-length lambda GpC does not activate abortive infection

I hypothesized that higher-order capsid structures — not free monomeric capsid components — are required to upregulate antiphage activity.[10,47,56,57] To test this, I leveraged the lambda capsid self-assembly strategy[36,64] to ectopically express phage capsid genes in the cytosol with the expectation that resulting viral "ghost capsids" would, in conjunction with ecC+ecE, induce abortive infection.

Although portal GpB was likely not expressed efficiently from this construct, native-promoter expression of GpC through GpFii did not induce abortive infection (Figure 4.2D). This finding is consistent with the hypothesis that protease activation is contingent on higher-order capsid assembly — specifically GpB polymerization into the portal ring — rather than on the presence of free, monomeric GpC or GpNu3. These data further support the interpretation that the protease domain itself, and not accessory sequences within the capsid operon, is the functional target required for ecC+ecE activation, and that procapsid assembly is a strict prerequisite for that activation.

4.3.6 Mutated residues likely modulate protease scaffold autocleavage

Because ectopic expression of phage capsid genes did not recapitulate antiphage activity, and because the mechanistic understanding of P1 Pro protease activity — including its catalytic-site residues — remains limited,[20,37,65] I leveraged structural and sequence alignments of P1 Pro to determine how the N24S and R135C substitutions might affect Pro architecture (Figure 4.3).[16,25,28,44]

HHPRED analysis revealed that the protease domain of Pro (residues 19–164) shares 99% structural identity with T4 Gp21 (PDB: 5JBL_D).[19] Structural alignment of Gp21 with the predicted Pro parent monomer revealed that Pro residues His57, Ser129, and

Asp150 occupy a near-identical orientation to the T4 Gp21 active-site residues His85, Ser140, and Asp168, recapitulating the canonical serine-protease catalytic triad (Figure 4.3C).[28,35,48,61] Conservation of this triad was confirmed by UniProtKB BlastP (default parameters; E-value threshold: 1) followed by sequence alignment with five iterations across the top 15 identified homologs: residues Ser129, His57, and Asp150 are conserved across all identified homologs, denoted by three solid black dots in the alignment (Figure 4.3D). The Pro query (UniProt: Q71TF6) is denoted by an asterisk; mutations are denoted by a circled cross (Figure 4.3D). Critically, the active-site residues of the N24S and R135C mutants overlap with those of wildtype Pro, indicating that neither substitution disrupts the catalytic triad itself (Figure 4.3C).

The N24S substitution likely modulates the α-helix that occludes the active site, dampening but not ablating proteolytic function — an interpretation consistent with the essentiality of Pro for P1 progeny production.[20,37] The R135C substitution falls within a ϕKZ Gp175 (a distant homolog of T4 Gp21) cleavage site, suggesting that this mutation modulates autocatalytic activation or substrate recognition.[59,60] Mapping the active site (cyan spheres) and the mutations (yellow spheres) onto the parent pentameric structure revealed that the mutations are peripheral to the predicted active site (Figure 4.3A–B). AlphaFold3[1] was employed to predict parent and mutant pentamers (parent metrics: pTM = 0.34, ipTM = 0.29, overall score = 0.37; Figure 4.3E). The wildtype Pro pentamer adopts the canonical serine-protease architecture.[19] The N24S mutant pentamer reveals a shallower, concave structure with a modified lid, while the R135C mutant adopts a jellyfish-like topology with an elongated scaffold (Figure 4.3E), supporting the interpretation that both substitutions perturb pentameric conformation without eliminating catalytic activity.

4.3.7 The E. coli cyclase amino-terminus is a phage-sensing module (preliminary, n = 1)

The *P. fluorescens* antiphage defense (pfC+pfE) failed to protect from phages on agar lawns, in contrast to ecC+ecE (Figure 4.1A; Figure S4.1). I therefore hypothesized that the ecC N-terminus encodes a more sensitive phage-sensing module than that of pfC. I reasoned that if the N-terminal sequences of ecC (yielding $ec^{N}pf^{C}$) and pfC (yielding $pf^{N}ec^{C}$) were exchanged and the chimeras expressed with their cognate effectors, antiphage activity should be modulated during P1 and lambda infection (Figure 4.4; Figure S4.3). The exchange boundary was selected based on sequence and structural alignment to maximize divergence between the swapped termini (Figure S4.3).

An important caveat: the chimeric cyclase experiments described here represent a **single biological replicate**; error bars are therefore absent from Figure 4.4C–D, and quantitative comparisons should be interpreted as preliminary. The data presented establish proof-of-concept for N-terminal modularity; firm quantitative conclusions require independent replication. Defense potency was measured in liquid culture over six hours and corroborated by percent (%) phage survival via spot-test analysis. After six hours, one technical replicate per condition was pelleted, the supernatant diluted 1:10, and spotted on defenseless *E. coli* MG1655 lawns. MOI was calculated from working stock titers (PFU/mL) as previously described.

Cognate antiphage defense robustly protected hosts over the six-hour time course while reducing phage titer in the culture, confirming that liquid culture is a more sensitive measure of antiphage activity than the agar-based spot-test (Figure 4.4B–C, 4.4C3). The potency of pfC+pfE-mediated protection in liquid media was initially surprising given the absence of a defensive phenotype on agar (Figure S4.1). Notably, pfC+pfE are expressed from two separate plasmids under independent inducible promoters, whereas ecC+ecE and the chimeric cyclase constructs are expressed from a single plasmid under one promoter. Independent promoter control of pfC and pfE likely permits higher combinatorial expression and a more rapid induction kinetics, contributing to the robust antiphage activity observed with pfC+pfE in liquid culture.

Cultures expressing $ec^{N}pf^{C}$+ecE survived P1 and lambda infection, with only 0.66% and 13.0% phage survival, respectively (Figure 4.4A1–A2, 4.4C1). Cultures expressing $ec^{N}pf^{C}$+pfE collapsed against both P1 and lambda, but residual antiphage activity was apparent: only 1.77% and 46.39% of P1 and lambda phages survived, respectively

(Figure 4.4A1–A2, 4.4C1). Cultures expressing $pf^{N}ec^{C}$+pfE or $pf^{N}ec^{C}$+ecE survived P1 infection but collapsed against lambda (Figure 4.4B1–B2); phage populations were reduced to 4.58–19.52% across these conditions (Figure 4.4C2). P1 was consistently more sensitive than lambda (4.58–5.73% vs. 17.68–19.52% survival; Figure 4.4C2). Even when fused cyclases were paired with an empty vector, antiphage activity was retained against P1 (1.15–11.50% phage survival; Figure 4.4A1–B1, 4.4C1–C2), while lambda was substantially more resistant under the same conditions (56.72–81.33% survival; Figure 4.4A2–B2, 4.4C1–C2).

Taken together — and with the explicit caveat that these data derive from a single biological replicate — the chimeric cyclase data are consistent with the ecC N-terminus encoding a phage-sensing module whose transfer to pfC is sufficient to confer phage sensitivity, and whose removal from ecC attenuates antiphage potency. These data nominate the cyclase N-terminus as the primary determinant of differential phage detection between the two cyclase systems and motivate the structural analyses described in Section 4.3.8. Independent replication is required before firm quantitative conclusions are drawn.

4.3.8 The cyclase N-terminus harbors localized redundant cleavage sites and a putative TBK1-like phosphorylation loop

To probe the structural basis for differential defense potency between ecC and pfC, an *in silico* approach was undertaken that integrated structural alignment, sequence alignment, and remote homology detection to characterize cyclase N-terminal regulatory architecture (Figure 4.5). Given 43% sequence identity between ecC and pfC (Figure S4.3C), I reasoned that primary-sequence differences propagate to distinct secondary and tertiary structures that underlie the divergence in phage sensitivity. Sequence motifs in the ecC N-terminus were identified by literature-guided manual annotation against previously characterized cleavage and regulatory motifs (Figure 4.5A).[3,18,33,41,42,45]

The ecC N-terminus contains multiple localized sequence motifs, several of which converge in a single region of low conservation (Figure 4.5A): an SppA signal-peptide motif[45] A-X-A (^{38}AKA^{40}; sky blue); a P21 cleavage site[41,42] [L/I]-X-E (^{70}ISE^{72}; royal blue); an inverted P21 cleavage site[41,42] (^{37}LSE^{35}; blue); a ϕKZ Gp175 cleavage site[60] [SAG]-X-E

([43]GME[45]; cyan); a ClpX-recognition signal[18] polar-[T/ϕ]-ϕ-basic-ϕ ([36]SLAKA[40]; sky blue); and two secondary HEPN motifs[3] of the form Exxx[R/K] ([25]ENGGK[29] and [35]ESLAK[39]; gold; Figure 4.5A, 4.5C).

Sequence alignment of ecC and pfC revealed nine residues that are absent from pfC between consensus positions 30 and 50 (Figure 4.5B). This nine-residue gap bisects the distal N-terminal CARF-like motif of pfC (consensus: [DN]-X-[ST]-X-X-X-[RK];[33] [35]NASMESR[41]; maroon), which itself harbors a ϕKZ Gp175 cleavage site ([37]SME[39]) followed by a second ([43]GNE[45]; Figure 4.5B). Predicted structures of the ecC (dark grey) and pfC (light grey) cyclase monomers were generated with AlphaFold3[1] and aligned in ChimeraX (alignment score: 670.3 across 171 pruned atom pairs at 0.660 Å RMSD across all 252 pairs; Figure 4.5C).

The nine-residue gap in pfC displaces an α-helix to the opposite face of the monomer, away from the α-helix triplet observed in ecC (Figure 4.5C). The additional residues in ecC permit the N-terminus to traverse the dimerization interface, forming a short α-helix on an exposed structural "spine" (sky blue helix; Figure 4.5C, S4.3). The pfC monomer, lacking these residues, adopts a "bunny-ears" tertiary configuration that does not traverse the dimerization interface (Figure 4.5C, S4.3).

To probe the functional identity of this structural spine, the ecC sequence was analyzed using HHPRED via the MPI Bioinformatics Toolkit.[70] HHPRED identified the ecC N-terminus as structurally homologous to the serine–threonine kinase TBK1 phosphorylation site within a transferase–inhibitor complex (PDB: 4EUT_B; 94.83% probability, E-value 0.027; Figure 4.5D).[31] The structural correspondence localizes to an intrinsically disordered region of the ecC N-terminus anchored by Ser36 within the inverted P21 cleavage site ([37]LSE[35]) or by Lys39 within the SppA cleavage site ([38]AKA[40]), which forms a short α-helix.[31] This interpretation is consistent with the deep evolutionary kinship between bacterial CBASS systems and the eukaryotic cGAS–STING innate immune pathway,[10] in which intrinsically disordered regions and lysine residues commonly serve as sites of regulatory modification. As demonstrated by Tsugita and colleagues, defined secondary structure at cleavage-site junctions is critical for proteolytic activity;[63] I therefore propose that the ecC N-terminal spine functions as an autoinhibitory element that limits basal cGMP production.[38] Relief of this autoinhibition

— either by protease cleavage of the spine or by kinase-directed phosphorylation at Lys39 and neighboring residues — would license cGMP production and downstream effector activation.

4.4 Discussion

This study sought to identify how bacteriophages induce abortive infection through cooperation between a cGMP-producing guanylyl cyclase and a ClyA-like α-pore-forming effector encoded by Uropathogenic *E. coli* str. KTE100. At least three phages — P1, lambda, and T6 — were identified as sensitive by spot-analysis, and their defensive potency was quantified by classical phage characterization assays. I demonstrated that ectopic expression of phage capsid genes in the absence of higher-order assembly does not induce inadvertent abortive infection — a phenotype consistent with prior observations from other Type-I CBASS systems.[49] Because ectopic phage gene expression failed to recapitulate antiphage activity, I hypothesized that a post-translational event during productive infection sensitizes ecC to phage challenge more acutely than does pfC. Chimeric cyclases between pfC and ecC were therefore constructed to identify the structural determinants of phage sensitivity. Several sequence motifs converge within the same region of low conservation on the ecC N-terminus. When ecC and pfC sequences were aligned, a nine-residue gap in pfC bisected the distal N-terminal CARF-like motif of pfC and the ecC secondary HEPN domain, generating a predicted structural "spine" at the dimerization interface of ecC that is absent in pfC.

Classical phage assays demonstrated that P1 and lambda are strongly sensitive to ecC+ecE-mediated defense, while T6 produced mature progeny at attenuated but detectable levels. P1 and lambda share several biologically relevant properties:[19,37,46,65] both are temperate phages,[8,19,51,65] both encode phosphatases,[20,65] both have latent periods of ∼45 minutes,[19,65] and both have burst sizes of 100–200 PFU.[19,65] In contrast, T6 displayed only a ∼1.88-fold reduction in burst size, consistent with its shorter latent period enabling progeny release before abortive infection is fully engaged. Notably, the laboratory stock originally labeled "T4" was determined to be *E. coli* φ80: short-read sequencing revealed ∼2.4 million reads mapping to the φ80 genome (NC_021190; data not shown), supported by plaque morphology and downstream phage-characterization

assays (denoted *; Figure 4.1). The attenuated T6 defense phenotype is further explained by the presence of an anti-CBASS protein within the T6 genome: a protein with 100% sequence identity to Acb1 was identified by UniProtKB BlastP, while Acb2 was absent.[21] Acb1 hydrolyzes 3′,3′-cGAMP, 3′,3′-cUA, and 3′,3′,3′-cAAA, but not cGMP;[21] T6 therefore likely encodes additional, presently uncharacterized counter-defense mechanisms that suppress cGMP-dependent antiphage activation — consistent with the broad anti-defense repertoire of T-even-family phages.

Both P1 and lambda mutated essential protease genes to evade ecC+ecE defense. P1 escapers exclusively mutated Pro (N24S or R135C), while lambda escapers mutated GpC/GpNu3, with additional mutations distributed across cII, gpV, and the gpE promoter (Figure 4.2A; Figure S4.2a, S4.2b). Lambda GpC belongs to the S49 superfamily of serine/lysine-dyad proteases,[9,16,26,64] while Pro is a U9- or U35-like aspartate/histidine/serine catalytic triad enzyme homologous to T4 Gp21.[41,42,48,55] Under endogenous lambda promoters, GpC alone failed to induce abortive infection (Figure 4.2D), supporting the model that procapsid assembly and GpB portal polymerization are prerequisites for protease activation. The natural translated form of full-length serine proteases is typically inactive prior to N-terminal autocleavage,[45] explaining why expression of the full-length protease gene in isolation does not trigger the defense.

Convergent targeting of capsid maturation proteases by both P1 and lambda — through chemically distinct active-site chemistries — places head-processing proteases at the center of the phage–ecC+ecE interaction. The distinct Pro-triad (U9/U35) versus GpC-dyad (S49) chemistry may explain why P1 elicits stronger immunity than lambda: the triad-mediated catalytic mechanism may be more efficient at cleaving the ecC N-terminal redundant cut-site junction. Conversely, the partial T6 resistance phenotype — despite T6 encoding a Gp21-like U9 triad protease — suggests either functional divergence in T6 Gp21 or active suppression by Acb1 and additional T6 counter-defenses.

Because *P. fluorescens* pfC+pfE did not defend from P1 or lambda on agar (Figure S4.1), I hypothesized that the ecC N-terminus encodes a phage-sensing module absent from pfC.[49,53,66] Chimeric cyclases with swapped N-termini were therefore expressed with cognate effectors to test this prediction. Both cognate and chimeric cyclase combinations

robustly reduced P1 from culture (Figure 4.4), reaffirming that liquid-culture assays are more sensitive than spot-test analysis. Replacing the ecC N-terminus with the pfC N-terminus produced substantially more lambda than P1 progeny phage, indicating a loss of antiphage activity against lambda specifically (Figure 4.4D). The $pf^{N}ec^{C}$ chimera collapsed cultures earlier than negative controls, suggesting that the pfC N-terminus may contain structural elements that constitutively destabilize effector interaction or promote premature cGMP production. I reiterate, however, that these chimeric cyclase observations derive from a **single biological replicate** and require independent replication before firm conclusions are drawn.

Together, these data argue that antiphage activity is engaged late in the phage replication cycle, that phages alter their proteases through point mutations or clustered deletions to evade defense, and that the cyclase N-terminus is the primary sensor element directing phage-dependent activation. The proposed mechanism involves phage head-processing proteases, translated in the host cytosol from the infecting phage genome, that undergo autocleavage during procapsid assembly and are subsequently released to cleave ecC at the N-terminal redundant cut-site junction. This interpretation is well-supported for T4 Gp21, to which P1 Pro shares 99% structural identity by HHPRED,[19,63] and it reconciles the observation that active procapsid assembly — not free protease expression — is required for ecC+ecE activation.

On the basis of these findings, I propose two non-mutually-exclusive models for phage-mediated cyclase activation (Figure 4.6). In Model 1 (Figure 4.6B), the redundant cut-site junction of the ecC N-terminus localizes the cyclase to the interior capsid space via the P21 Capsid Targeting Sequence.[41,42] The phage protease autocleaves from the scaffold, re-enters the cytosol, and cleaves the cyclase at the exposed spine (Figure 4.6B2). Removal of the autoinhibitory N-terminal spine disinhibits the cyclase, increasing cGMP production above the threshold required to oligomerize the ClyA-like effector and kill the host before viral replication is complete (Figure 4.6B3). In Model 2 (Figure 4.6C), phage-encoded kinases such as the P1 Doc toxin[11,17] phosphorylate ecC N-terminal residues during infection. The intrinsically disordered region bearing the TBK1-like phosphorylation site[31] is a plausible substrate; phosphorylated residues (likely Thr14,[11,17] Thr30,[11,17] or Arg19[5,62]) recruit the protease to the redundant cut-site junction

following its release from the procapsid interior. Cleavage disinhibits the cyclase and drives cGMP production above the threshold required for effector oligomerization (Figure 4.6C2–C3). In both models, basal cGMP production by ecC under ectopic expression in the absence of phage infection does not reach the effector oligomerization threshold, consistent with the absence of inadvertent host killing under homeostatic expression conditions (Figure 4.1D). Protease-mediated N-terminal cleavage is therefore the proposed switch that drives cGMP above this threshold and commits the host to abortive infection.

Independent and prior completion. The work in this chapter was conceived, executed, and interpreted independently and was completed prior to the bioRxiv preprint of Hobbs and Kranzusch (2026), which arrived at a mechanistically convergent conclusion through orthogonal *in vitro* biochemical reconstitution.[71] Hobbs and Kranzusch demonstrated that the phage T4 prohead protease gp21 is necessary and sufficient to activate the *E. coli* CD-NTase EcCdnG by cleaving a surface-exposed disordered loop, and that approximately 24% of CBASS operons across Clades A, D, and G harbor analogous activation loops. The genetic, chimeric, and structural lines of evidence presented here independently converge on the same fundamental principle — that protease-dependent cleavage of an N-terminal regulatory loop licenses CBASS cyclase activation — and extend it to a guanylyl cyclase–α-PFT system biochemically distinct from EcCdnG. The independent convergence of these findings strengthens, rather than diminishes, the conclusion that proteolytic activation-loop cleavage is a conserved organizational principle of cyclic-nucleotide-based bacterial innate immunity.

4.5 Materials and Methods

4.5.1 General cloning

Plasmids were provided by Dr. Raven Huang. All pBAD33 constructs were cloned with restriction enzymes XbaI and KpnI-HF under an arabinose promoter with a chloramphenicol selection marker. The complete antiphage operon was cloned onto pQE-60 by linearization with HindIII-HF and BamHI-HF under an IPTG-inducible promoter with an ampicillin selection marker.

4.5.2 Spot-test

These methods were adapted from Ofir and colleagues.[43] Phages screened: T3, T4, T5, T6, T7, φ80, P1, lambda, HK97, HK75, and HK544. **Day 0 — bacterial preparation:** Strains KF111 and KF114 were streaked from −80°C glycerol stocks onto LB agar and incubated at 37°C overnight. **Day 1 — liquid-culture initiation:** Plates were removed in the morning. In the late afternoon, a single colony of each strain was inoculated into 2 mL LB in a 10 mL glass culture tube and incubated at 37°C overnight. **Days 2–4 — spot assay (three independent biological replicates, one per day):** Overnight cultures were back-diluted 1:100 in 6 mL LB in a 25 mL glass culture tube and incubated at 37°C on the pinwheel for 3–6 hours. Strains were induced with 0.2% arabinose and 100 µM IPTG for at least 20 minutes. Phage stocks were serially diluted 1:10 in 96-well plate format (20 µL phage into 180 µL LB), changing tips at each dilution and discarding the final 20 µL. Eight milliliters of 0.8% LB top agar was aliquoted into a 14 mL round-bottom culture tube supplemented with 0.2% arabinose and 100 µM IPTG and maintained in a hot-water bath until use. The OD of each strain was measured by 1:10 dilution in 1 mL LB. Each strain was diluted into 8 mL pre-warmed 0.8% LB top agar and poured onto a rectangular LB agar plate. Top agar was allowed to solidify at room temperature without the lid for 5–10 minutes. Using a 100 µL multichannel pipette, 2 µL of each phage dilution was spotted onto the solidified top agar. Spots were allowed to evaporate at room temperature without the lid (∼5–10 minutes), and plates were incubated at 37°C overnight.

4.5.3 Infection dynamics

These methods were adapted from Ofir and colleagues.[43] Overnight cultures were back-diluted in 4 mL LB^{+} and incubated at 37°C for 1.5–2 hours until OD 0.1–0.2. Strains were induced immediately prior to infection with 0.2% arabinose and 100 µM IPTG. 180 µL of induced strains were inoculated into phage or media wells (MOI 0.02 or 2) in a 96-well plate and incubated at 37°C with orbital shaking for six hours, with OD measured every 10 minutes. After six hours, clarified supernatants were serially diluted 1:10 and phage titers enumerated from lawns of defenseless MG1655 at the most dilute spot yielding ≥10 PFU; two countable dilutions were averaged. Percent phage survival was

calculated by dividing the titer from the antiphage-expressing strain by that of the defenseless control and multiplying by 100.

4.5.4 Burst-size assays

These methods were adapted from Hyman and Abedon.[22,23] Strains were back-diluted to OD 0.02 in LB^+ and grown to early log phase (∼2 hours) at 37°C. Strains were induced 20 minutes before infection. Phages were seeded at MOI 0.02; 1 mL of normalized *E. coli* was added, briefly vortexed, and allowed to adsorb for 3 minutes (T6) or 5 minutes (P1, lambda) at room temperature. Infected cultures were pelleted (1 min, top speed), resuspended in fresh wash buffer, and the wash repeated once. The first time point was taken by adding 100 µL aliquots to 10 µL chloroform. Time points were taken every 5 minutes for 25 minutes (T6) and every 15 minutes for 120 minutes (P1, lambda). After all time points were collected, samples were vortexed, centrifuged, diluted 1:10 in LB, and spotted onto MG1655 lawns. Burst size was calculated as the average titer at 45–75 minutes divided by the titer 30 minutes pre-burst. Fold-change was calculated by dividing the empty-vector burst size by the ecC+ecE burst size.

4.5.5 Escaper-phage generation, purification, and high-titer-lysate (HTL) preparation

Methods were adapted from Stokar-Avihail and colleagues.[56] Overnight cultures expressing ecC+ecE were back-diluted in LB^+ and grown to late log phase (5 hours). Cultures were normalized and immediately induced. 100 µL of HTL P1 or lambda lysate was seeded into 3 mL of enriched 0.8% LB top agar, overlaid on an LB plate, cooled for 5–10 minutes, and incubated at 37°C overnight. **Plaque purification:** Filtered tips were clipped with sterile scissors. Individual plaques were plugged and eluted in 50 µL TM buffer + 5 µL chloroform in a 1.5 mL centrifuge tube. Eluates were centrifuged (10,000 rpm, 1 min) and streaked to individual plaques on MG1655 top-agar lawns. Purification was repeated three times. **HTL generation:** MG1655 was grown to early log phase (OD ∼0.1) in 25 mL LB^+ in a 50 mL conical tube on a floor shaker. Three purified plaques were inoculated and cultures incubated at 37°C for 5–7 hours until visibly lysed. Lysates were centrifuged (top speed, 10 min) and filtered through a 0.22 µm PES syringe filter.

Titer and resistance were confirmed by 1:10 dilution spot on MG1655 empty-vector lawns.

4.5.6 Escaper-phage DNA sequencing preparation

MG1655 was back-diluted to OD 0.02 in 260 mL LB + 100 µg/mL ampicillin and grown to early log phase. The culture was split into 10 × 25 mL aliquots; three isolated plaques per aliquot were inoculated and cultures grown for 6 hours until visibly lysed. Lysates were centrifuged (7,893 rpm, 10 min) and filtered through a 0.22 µm PES filter. Titer and resistance were confirmed by spot assay. Lysates were treated overnight at room temperature with 4 µL DNase (50 mg/mL) and then precipitated with 10 mL of 30% PEG 8000 (w/v) + 1 M NaCl at 4°C overnight. Precipitated phage was collected by centrifugation (7,893 rpm, 10 min, 4°C), decanted, and resuspended in 250 µL of 5 mM MgSO. DNA was extracted with Promega Wizard Columns, washed twice with ∼98% isopropanol, dry-spun, and eluted with 50 µL preheated nuclease-free water. DNA was quantified by NanoDrop, stored at −20°C, and submitted to the University of Illinois for Oxford Nanopore long-read sequencing.

4.5.7 Phage-mutation identification

Short-read and long-read sequencing were performed on escapers and parents. Read quality was evaluated with FastQC. Raw reads were mapped to a parent consensus sequence with Minimap2 in Geneious Prime. A mutation was classified as true if both percent reads mapped and percent variant frequency exceeded 90% (Figure S4.2). Parent reads were mapped to reference genomes (lambda Accession: J01416.1; P1 Accession: AF234172) with Minimap2 to rule out false positives. Filtlong was used to filter raw reads to 15–60 kb before mapping where file size precluded whole-file mapping.

4.5.8 Gibson assembly of wildtype and mutant phage gene constructs

Lambda *nu3*, *C*, and P1 *pro* were cloned into pBAD33, and antiphage genes into pQE-60, by Gibson Assembly per the NEB protocol. Constructs were transformed into chemically competent DH5α *E. coli* (42°C, 25 sec), recovered in 1 mL SOC at 37°C with shaking (400 rpm, 45 min), and plated on LB. One colony per construct was grown overnight in 3

mL LB and stored at −80°C in 50% glycerol. Plasmids were miniprepped (Qiagen QIAspin) and confirmed by Sanger sequencing (Plasmidsaurus).

4.5.9 Growth curves

Strains (JMH48–60; KF111–KF114) were streaked onto LB agar and grown overnight at 37°C. Cultures were back-diluted to OD 0.02 in 4 mL LB^{+}, induced with 0.02% arabinose and 10 µM IPTG, and 300 µL of each strain seeded in triplicate into a 96-well plate. OD was measured every 10 minutes for 6 hours at 37°C with orbital shaking. Endpoint CFU/mL was determined by spot assay. Three independent biological replicates were performed.

4.5.10 Higher-order structure plasmid construction

Lambda *B* through *Fii* (parent and Escaper 1) were cloned via Gibson Assembly into pBAD33 as described above and co-transformed with the complete antiphage operon on pQE-60 into MG1655. Growth curves were performed as described above.

4.5.11 Bioinformatic analyses

P1 Pro wildtype and mutant monomer and pentamer structures were predicted with AlphaFold3.[1] Pro wildtype and mutant monomers were structurally aligned with T4 Gp21 (UniProt: P06807) in ChimeraX. Pro homologs were identified from UniProtKB Reference Proteomes and Swiss-Prot via BlastP (default parameters; E-value threshold: 1), followed by sequence alignment with five iterations across the top 15 identified proteins. Redundant cut-site sequence motifs were identified from previously published works[3,18,34,41,45,54,55] and mapped onto cyclase monomers in ChimeraX.

4.6 Author's Note on Data Maturity and Independent Completion

Two limitations of this chapter are acknowledged in the interest of full transparency. First, the chimeric cyclase experiments described in Section 4.3.7 (Figure 4.4) represent a **single biological replicate**; the quantitative comparisons should therefore be interpreted as preliminary, and Section 4.6.2 articulates the gain- and loss-of-function experiments required for definitive replication. Second, the P1 burst-size fold-change stated in Section

4.3.2 (163-fold) was recalculated from the stated average burst sizes (∼326 PFU without ecC+ecE; ∼2 PFU with ecC+ecE); the raw one-step growth data should be re-examined before publication to confirm these values match those presented in Figure 4.1C.

Additionally, as noted in the Introduction and Discussion, the work in this chapter was conceived, executed, and interpreted independently and was completed prior to the bioRxiv preprint of Hobbs and Kranzusch (2026). The independent convergence of these findings on the conclusion that protease-dependent cleavage of an N-terminal regulatory loop licenses CBASS cyclase activation strengthens the broader scientific claim and is acknowledged here in the spirit of full scholarly transparency.

4.7 Figures

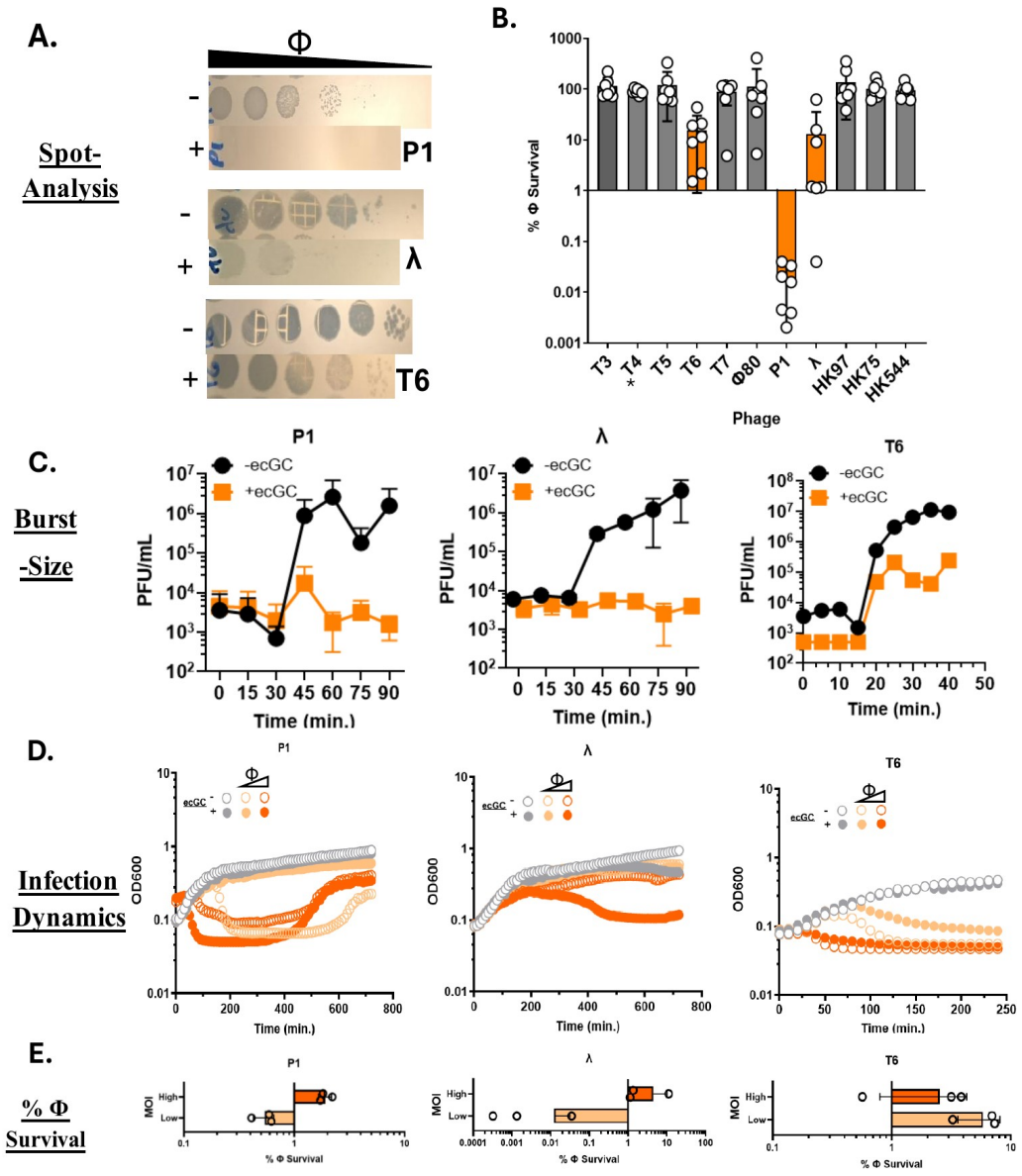


Figure 4.1. The KTE100 guanylyl cyclase–effector operon defends from phages P1, lambda, and T6. High-titer phage lysates (HTL) were diluted 1:10 and each dilution spotted onto lawns of *E. coli* MG1655 ectopically expressing the complete KTE100 guanylyl-cyclase antiphage operon (+ecGC) or the matched empty-vector controls (−ecGC). **(A)** Spot analysis: ectopic expression of the complete antiphage system (ecC+ecE) visibly reduces plaque size and titer of phages P1, lambda, and T6. **(B)** Quantification of percent phage survival across 11 phages screened. Phage

survival is ∼10% for lambda and T6 and <1% for P1. Data are representative of seven biological replicates. **(C)** One-step growth (burst-size) curves over 90 minutes for P1 and lambda, and 50 minutes for T6. Filled circles, with ecGC; filled squares, without ecGC. Data are representative of three biological replicates. **(D)** Liquid-culture infection dynamics. Open symbols, no antiphage defense; filled symbols, +ecC+ecE. Color intensity scales with MOI (light to dark = increasing MOI). Data are representative of three biological replicates. **(E)** Enumeration of total phage survival following the time-course. The asterisk denotes a contaminant in the original "T4" stock subsequently identified by short-read sequencing as *E. coli* φ80.

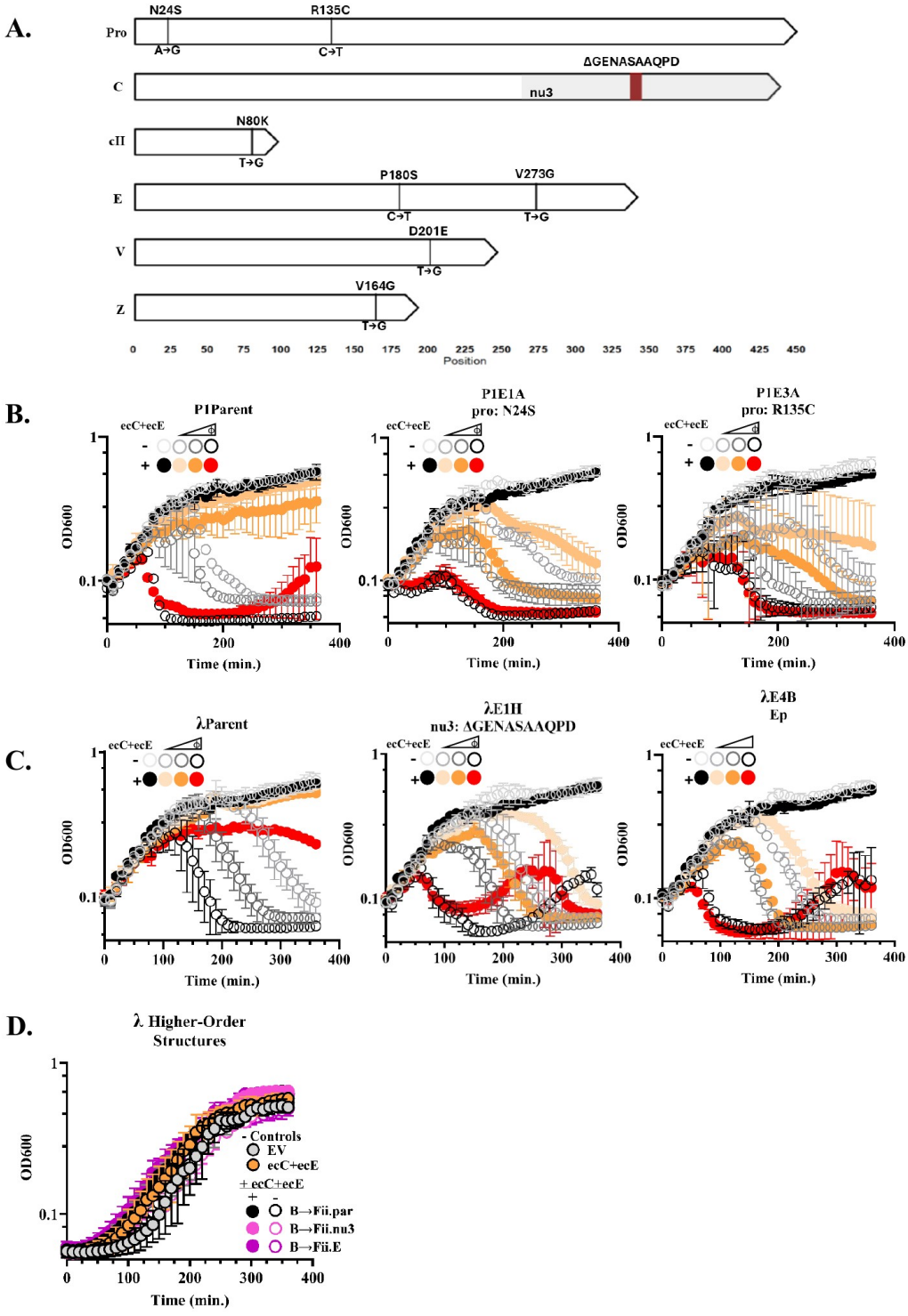


Figure 4.2. Ectopic expression of phage parent genes does not recapitulate abortive infection. (A) Schematic of point mutations and deletions identified in escaper phages P1 (*pro*) and lambda (*C*, *cII*, *E*, *V*, *Z*). Top of arrow: residue substitution; bottom: nucleotide change. The red bar in *C* indicates a clustered deletion ($\Delta^{337/98}$GENASAAQPD$^{345/107}$). **(B)** Infection dynamics of

the P1 parent and two escapers (P1E1A, P1E3A) challenged against the *E. coli* KTE100 antiphage system (ecC+ecE) over six hours. Open circles, no defense; filled circles, active defense. No-phage controls in light grey/black; MOI 0.02 in pink/grey; MOI 0.2 in orange/dim grey; MOI 2 in red/black. Data are representative of two biological replicates. **(C)** Infection dynamics of the lambda parent and two escapers (λE1H, λE4B) challenged against ecC+ecE over six hours. Color scheme as in (B). Data are representative of two biological replicates. **(D)** Infection dynamics of ectopically expressed lambda portal-to-collar complex (B → Fii) from the parent (.par), nu3 mutant (.nu3), and E mutant (.E) over six hours. Open symbols indicate genes expressed without ecC+ecE; filled symbols indicate co-expression with ecC+ecE. Data are representative of three biological replicates.

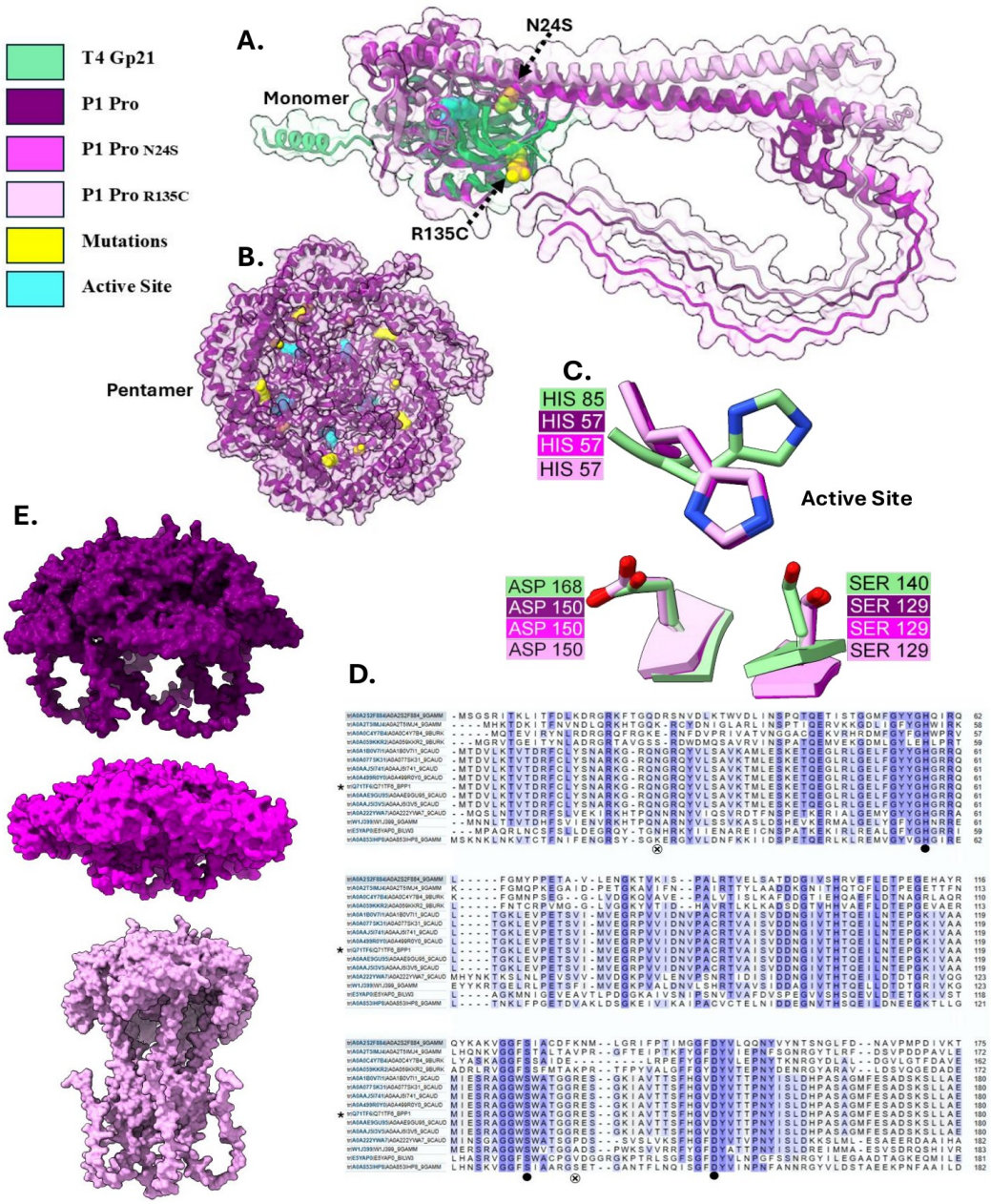


Figure 4.3. P1 Pro escaper mutations alter pentameric scaffold without disrupting the catalytic triad. (A) Structural alignment of phage protease monomers: T4 Gp21 (green), P1 Pro wildtype (purple), Pro N24S mutant (magenta), and Pro R135C mutant (light purple). Mutations are denoted by yellow spheres; active-site residues by cyan spheres. **(B)** Mapping of mutations

and active site onto the parent pentameric structure. **(C)** Active-site residues of the wildtype and mutant proteases. The catalytic triad (His57, Ser129, Asp150 in P1 Pro; His85, Ser140, Asp168 in T4 Gp21) is conserved across all four structures. **(D)** Sequence alignment of the top 15 Pro homologs from UniProt BlastP. Asterisk: Pro query (UniProt: Q71TF6); filled black circles: catalytic triad residues; circled cross: residues mutated in escapers. Blue shading denotes residue identity. **(E)** Predicted Pro pentamers: top, wildtype; middle, N24S mutant; bottom, R135C mutant. The N24S mutant adopts a shallower, concave configuration with a modified lid; the R135C mutant adopts a jellyfish-like topology with an elongated scaffold. Structures were predicted with AlphaFold3 and rendered in UCSF ChimeraX (parent pentamer metrics: pTM = 0.34, ipTM = 0.29, overall rank = 0.37).

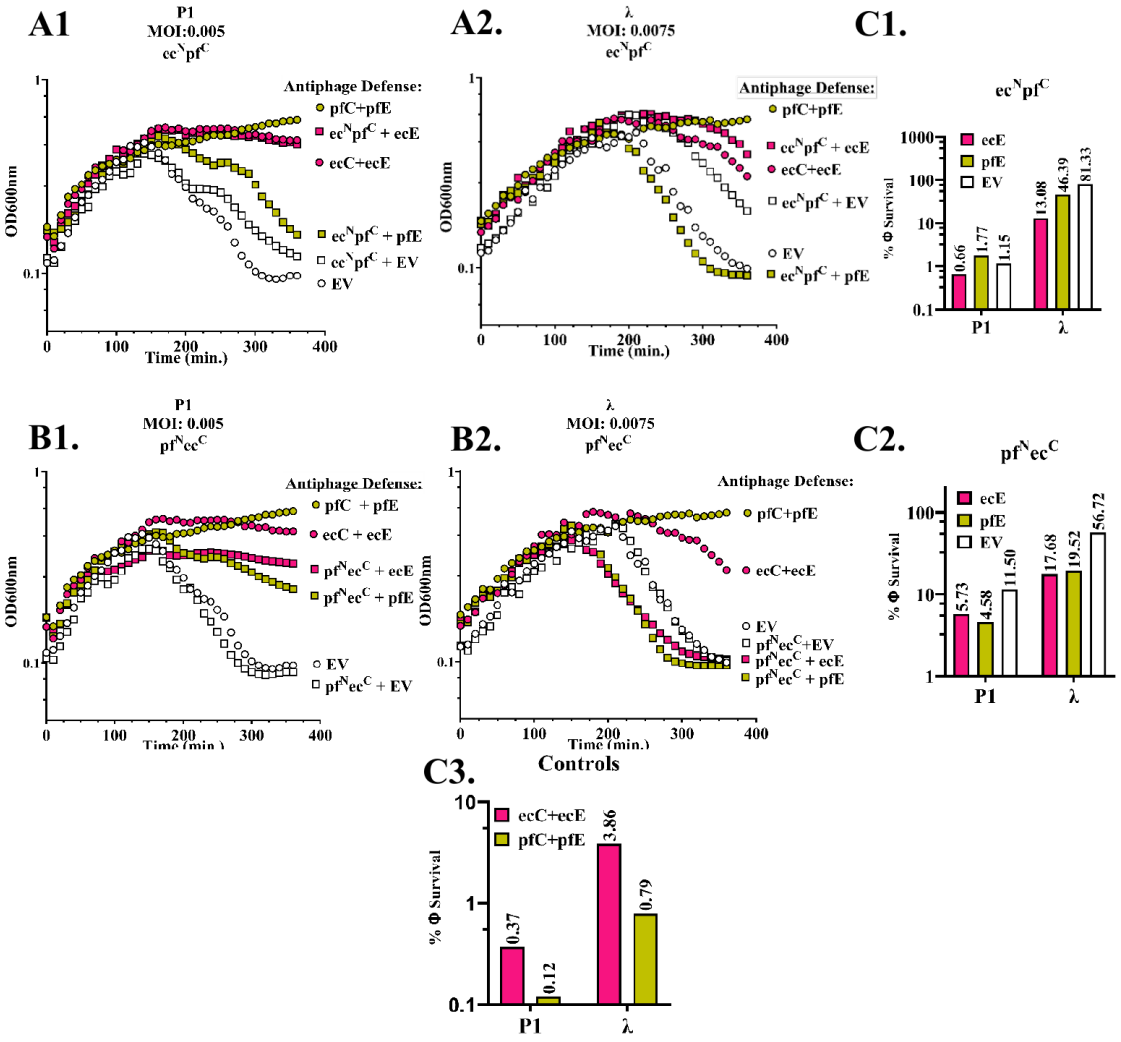


Figure 4.4. The cyclase amino-terminus determines phage sensitivity (preliminary, n = 1). Important caveat: the experiments shown represent a single biological replicate; quantitative comparisons should be interpreted as preliminary, and independent replication is required before firm conclusions are drawn. **(A1)** Infection dynamics of the $ec^{N}pf^{C}$ chimeric cyclase against phage P1 at low MOI (0.005) over six hours. **(A2)** Infection dynamics of the $ec^{N}pf^{C}$ chimeric cyclase against phage lambda at low MOI (0.0075) over six hours. **(B1)** Infection dynamics of the $pf^{N}ec^{C}$ chimeric cyclase against phage P1 at low MOI (0.005) over six hours. **(B2)** Infection dynamics of the $pf^{N}ec^{C}$ chimeric cyclase against phage lambda at low MOI (0.0075) over six hours. **(C1)** Percent phage survival of the $ec^{N}pf^{C}$ fusion cyclase paired with the indicated effector or empty vector. **(C2)** Percent phage survival of the $pf^{N}ec^{C}$ fusion cyclase paired with the indicated effector or empty vector. **(C3)** Percent phage survival of the cognate antiphage defense systems (ecC+ecE; pfC+pfE) compared with empty-vector controls. Percent phage survival was

calculated by dividing the phage titer following six-hour incubation with the chimeric cyclase + effector (or empty vector) combinations by the empty-vector titer.

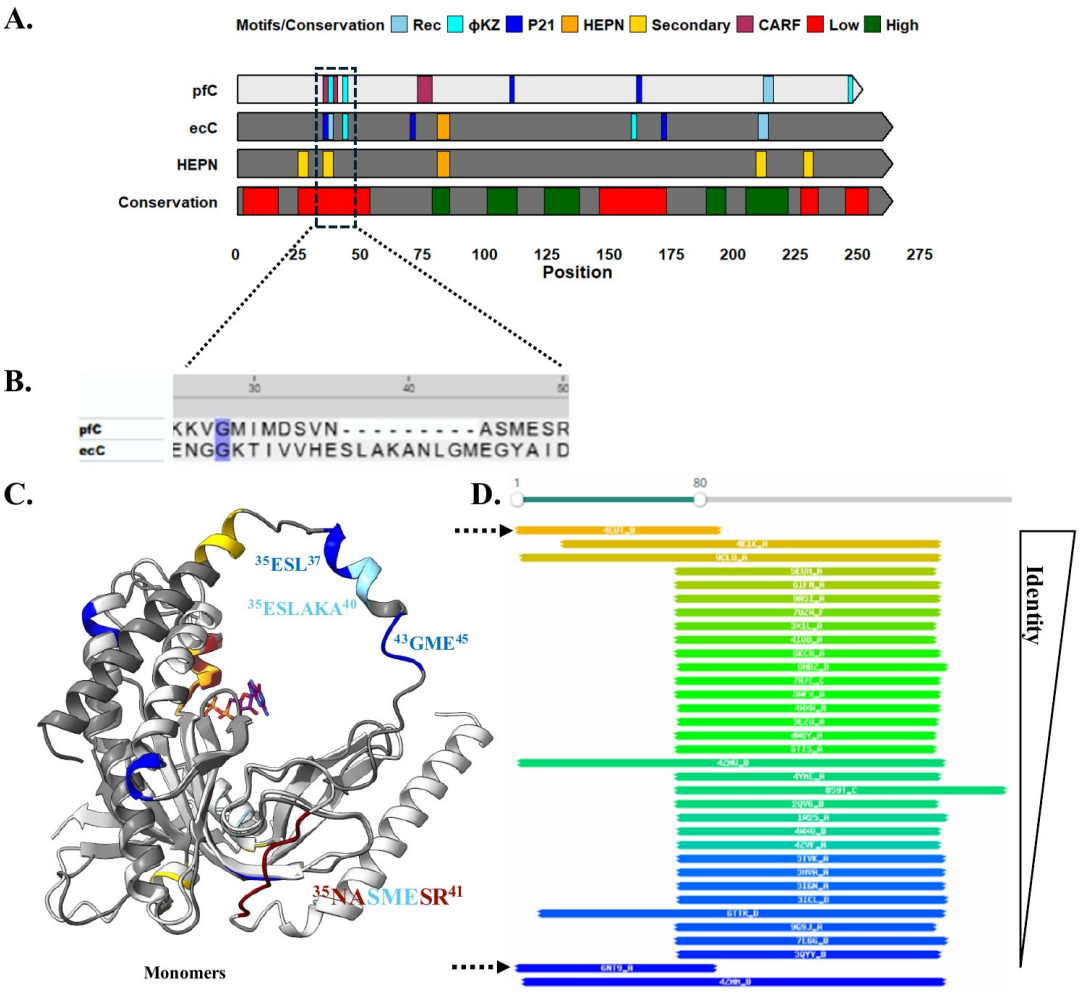


Figure 4.5. The *E. coli* cyclase harbors a putative TBK1-like phosphorylation/cleavage loop on its N-terminus. (A) Linear domain map of the *P. fluorescens* (pfC; light grey) and *E. coli* (ecC; dark grey) cyclase protein sequences, annotated with CARF motifs (maroon), ClpX-recognition motif (Rec; sky blue), ϕKZ Gp175 cleavage sites (cyan), and T4 P21 cleavage sites (royal blue). The ecC cyclase harbors primary (orange) and secondary (gold) HEPN motifs that fall within a region of low conservation on the N-terminus (red conservation track). **(B)** Detailed sequence alignment of the first 76 residues of pfC and the first 85 residues of ecC. Horizontal dashes denote a nine-residue gap in pfC relative to ecC; blue shading denotes residues identical between the two cyclases. **(C)** Bird's-eye view of the predicted cyclase monomer structural alignment with one bound GTP molecule (purple). Colored regions correspond to the motifs identified in the cyclase sequences (panel A). The nine-residue insertion in ecC (sky blue helix) generates a structural "spine" that traverses the dimerization interface and is absent from pfC. **(D)** HHPRED hits-analysis of the *E. coli* cyclase. Dashed black box indicates the swapped residues. Black dashed arrows indicate a phosphorylation sensor identified by the HHPRED analysis.

HEPN and cleavage motifs are highlighted in orange and blue, respectively. Motifs were identified using Perplexity, Prosite, and the literature.

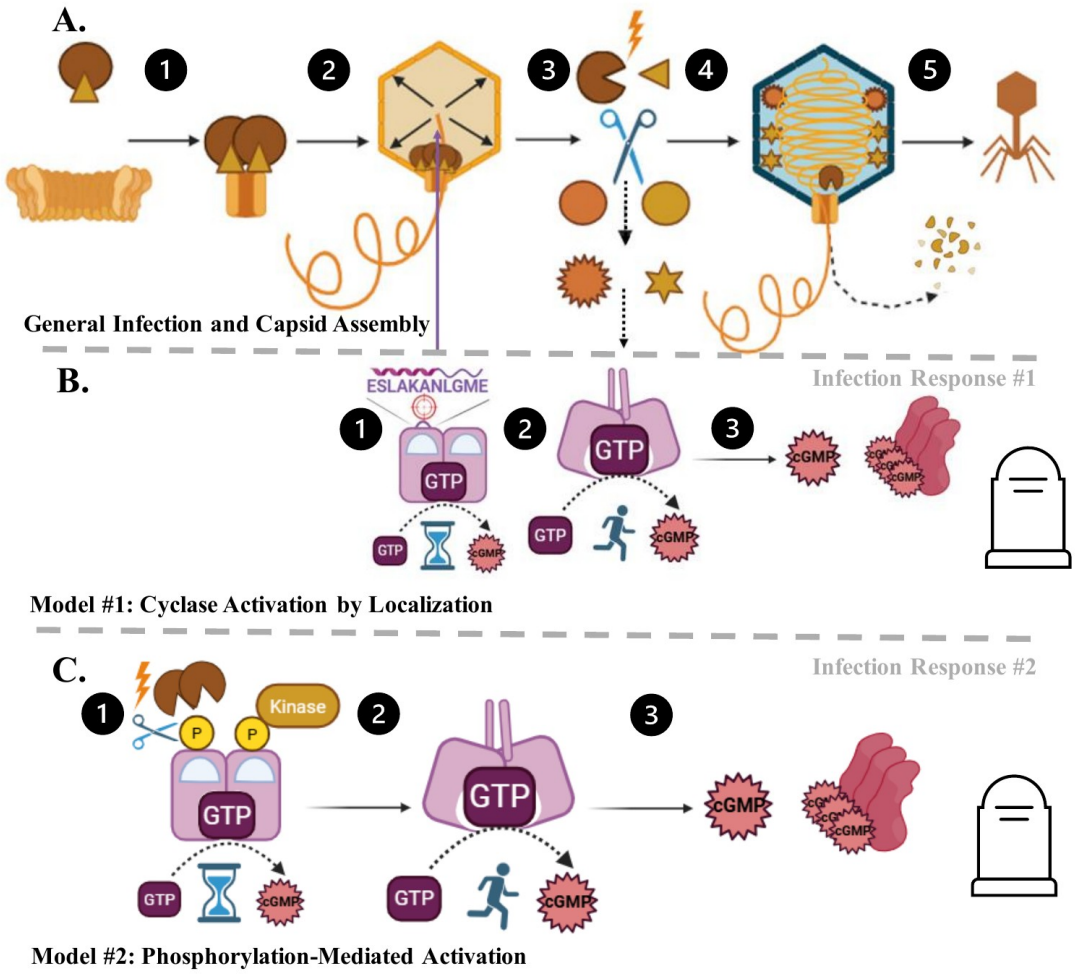


Figure 4.6. Two non-mutually-exclusive models for guanylyl cyclase activation by P1 phage infection. (A) General phage infection and capsid assembly. **(A1)** Virion assembly initiates with portal-complex formation. **(A2)** The scaffold and protease oligomerize onto the portal complex to generate the prohead. **(A3)** Major capsid protein is recruited and DNA packaging expands the procapsid into the mature capsid, activating the protease via autocleavage. **(A4)** The activated protease processes viral structural and internal proteins for packaging into the capsid. **(A5)** The mature head connects to the tail to form the complete virion. **(B)** Model #1: Cyclase Activation by Localization. **(B1)** The inverted P21 cleavage site within the ecC N-terminus localizes the cyclase to the interior capsid space via the P21 Capsid Targeting Sequence. **(B2)** The activated protease cleaves the redundant cut-site junction on the cyclase N-terminus. **(B3)** Cleavage relieves autoinhibition, increases cGMP production, and oligomerizes the ClyA-like α-PFT effector to commit the host to abortive infection. **(C)** Model #2: Phosphorylation-Mediated Activation. **(C1)** A phage-encoded kinase (e.g., the P1 Doc toxin) phosphorylates ecC N-terminal residues during infection, recruiting the activated phage protease to the redundant cut-site

junction. **(C2)** The cyclase spine is cleaved. **(C3)** Cleavage drives cGMP above the effector oligomerization threshold, committing the host to abortive infection.

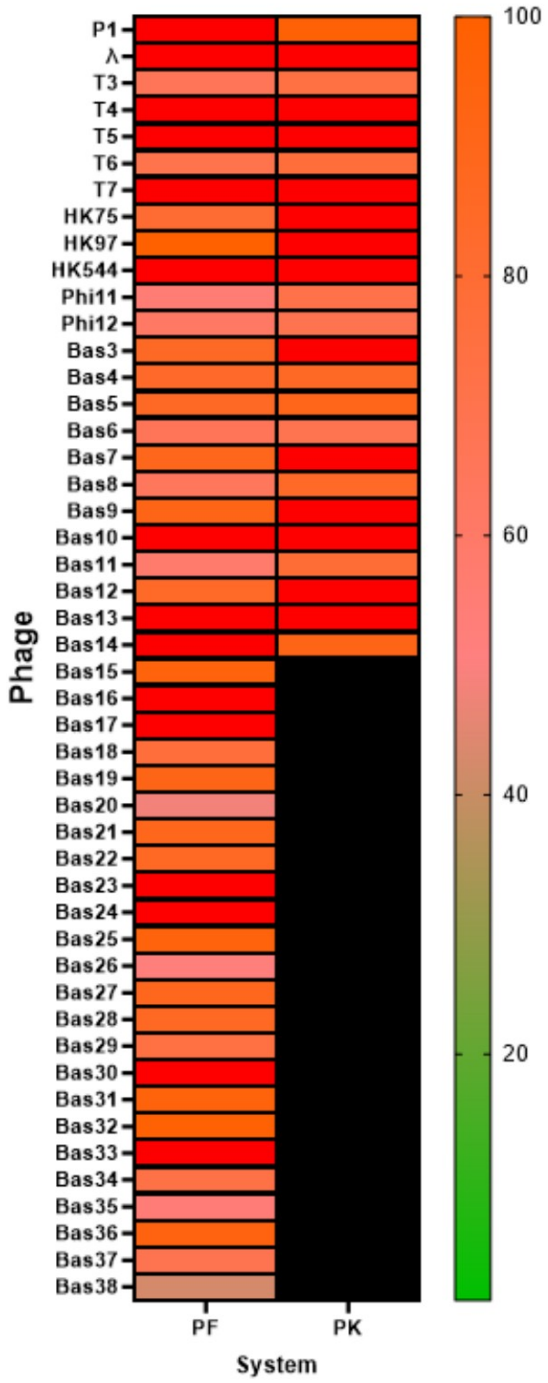


Figure S4.1. Spot-analysis of *P. fluorescens* and *P. koreensis* antiphage defenses against 40 phages. Heatmap of phage spot-plate efficiency-of-plating (EOP) on lawns of *E. coli* MG1655 ectopically expressing the putative antiphage defense from *P. fluorescens* (PF; left column) or *P. koreensis* (PK; right column). Classic laboratory phages and BASEL collection phages were

diluted 1:10 and spotted on lawns of MG1655 ectopically expressing the cyclase, the effector, or the complete defense system. Percent phage survival was determined by dividing the titer of each phage on the +GC strain by that on the −GC control and multiplying by 100, for both low and high MOI. Black tiles denote phages not tested against the *P. koreensis* system. Color scale indicates percent phage survival; red, low survival (strong defense); orange, intermediate; green, high survival (weak or no defense).

% Variant Frequencies

Genes→ Mutation→ ΔNucleotide→	Nu3 (del)	cII N80K (T=>G)	Ep N/A (A=>T)	E P180S (C=>T)	E V273G (T=>G)	V D201E (T=>G)	Z V164G (T=>G)
λE1F.LR				100			
λE1H.LR	99.3						
λE1H.SR	77-99.8	24.7				31.9	20.9
λE4B.SR			99.7				
λE4C.SR		23.9	98.6			27.3	
λE5B.SR		25.5			99.9	32.1	20.4
λE5C.SR		24.9	99.3			31.4	

Genes→ Mutation→	Pro (N24S) A=>G	Pro (R135C) (C=>T)
P1E1A.LR	100	
P1E2A.SR	100	
P1E2B.SR	100	
P1E3A.SR		100
P1E3B.SR		100

Abbreviations:
(e.g. P1E1A.SR)
Phage = P1/λ
E = Escaper,
= Biological Replicate,
Letter = Technical Replicate,
LR/SR = Long/Short Read

Figure S4.2a. Variant frequencies of escaper phage genomes. Long-read or short-read escaper phage genomes were mapped to the lambda (NCBI Accession: NC_001416) or P1 (NCBI Accession: AF234172) reference sequence. Ep denotes the promoter region of gene *E*. Naming convention (e.g., λE1H.SR): phage = lambda or P1; E = escaper; numeral = biological replicate; letter = technical replicate; LR/SR = long-read/short-read. Variant frequencies are reported as percentages; cells without entries indicate variant frequency below the threshold for true-mutation classification (90%) for that gene–replicate combination. Convergent mutations across biological replicates: P1 escapers exclusively mutated *pro*; lambda escapers consistently acquired mutations in *cII* (N80K) and *gpV* (D201E) across all three biological replicates.

	Phage	% Reads Mapped	Total Raw Reads	Reads Mapped	Reads Not Mapped
λ	Parent.LR	99.99	160,228	160,228	21
	Parent.SR	98.67	7,484,852	7,484,852	100,740
	λE1H.SR	98.09	6,691,626	6,691,626	127,696
	λE1H.LR	99.96	202,394	202,394	72
	λE1F.LR	98.97	867	867	9
	λE4B.SR	97.87	6,690,868	6,690,868	145,510
	λE4C.SR	97.72	3,808,003	3,808,003	88,865
	λE5B.SR	97.91	6,368,610	6,368,610	135,820
	λE5C.SR	97.14	6,565,760	6,565,760	193,220

	Phage	% Reads Mapped	Total Raw Reads	Reads Mapped	Reads Not Mapped
P1	Parent.LR	98.29	29,583	29,077	506
	Parent.SR	94.83	6,814,320	6,462,209	352,111
	P1E1A.LR	98.13	1,122	1,101	21
	P1E2A.LR	92.62	2,763	2,559	204
	P1E2A.SR	34.88	9,909,110	3,456,401	6,456,709
	P1E2B.LR	86.23	3,318	2,861	457
	P1E3A.LR	94.81	27,067	25,662	1,405
	P1E3A.SR	56.46	5,030,842	2,840,242	2,190,600
	P1E3B.LR	95.16	36,402	34,640	1,762

Abbreviations:

(e.g. λE1H.SR) Phage= λ/P1, E = Escaper, # = Biological Replicate, (A-H) = Technical Replicate . LR/SR = Long/Short Read

Figure S4.2b. Coverage and mapping efficiency of short-read or long-read phage genomes. Phage genomes consist of parent phages lambda or P1 with their corresponding escapers (E), enumerated by biological replicate (numeral) and technical replicate (letter). LR = long-read; SR = short-read. Percent reads mapped was calculated by dividing total raw reads by reads successfully mapped and multiplying by 100. Raw reads were mapped with Minimap2 in Geneious Prime against the appropriate parent reference genome.

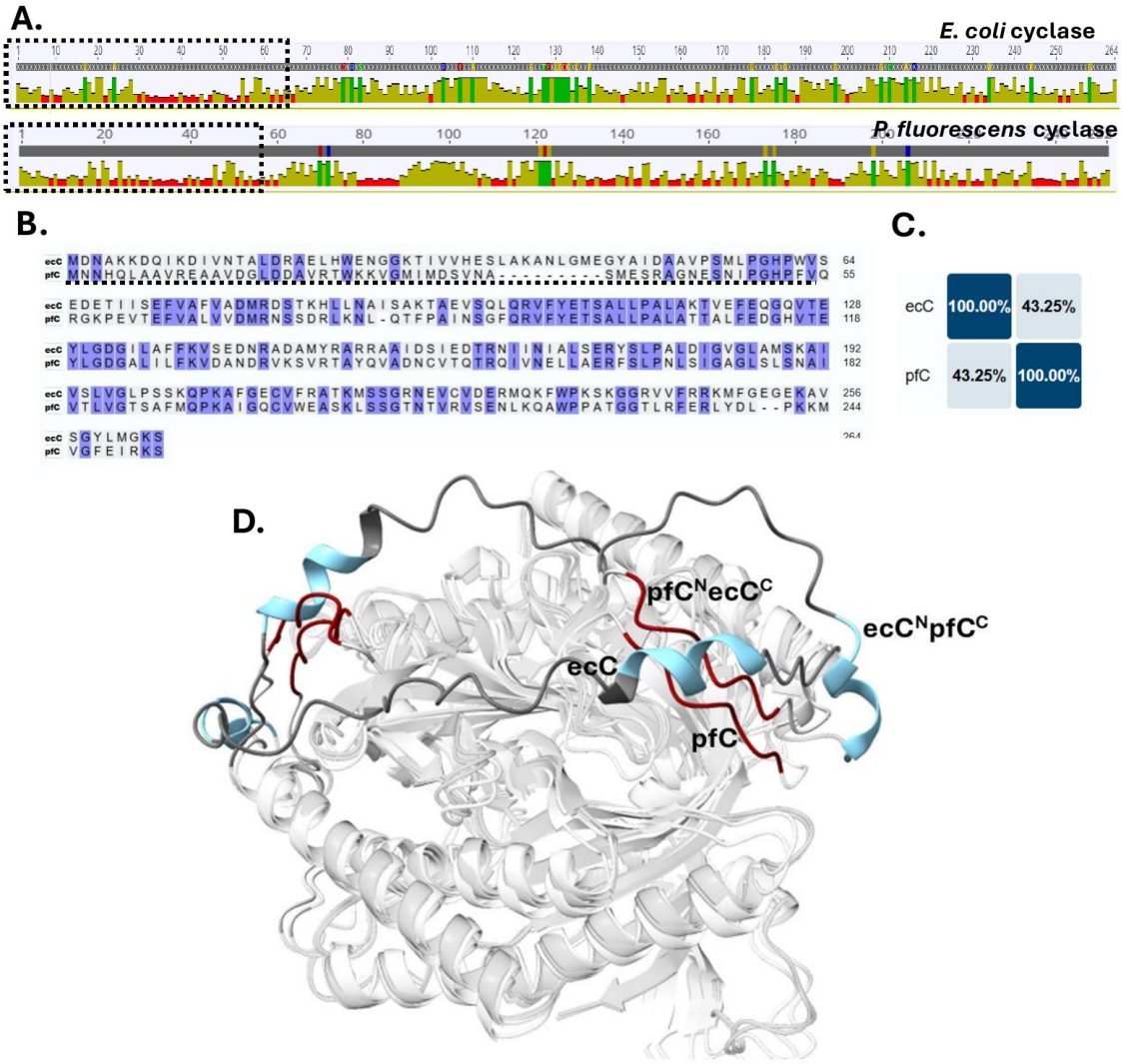


Figure S4.3. Complete alignment of *E. coli* and *P. fluorescens* cyclases. (A) NCBI BlastP alignment of the *E. coli* (ecC) and *P. fluorescens* (pfC) cyclase protein sequences. Numbers denote residue positions. Hashed black boxes denote the N-terminal regions exchanged in the chimeric constructs. Colored vertical bars denote per-residue identity (high–medium–low: green–red–yellow). **(B)** Sequence alignment of the *E. coli* (ecC) and *P. fluorescens* (pfC) cyclases generated in UniProt. Residue identity is rendered by depth of blue shading. The black dashed line denotes the N-terminal regions exchanged in the chimeras. Floating horizontal dashes denote a nine-residue gap in pfC. **(C)** ecC/pfC residue-identity matrix (43.25% pairwise identity). **(D)** Predicted structural alignment of ecC, pfC, and the two chimeric cyclases ($ec^{N}pf^{C}$, $pf^{N}ec^{C}$). Sky blue: residues ESLAKA from the ecC N-terminus. Maroon: residues NASMESR from the pfC N-terminus. Structures were predicted with AlphaFold3 and rendered in UCSF ChimeraX.

4.8 Abbreviations

CARF CRISPR-Associated Rossmann Fold
CBASS Cyclic-Oligonucleotide-Based Antiphage Signaling Systems
CD NCBI Conserved Domains tool
cGMP Cyclic Guanosine Monophosphate
ClyA Cytolysin A
ecC *E. coli* KTE100 Cyclase
ecC+ecE *E. coli* MG1655 with pQE-60.ecC + pBAD33.ecE inducible expression system
ecC^{N}pfC^{C} ecC N-terminus fused to pfC C-terminus
ecE *E. coli* KTE100 Effector
GTP Guanosine Triphosphate
HEPN Higher Eukaryote–Prokaryote Nucleotide-Binding domain
HTL High-Titer Lysate
LB Luria Broth
LB^{+} LB + 5 mM $MgCl_2$ + 0.1 mM $MnCl_2$
LB$_a$χ LB + 100 µg/mL ampicillin + 50 µg/mL chloramphenicol
LB$_a$χ^{+} LB$_a$χ + 5 mM $MgCl_2$ + 0.1 mM $MnCl_2$
LB$_a$χ^{+I} LB$_a$χ^{+} + 0.2% arabinose + 100 µM IPTG
MOI Multiplicity of Infection
NF-H_2O Nuclease-Free Water
pfC *P. fluorescens* Cyclase
pfC+pfE *P. fluorescens* Cyclase + Effector (two plasmids; two independent promoters)
pf^{P}ec^{c} pfC N-terminus fused to ecC C-terminus
pfE *P. fluorescens* Effector
PFT Pore-Forming Toxin
PFU Plaque-Forming Units
pkC *P. koreensis* Cyclase
pkC+pkE *P. koreensis* Cyclase + Effector
PYCSAR Pyrimidine Cyclase System for Antiphage Resistance

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Future Directions:

Demonstrate Phage head-processing protease cleave N-terminal cyclase

Increase in cyclic guanosine monophosphate (cGMP) should be demonstrated with and without phage infection via ELISA[70]. Once established site-directed mutagenesis assay of the cyclase redundant cut-sites junction should be investigated to determine sensitivity of mutated residues. cGMP ELISA should be performed in tandem with site-directed mutagenesis to support functionality cGMP synthesis of the cyclase. Western blots should be performed to demonstrate N-terminal cleavage and/or phosphorylation of the cyclase. A bacterial two-hybrid assay should be performed to demonstrate the localization of the antiphage cyclase and phage protease. Mutations to the cGMP-catalytic core HEPN domain of the *E. coli* cyclase should be performed to demonstrate the necessity of cGMP to oligomerize the downstream ClyA-like pore-forming effector.

Investigations into the role cGMP plays in cell wall remodeling

Ectopic expression of the *E. coli* str. KTE100 guanylyl cyclase has been shown to modulate cell wall function via compound-microscopy and gram-staining. I surmise cGMP synthesis is either modulating peptidoglycan synthesis or inhibiting cell wall recycling. Immediate priorities should, I. measure peptidoglycan synthesis via transmission electron microscopy, then quantification of total peptidoglycan content can be confirmed via extracted and muropeptides measured by HPLC or LC-MS. II. Cyclic GMP synthesis via ELISA should be performed to demonstrate elevated levels compared to a negative control is needed to elicit cell wall modularity. Cyclic GMP analogs or

inhibitors should be tested to ensure the cGMP-dependent phenotype. III. Outer-membrane integrity assays investigating SDS sensitivity should be performed to test permeability. IV. RNA-seq or qRT-PCR should be used to identify which cell wall genes are differentially regulated during transcription, with particular focus on penicillin-binding proteins, peptidoglycan synthetases, outer membrane biogenesis proteins (Lpt system, Bam complex), LPS enzymes, and amidases such as ampD, ampG, nagZ, anmK, murQ, mpl. Potential cGMP-binding proteins can further be identified by pull-down assays. AmpD amidase activity should be measured from cell lysates or assays that investigate transglycosylase activity.